



Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich



Energy Systems Analysis

Konstantinos Furlaris

System-level optimization for the integration of heat pumps in district heating and cooling

MSc Thesis Report

Master in Energy Science and Technology

Chair of Energy Systems Analysis (ESA)
ETH Zurich

Supervising Professor: Prof. Dr. Russell McKenna
Project Supervisors: Dr. Tom Mike Terlouw and Dr. Tarek Alskaf

Zurich, September 2, 2026

Abstract

This Master’s thesis presents a two-stage optimization framework to evaluate the techno-economic viability and decarbonization potential of integrating large-scale heat pump (LSHP) and thermal energy storage (TES) into transitioning 4th generation district heating and cooling networks (DHCNs) within a multi-energy system (MES) paradigm. The first stage employs a deterministic Mixed-Integer Linear Programming (MILP) of Total Annual Cost (TAC) minimization model at an hourly resolution to determine the optimal asset sizing and baseline operational dispatch across greenfield and brownfield investment pathways, benchmarked against pre-existing Biomass Boiler (BB) and Combined Heat and Power (CHP) units in Zurich and Amsterdam for the brownfield approach. The second stage implements a stochastic MILP scenario-based co-optimization model at a 15-minute resolution, utilizing ten-year historical balancing market (BM) price clustering via K-Medoids to capture the potential to further diminish the total annualized system costs by offering automatic Frequency Restoration Reserve (aFRR) energy flexibility in secondary BM alongside Day-Ahead Market (DAM) procurement. Also, a comprehensive ecological footprint Life Cycle Assessment (LCA) tracks both embedded and operational greenhouse gas (GHG) emissions of the optimization outcomes. Results demonstrate that coupling LSHP with TES significantly lowers the TAC and levelized cost of provided heating and cooling energy through the DHCN while reducing lifecycle GHG emissions—achieving up to a 57% emissions reduction in Zurich under a greenfield pathway and lowering annualized costs by 11–16% in brownfield settings compared to fossil fuel-based supply strategies. Furthermore, active participation in the aFRR balancing market unlocks an additional 9–13% reduction in total expected annualized costs across both cities, driven primarily by remunerative downward and upward balancing flexibility and the physical decoupling heat production from instantaneous thermal consumption capabilities provided by the TES, with downward flexibility activations strongly prevailing over upward flexibility activations.

Acknowledgments

I would like to thank Prof. Dr. Russell McKenna and the Chair of Energy Systems Analysis (ESA) at ETH Zurich for providing me with the opportunity to carry out this research. I am especially grateful to my supervisors, Dr. Tom Mike Terlouw and Dr. Tarek Alskaif, for their consistent support, thoughtful feedback, constant motivation, and guidance throughout this thesis. Their commitment and the time they dedicated to supervising my work were invaluable.

I would also like to thank Sammit Vartak, for explaining me how to create heating and cooling demand profiles through the Hotmaps project. Special thanks go to Julian Hermann for his valuable comments on my mid-term presentation and his proposal of literature papers for modeling heat pumps, which proved particularly useful, and to Martin Densing for his initial feedback on my second-stage optimization architecture.

As this thesis marks the final step of my Master of Science in Energy Science and Technology, I would like to express my deepest gratitude to my parents and friends for their unwavering support and encouragement throughout my academic journey. I am also grateful to the HELLENiQ ENERGY Group for covering my tuition fees through the scholarship they awarded to me.

Furthermore, the code and input data used to execute this research are publicly available in the following GitHub repository:

`https://github.com/kfourlaris/MSc-thesis---Konstantinos-Fourlaris-`

Finally, during the writing of this thesis, Gemini was used to improve the readability and language of the report. The content was thoroughly reviewed, edited, and verified, and I take full responsibility for the final text.

Contents

Abstract	i
Acknowledgments	ii
List of Figures	v
List of Tables	vii
List of Acronyms	viii
Nomenclature	ix
1 Introduction	1
1.1 Literature Gaps and Novelty	3
1.2 Research Questions	4
1.3 Structure of the Thesis	5
2 Background	6
2.1 Industry State of the Art	6
2.2 Modeling Methods and Optimization Frameworks	8
3 Optimization Methodology	12
3.1 First Stage Optimization	16
3.1.1 Decision Variables	16
3.1.2 Objective Function	18
3.1.3 Global Energy Network Balance Constraints	18
3.1.4 Technology Specific Constraints	19
3.2 Second Stage Optimization	21
3.2.1 Decision Variables	22
3.2.2 Objective Function	23
3.2.3 Global Energy Network Balance Constraints	25
3.2.4 Technology Specific Operational Constraints	26
3.2.5 Expected BM Energy Volumes Metrics	29

4	Input Data & Post-Optimization Analysis	30
4.1	Input Data and Parameters	30
4.1.1	Heating and cooling yearly demand profiles ($D_{H,t}$ and $D_{C,t}$)	30
4.1.2	District network's temperature levels and LSHP's heating and cooling coefficient of performance (COP_H and COP_C)	34
4.1.3	Energy efficiencies of BB, CHP and TES	37
4.1.4	Deterministic fuel prices parameters	38
4.1.5	Stochastic energy balancing prices clustering	40
4.1.6	Capital Recovery Factor	42
4.1.7	Technology specific costs, market sizes and minimum load fractions	42
4.2	Environmental Perspective: the Assessment of GHG Emissions	47
4.2.1	Operational GHG Emissions	47
4.2.2	Technology Embedded Emissions	48
4.2.3	Total System Emissions Aggregation	49
4.3	MES Configurations considered and $LCOE_{DN}$	50
5	Optimization Results & Associated GHG Emissions	51
5.1	Brownfield Investment Pathway	51
5.1.1	First-Stage Optimization Results	52
5.1.2	Second-Stage Optimization Results	62
5.2	Greenfield Investment Pathway	67
5.2.1	First Stage Optimization Results	68
5.2.2	Second Stage Optimization Results	72
6	Discussion	76
6.1	Results Reflection	76
6.2	Limitations of the Study	77
6.3	Recommendations for Further Research	79
7	Conclusions and Outlook	80
8	Appendix	82
8.1	Literature review Tables	82
8.2	Lake/Seawater monthly temperature fluctuations for the calculation of COP	88
8.3	TTF natural gas prices	88
8.4	K-Medoids scenario clustering formulation	88
8.5	Full year optimization results plots	89
	Bibliography	91
	Declaration of Originality	100

List of Figures

2.1	The main structural components and the physical layout of the Everlence’s mega HP project adopted by [32].	7
2.2	The historic transition towards 4GDH networks adopted by [12] licensed by CC BY-NC 4.0.	10
3.1	A schematic of the proposed MES system configuration assumed for this study with the following energy carriers: Biomass (B), Natural Gas (NG), Electricity (E), Heat (H) and Cooling (C).	15
5.1	Baseline configuration seasonal heating and cooling dispatch operational profiles for the Zurich brownfield network approach.	54
5.2	Baseline configuration seasonal heating and cooling dispatch operational profiles for the Amsterdam brownfield network approach.	55
5.3	HP w/o storage configuration seasonal heating and cooling dispatch operational profiles for the Zurich brownfield network approach.	57
5.4	HP w/o storage configuration seasonal heating and cooling dispatch operational profiles for the Amsterdam brownfield network approach.	58
5.5	HP w/ storage configuration seasonal heating and cooling dispatch operational profiles for the Zurich brownfield network approach.	60
5.6	HP w/ storage configuration seasonal heating and cooling dispatch operational profiles for the Amsterdam brownfield network approach.	61
5.7	Seasonal heat dispatch profiles of Scenario 1 for Zurich with full participation in the BM during a brownfield network approach.	65
5.8	Seasonal heat dispatch profiles of Scenario 1 for Amsterdam with full participation in the BM during a brownfield network approach.	66
5.9	“HP w/ storage” configuration seasonal heating and cooling dispatch operational profiles for the Zürich greenfield network approach.	70
5.10	“HP w/ storage” configuration seasonal heating and cooling dispatch operational profiles for the Amsterdam greenfield network approach.	71
5.11	Seasonal heat dispatch profiles of Scenario 1 for Zurich with full participation in the BM during a greenfield network approach.	74
5.12	Seasonal heat dispatch profiles of Scenario 1 for Amsterdam with full participation in the BM during a greenfield network approach.	75

8.1	HP w/ storage configuration full year heating and cooling dispatch operational profiles for the Zurich brownfield network approach.	90
-----	--	----

List of Tables

4.1	Structural energy demand and sector breakdown parameters	33
4.2	Summary of steady-state energy efficiencies	38
4.3	Investment economic parameters used for the examined technologies in (M€) during the Greenfield approach	44
4.4	Technology capacity sizing limits	45
4.5	Minimum load fractions by technology during heating operation	46
5.1	First stage optimization results for Zurich brownfield approach	52
5.2	First stage optimization results for Amsterdam brownfield approach . . .	52
5.3	Second stage optimization results for Zurich brownfield approach	63
5.4	Second stage optimization results for Amsterdam brownfield approach . .	63
5.5	Expected operational flexibility provided by the LSHP across the opti- mization horizon under a brownfield approach.	63
5.6	First stage optimization results for Zurich greenfield approach.	68
5.7	First stage optimization results for Amsterdam greenfield approach. . . .	68
5.8	Second stage optimization results for Zurich greenfield approach	72
5.9	Second stage optimization results for Amsterdam greenfield approach . . .	73
5.10	Expected operational flexibility provided by the LSHP across the opti- mization horizon under a greenfield approach.	73
8.1	Literature review summary (Papers 1-11)	83
8.2	Literature review summary (Papers 12-20)	84
8.3	Literature review summary (Papers 21-29)	85
8.4	Literature review summary (Papers 30-40)	86
8.5	Literature review summary (Papers 41-43)	87
8.6	Seawater temperatures in Denmark	88
8.7	TTF monthly natural gas wholesale prices for 2025	88

List of Acronyms

4GDH	4th-Gen. District Heating	ID	Intraday
aFRR	Auto. Frequency Restoration Reserves	IEA	International Energy Agency
AM	Amsterdam	IPCC	Intergov. Panel on Climate Change
ASHRAE	Amer. Soc. of Heat., Refrig. & AC Eng.	IQR	Interquartile Range
BB	Biomass Boiler	LCA	Life Cycle Assessment / Analysis
BDEW	German Assoc. of Energy & Water Mgmt.	LCOE_{DN}	Lev. Cost of Energy of the DN
BM	Balancing Market	LHV	Lower Heating Value
CAPEX	Capital Expenditure	LP	Linear Programming
CCGT	Combined Cycle Gas Turbine	LSHP	Large-Scale Heat Pump
CDD	Cooling Degree Days	MES	Multi-Energy System
CF	Conversion Factor	MILP	Mixed-Integer Linear Programming
CH	Electric Chiller	MINLP	Mixed-Integer Non-Linear Program.
CHP	Combined Heat and Power	MIQCP	Mixed-Integer Quadratically Constrained
COP	Coefficient of Performance	O&M	Operation and Maintenance
CRF	Capital Recovery Factor	OPEX	Operational Expenditure
DA / DAM	Day-Ahead / Day-Ahead Market	P2H	Power-to-Heat
DCN	District Cooling Network	PAM	Partitioning Around Medoids
DH / DHN	District Heating / Network	PICASSO	Platform for aFRR Coordination
DHCN	District Heating and Cooling Network	PTES	Pit Thermal Energy Storage
DHW	Domestic Hot Water	PV	Photovoltaic
DN	District Network	SC	Space Cooling
ENTSO-E ..	Eur. Network of TSOs for Electricity	SH	Space Heating
ETS	EU Emissions Trading System	SLP	Synthetic Load Profile
EU	European Union	SoC	State of Charge
EV	Electric Vehicle	SWM	Stadtwerke München
FCR	Frequency Containment Reserves	TAC	Total Annualized Cost
GHG	Greenhouse Gas	TEAC	Total Expected Annualized Cost
GT	Grid Tariff	TES	Thermal Energy Storage
HDD	Heating Degree Days	TSO	Transmission System Operator
HP	Heat Pump	TTF	Title Transfer Facility
HTHP	High-Temperature Heat Pump	WC	Water Chiller
HVAC	Heating, Ventilation, and AC	WtE	Waste-to-Energy
HWS	Hot Water Storage	ZH	Zurich

Nomenclature

Indices

c	Set containing examined cities district networks $\{ZH, AM\}$
j	Set containing examined energy carriers $\{NG, B, E, H, C\}$
k	Set containing examined energy generation technologies $\{BB, CHP, HP, CH\}$
s	Stochastic scenarios in the second-stage optimization formulation ($s \in S$)
t	Time resolution intervals ($t \in T$ for hourly, $t \in T_{15}$ for 15-minute)

Parameters

α	Capital recovery factor
δ_k	Minimum load fraction of technology k ($\delta_{BB}, \delta_{CHP}, \delta_{HP}$)
Δt	Time step integration factor (1 h for first-stage optimization, 0.25 h for second-stage optimization)
$e_{em,k}$	Standardized embedded capacity emission factor for conversion technology k [t CO ₂ -eq/MW]
$e_{em,TES}$	Standardized embedded capacity emission factor for the TES [t CO ₂ -eq/MWh]
$e_{op,j}^c$	Life cycle operational emission factor for input energy carrier j in district network c [t CO ₂ -eq/MWh]
$\eta_{el,CHP}$	Electrical efficiency of the combined heat and power plant
$\eta_{th,CHP}$	Thermal efficiency of the combined heat and power plant

Continued on next page

$\eta_{c, TES}, \eta_{d, TES}$	Thermal charging and discharging efficiency of the pit thermal energy storage
$\lambda_{TES}, \lambda_{15min}$	Standing hourly and 15-minute self-discharge loss rates of the pit thermal energy storage
$\Lambda_{revenue}$	Fixed contract electricity export tariff / feed-in tariff of the combined heat and power plant [€/MWh]
$\Lambda_{gas,t}$	Monthly varying TTF gas price for the combined heat and power plant [€/MWh]
$\kappa_{gas,t}$	Daily varying EU ETS carbon prices for the combined heat and power plant [€/tCO ₂]
μ	Acceptance ratio of upward balancing energy activation bids by the LSHP
ν	Acceptance ratio of downward balancing energy activation bids by the LSHP
$\pi_{biomass}$	Constant yearly biomass fuel price for the biomass boiler [€/MWh]
$\pi_{gas,t}$	Daily varying input energy price for the combined heat and power plant [€/MWh]
$\pi_{elec,t}, \pi_{elec,t,s}$	Time-varying, day-ahead wholesale electricity market price including grid tariffs [€/MWh]
$\pi_{bal,up,t,s}, \pi_{bal,down,t,s}$	Time-varying upward and downward grid balancing clearance prices [€/MWh]
ρ_s	Historical probability of occurrence for stochastic scenario s
$CAPEX_k, OPEX_k$	Capital expenditure and fixed operational expenses of technology k [€/MW]
$CAPEX_{TES}, OPEX_{TES}$	Capital expenditure and fixed operational expenses of the TES [€/MWh]
$COP_{H,HP,t}, COP_{C,HP,t}$	Time-dependent coefficient of performance for heating and cooling provided by the heat pump
$COP_{C,CH,t}$	Time-dependent coefficient of performance for cooling provided by the electric water chiller
$D_{H,t}, D_{C,t}$	Hourly heating and cooling network demand of the examined year [MWh]
$D_{SH,t}, D_{SC,t}$	Hourly space heating and space cooling network demand of the examined year [MWh]
$D_{DHW,t}$	Hourly demand for domestic hot water of the examined year [MWh]
$P_{k,min}, P_{k,max}$	Minimum and maximum market capacity size boundaries of technology k [MW]

Continued on next page

Nomenclature – continued from previous page

$P_{TES,min}, P_{TES,max}$	Minimum and maximum market capacity size boundaries of the TES [MWh]
$P_{demand,peak}$	Maximum absolute peak network district heating requirement [MW]
Variables	
<i>Objective function</i>	
C_{inv}	Total annualized investment and fixed operational overhead costs [€]
$C_{fuel}, C_{base,s}$	Dynamic operational fuel commodity and day-ahead electricity market expenditures [€]
$C_{bal,down,s}$	Multi-scenario operational financial flow from providing downward balancing market services [€]
$C_{OPS,s}$	Scenario dependent total operational system costs [€]
$R_{elec}, R_{base,s}$	Baseline co-generation energy export electricity sales feed-in market revenues [€]
$R_{bal,up,s}$	Multi-scenario operational financial revenues from providing upward balancing market services [€]
$TAC, TEAC$	Objective functions: Total Annualized Cost and Total Expected Annualized Cost [€]
<i>Decision variables</i>	
Design	
$b_k \in \{0, 1\}$	Binary selection choice variable for asset allocation installation (b_{BB}, b_{CHP}, b_{HP})
$P_k \in \mathbb{R}_{\geq 0}$	Designed nominal capacity of technology k [MW]
$P_{TES} \in \mathbb{R}_{\geq 0}$	Designed nominal capacity of the TES [MWh]
Operation	
$b_{market_dir,t,s} \in \{0, 1\}$	Binary variable for preventing simultaneous bidding of upwards and downwards flexibility
$c_{TES,t} \in \{0, 1\}$	Binary variable for preventing simultaneous charging and discharging of TES
$E_{state,TES,t}, E_{state,TES,t,s} \in [0, P_{TES}]$	Heat stored inventory level inside the storage boundary [MWh]
$U_{i,k,t}, U_{i,k,t,s} \in \mathbb{R}_{\geq 0}$	Input energy consumption rate of energy carrier i by technology k at interval t under scenario s [MW]
$U_{charge,TES,t}, U_{charge,TES,t,s} \in \mathbb{R}_{\geq 0}$	Input operational thermal energy charging rate into the storage pit [MW]

Continued on next page

Nomenclature – continued from previous page

$V_{i,k,t}, V_{i,k,t,s} \in \mathbb{R}_{\geq 0}$	Output energy generation rate of energy carrier i by technology k at interval t under scenario s [MW]
$V_{discharge,TES,t}, V_{discharge,TES,t,s} \in \mathbb{R}_{\geq 0}$	Output operational thermal energy discharging rate into the storage pit [MW]
$V_{H,DA,HP,t,s} \in \mathbb{R}_{\geq 0}$	Scenario dependent heat output rate produced in the HP by electricity dispatched exclusively from the day-ahead electricity market [MW]
$V_{H,bal,down,HP,t,s} \in \mathbb{R}_{\geq 0}$	Scenario dependent heat output rate produced in the HP by electricity dispatched exclusively from the balancing electricity market [MW]
$V_{balancing,up,t,s} \in \mathbb{R}_{\geq 0}$	Active real-time upward electricity bidding reduction commitment rate [MW]
$V_{balancing,down,t,s} \in \mathbb{R}_{\geq 0}$	Active real-time downward electricity bidding consumption commitment rate [MW]
$y_{k,t}, y_{k,t,s} \in \{0, 1\}$	Binary variable indicating whether a technology k is operating at each time interval (y_{BB}, y_{CHP}, y_{HP})
$y_{H,HP,t}, y_{C,HP,t} \in \{0, 1\}$	Binary variable for preventing providing heating and cooling at the same time interval by the HP

Chapter 1

Introduction

Global warming is among the largest threats that pose a risk to the survival of civilization, since it can cause “dangerous and widespread disruption” affecting billions of people [1]. Global warming can lead to severe impacts, including extreme weather events, extinction of land animals and plants, disruptions to agriculture, wildlife, water, and energy resources, and even far reaching impacts on human health and existence [2]. To mitigate the effects of global warming, the Paris Agreement was signed in 2015. Its goal is to limit the increase in global average temperature to well below 2°C above pre-industrial levels, while also pursuing efforts to restrict this temperature rise to 1.5°C. Since anthropogenic global warming is a direct result of emitting additional greenhouse gas (GHG) emissions, such as carbon dioxide (CO₂), reducing such emissions is essential to reduce its impact on the planet.

Energy-related GHG emissions accounted for 77% of the worldwide GHG emissions in 2023 [3]. Of these, heat and electricity generation contributed to approximately one-third [3]. Europe has made comparatively strong progress in reducing energy-related GHG emissions relative to other regions. This has been achieved notably through the implementation of stringent climate policies such as the “*Fit for 55 package*”, which establishes legally binding obligations to achieve the EU’s climate targets [4]. Nevertheless, according to the most recent statistics published in 2025 by Eurostat [5], households still account for approximately 27% of Europe’s total energy consumption. Within this sector, 78% is attributed to space heating (SH) and domestic hot water (DHW) and approximately 5% to space cooling (SC) [6]. Despite the European Union’s (EU) efforts towards energy decarbonization and the progress of replacing fossil fuels with renewables, zero-carbon solutions in heating and cooling has been slower than in the case of electricity generation: with fossil fuels remaining the dominant energy source, supplying more than 60% of the residential heating and cooling demand in 2025 [6]. These numbers highlight the significant contribution of CO₂ emissions from the heating and cooling sector, emphasizing the need for sustainable and low-carbon alternatives.

The electrification of thermal and cooling end-uses presents a viable dual solution, delivering both economic efficiency and environmental sustainability [7–11]. In this context, Multi Energy Systems (MES) coupled with District Heating and Cooling Networks

(DHCNs) emerge as one of the most promising paradigms for sector coupling and electrification [7]. By integrating energy conversion, storage, and distribution across multiple energy carriers, MES allow for the simultaneous supply of electricity, heating, and cooling. This integrated architecture offers significant advantages over decentralized, building systems: it facilitates the deployment of diverse small-scale generation and storage technologies (e.g., solar photovoltaic panels, combined heat and power plants, battery energy storage, and Thermal Energy Storage (TES)) while enabling the exploitation of local renewable and ambient energy resources, such as industrial waste heat and solar thermal energy [7, 8].

On the matter, district thermal networks are currently undergoing a structural shift. As highlighted by Lund et al. [12], the heating sector is actively transitioning from 3rd- to 4th-generation district heating networks (4GDH). These 4GDH networks, characterized by significantly lower supply temperatures, may reduce distribution heat losses, accommodate a broader variety of low-grade heat sources, and offer superior economic and environmental performance compared to individual building-level heating systems [12, 13].

Concurrently, rising ambient temperatures and more frequent extreme heat events driven by climate change have accelerated urban cooling demand, a trend projected to escalate significantly in the coming decades [14]. District cooling networks (DCNs) have proven to be an exceptionally energy efficient mechanism to meet this expanding demand [15] both in research and on actual projects. Electrically driven heat pumps (HPs) – especially when coupled with TES – serve as a key enabling technology within DHCNs, efficiently linking thermal and cooling networks with the power grid while leveraging both renewable electricity and low-temperature heat sources [16].

In this context, large-scale electrically driven heat pumps (LSHPs) and TES, mainly in the form of hot water storage (HWS), represent one of the most effective Power-to-Heat (P2H) options for coupling electrical with thermal and cooling grids [17, 18]. On the supply side, LSHPs enable the integration of higher shares of variable renewable power sources, such as wind and solar energy, by serving as flexible electrical loads [19]. On the source side, LSHPs coupled with thermal storage efficiently elevate low-grade sustainable heat from ambient water, sewage systems, and industrial effluent into useful thermal supply, thereby displacing fossil fuel-based generation [20]. Furthermore, the lower operating supply temperatures of 4GDH networks perfectly align with the optimal thermodynamic operating range of LSHPs, maximizing their Coefficient of Performance (COP) and enabling them to reliably satisfy urban thermal and cooling demand [20].

In addition to the fundamental decarbonization and cost optimization benefits of a MES coupled with a DHCN, where technologies such as LSHP and TES play a pivotal role in electrifying urban thermal and cooling demand, these integrated architectures inherently function as flexible loads capable of delivering vital grid stabilization services. In this context, the comprehensive review by Golmohamadi et al. [21] emphasizes the substantial potential of district energy networks to provide demand-side flexibility to power systems, including load shifting, peak shaving, and grid congestion management. This flexibility potential is significantly augmented by the presence of TES, which physically decouples heat generation from instantaneous thermal consumption. By storing ther-

mal energy, TES grants LSHP the operational freedom to dynamically adjust electricity consumption in response to real-time power grid requirements without compromising end-user comfort [21].

Furthermore, research by Tan et al. [22] demonstrates that optimal operational management of such configurations can markedly reduce total investment and operational costs by strategically offering flexible capacity/energy in electricity balancing and ancillary service markets (e.g., primary and secondary reserves). This operational strategy aligns directly with the evolving European regulatory framework, which actively fosters the market participation of flexible assets to supply grid flexibility services and relieve network congestion [23–25]. Consequently, market-based monetization of demand-side flexibility creates additional revenue streams, further improving the economic viability of these P2H installations.

In summary, a MES that couples LSHP and TES within a DHCN may offer a highly promising paradigm for decarbonizing urban heating and cooling. By simultaneously achieving direct operational cost and emissions reductions and unlocking new revenue streams through grid stabilization and ancillary market participation, such integrated systems may provide a resilient, economically attractive solution for the future low-carbon energy landscape.

1.1 Literature Gaps and Novelty

While existing literature has extensively demonstrated the techno-economic and environmental benefits of coupling LSHPs and TES within DHCNs (e.g., [26, 27]) as well as optimized their operational dispatch strategies for electricity grid flexibility and participation into different balancing market segments (e.g., [28, 29]), the two following aspects remain insufficiently explored in the existing literature:

1. **Lack of regional and climatic sensitivity analysis:** Prior research overwhelmingly focuses on isolated, single electricity bidding zone case studies under fixed geographical and meteorological conditions, predominantly concentrated in Nordic regions such as Denmark and Finland [10, 28]. Consequently, the extent to which different system characteristics including differences in urban energy demand profiles and region-specific exogenous parameters are jointly considered remains insufficiently explored in the current literature.
2. **Prevalence of sequential market-clearing optimization:** A significant majority of flexibility integration studies employ a sequential optimization framework (e.g., [29, 30]), wherein individual market segments (e.g., spot markets followed sequentially by reserve or different (Balancing Market) BM gate openings) are cleared independently and chronologically. While sequential models reflect realistic bidding conditions, they fail to capture a holistic co-optimization of MES assets across market segments. Specifically, they ignore the strategic coupling between (Day Ahead Market) DAM commitment decisions and subsequent stochastic BM en-

ergy activations, particularly when uncertainty in BM prices and activations is considered.

To address these research gaps, this Master’s thesis introduces a novel, double-stage optimization framework that advances the current literature along two key dimensions.

First, this work conducts a modeling study for the design and operation of LSHP and TES coupled DHCNs and compares its techno-economic potential across two different regions/European energy networks, in countries which lead the energy transition of the heating and cooling sector but face significant regional differences in terms of their electricity and gas prices as well as their associated GHG emissions: “Zurich (Switzerland)” and “Amsterdam (Netherlands)”. By evaluating the existing DHCN infrastructure assets across the two distinct geographical regions, the framework systematically quantifies how regional climate conditions, local energy demand and dynamic differences in fuel input and electricity prices jointly shape the optimal design, sizing, and dispatch of a MES which includes LSHP-TES coupled with DHCN network. These findings should be useful to municipalities who already have or are about to have heat transition plans.

Second, this thesis formulates a co-optimization model that overcomes the limitations of traditional sequential market clearing. First a deterministic baseline is established to determine the optimal MES capacity and baseline dispatch without BM participation. Then, a stochastic operational optimization model is developed to simultaneously co-optimize DAM positioning and aFRR energy activation provisions. By utilizing historical price clustering and scenario-based stochastic formulations, the model explicitly accounts for uncertainty in market prices and energy activations. This approach captures the interactions, opportunity costs between meeting the energy demand of end-consumers and participating in BM, thereby providing insights into the revenue potential and operational implications of BM participation for urban energy investors and DHC operators.

1.2 Research Questions

As a result of the literature examined and of the literature gaps identified, the main research question of this Master’s thesis can be formulated as follows:

To what extent can heat pump-coupled district heating and cooling networks be cost-effective and simultaneously decarbonized?

To answer the main research question, the following sub-questions are formulated as well:

1. How cost-effective are LSHPs coupled DHCNs compared to fossil fuel-based district-only supply strategies?
2. To what extent does the presence of thermal storage technologies affect the integration of LSHPs within DHCNs?

3. How do location specific conditions influence the integration and economic benefits of LSHPs in district energy systems?
4. How does participation in secondary reserve electricity markets using LSHP-TES systems affect the total system costs and operational revenues of DHC network operators?

1.3 Structure of the Thesis

To address the formulated research questions, this thesis develops an optimization framework for a MES coupled with transitioning fourth generation DHCNs. Motivated by the literature and the industrial flagship applications followed in Chapter 2, the candidate technology portfolio incorporates LSHP and TES, alongside pre-existing network assets in Zurich and Amsterdam, including Biomass Boilers (BB) and Combined Heat and Power (CHP) plants.

A two-stage (Total Annualized Cost) TAC minimization optimization framework is implemented to model the MES:

- **Stage 1 (Deterministic design and baseline operation):** Sizing and baseline dispatch optimization are conducted under deterministic conditions using 2025 input data and exogenous parameters, directly addressing mainly Sub-Questions 1, 2 and 3.
- **Stage 2 (Stochastic operational optimization):** Constrained by the optimal capacities determined in Stage 1, this stage optimizes operational dispatch with allowing the MES to participate in aFRR energy activation provisions using stochastic BM price clusters, addressing mainly Sub-Questions 3 and 4.

Collectively, this two-stage optimization framework resolves the main research question. This framework compares also a brownfield approach (zero investment costs for existing BB and CHP assets) with a greenfield approach (all technologies built from scratch), while evaluating the post-optimization associated CO₂ emissions of all configurations from the optimization results. This optimization problem is implemented in Python and solved using the Gurobi Optimizer (v13.0.1) [31]. The remainder of this thesis is organized into six chapters: Chapter 2 presents the background on this topic both in real-world context and on research dimension. Chapter 3 details the mathematical formulation of the two-stage optimization methodology. Chapter 4 presents the input data and parameters for the Zurich and Amsterdam networks, along with the post-optimization assessment for GHG emissions and the defined MES configurations. Chapter 5 presents the optimization results and the corresponding emissions evaluation across both investment pathways. Chapter 6 discusses the key findings, reflects on the cross-regional insights against literature, and addresses the limitations and avenues for future research. Finally, Chapter 7 concludes the thesis by synthesizing the main conclusions and providing the final answers to the research questions.

Chapter 2

Background

This chapter provides an overview of the technological and methodological foundations underpinning the integration of HPs and TES into district energy systems. It reviews first real-world industry flagship implementations on this topic and then examines relevant mathematical modeling and optimization paradigms from the academic literature.

2.1 Industry State of the Art

The operational and economic benefits of MES coupling LSHP and TES within DHCNs are not limited to academic literature; they have been validated through completed projects, most notably in the Nordic countries.

Denmark, a pioneer in sector-coupling applications, offers prominent examples of commercial-scale LSHP integration designed for both baseload heat supply and electricity grid flexibility:

- **Everllence Energy Solutions (Esbjerg, Denmark):** Brought into operation in 2024, the Esbjerg project features a single LSHP of 70 MW thermal capacity, with a CO₂ based seawater heat source system, the largest of its kind globally. Harnessing ambient heat from seawater and powered by offshore wind energy, the plant satisfies the heating demand of approximately 25,000 households while mitigating roughly 120,000 tonnes of CO₂ emissions annually [32]. Figure 2.1 depicts the main structural components and physical layout of this LSHP.
- **Pit Thermal Energy Storage (PTES) Systems (Høje Taastrup and Dronninglund, Denmark):** To decouple real-time heat generation from thermal consumption, Denmark has deployed large-scale PTES installations utilizing water-filled storage reservoirs. The Høje Taastrup PTES (70,000 m³) is integrated into the Greater Copenhagen regional district heating network [33, 34], while the Dronninglund PTES (60,000 m³) serves local community thermal demand [27]. Both storage facilities completed by 2025 operate in synergy with HPs, enabling dynamic heat dispatch, peak shaving, and possibly optimized participation in power markets.

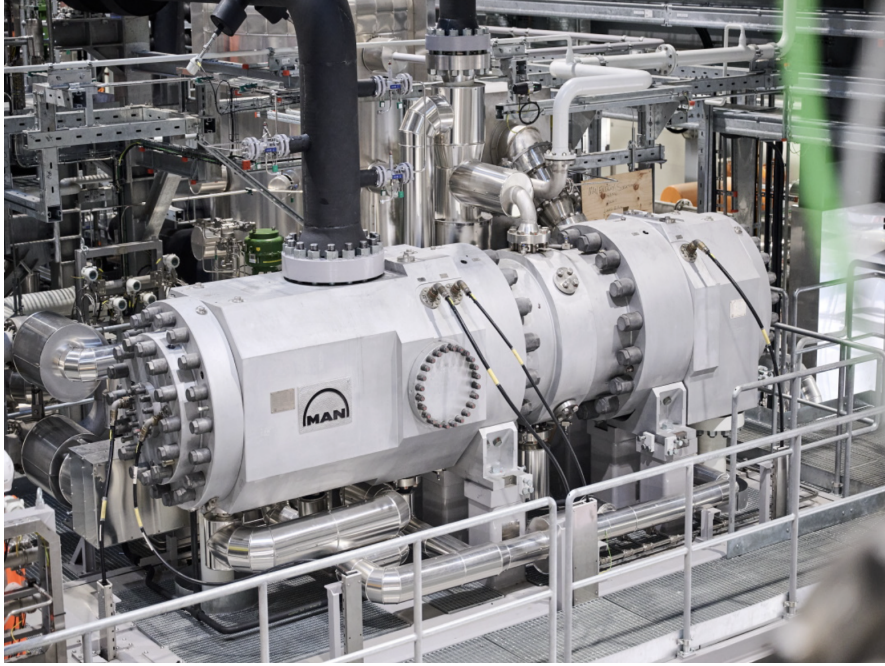


Figure 2.1: The main structural components and the physical layout of the Everllence’s mega HP project adopted by [32].

Similarly, major municipal utilities in Finland and Sweden have deployed multi-megawatt HP projects using diverse low-grade thermal sources and providing combined heating and cooling:

- **Helen (Helsinki, Finland):** The DHCN operator company of Helsinki commissioned the Eiranranta HP plant in 2025, which utilizes treated municipal wastewater as a low-temperature heat source. Delivering approximately 90 MW of district heating capacity and 60 MW of district cooling capacity, the facility replaces fossil fuel-based winter heating and provides urban summer cooling, cutting Helsinki’s carbon emissions by an estimated 80,000 tonnes per year [35].
- **Stockholm Exergi (Stockholm, Sweden):** The DHCN operator company of Stockholm commissioned in 2023 the Hammarbyverket facility, which represents one of the world’s largest HP installations extracting thermal energy from purified sewage water. Upgraded and expanded to deliver up to 225 MW of total heating and cooling capacity, the plant supplies space heating for approximately 95,000 households while simultaneously providing district cooling to the urban core [36].

As these full-scale flagship projects demonstrate, the integration of LSHP, TES, and DHC networks represents an established, commercially viable industrial standard rather than an unproven concept. Consequently, future design and optimization frameworks for existing or greenfield DHCNs can confidently consider these advanced and commercially mature technologies when defining candidate portfolios for heat and cooling generation.

2.2 Modeling Methods and Optimization Frameworks

Previous research has investigated the modeling and optimization of HP and energy storage integration into DHCNs, as well as the broader coupling of the power and thermal energy sectors. Existing literature primarily focuses on quantifying the tangible economic and environmental benefits of these integrated configurations, evaluating diverse MES architectures and optimization paradigms tailored to specific research scopes. Broadly, energy system models in the literature fall into two main categories: simulation and optimization models [7]. Simulation models are used to emulate the operation of a process or a system and provide insights into its operational behavior. In contrast, optimization models formulate key structural design and operational dispatch choices as decision variables, seeking to optimize specific objective functions subject to technical, operational, and physical constraints [7].

Mathematical programming is widely applied to identify optimal configuration designs and operation, employing either single or multi objective optimization to address economic and environmental targets. Linear Programming (LP) and Mixed Integer Linear Programming (MILP) represent the dominant methodologies in this domain. These formulations are inherently mixed-integer, non-linear, and non-convex because discrete binary decisions dictate asset installation and unit on/off states, while non-linear thermodynamics dictate component part-load efficiencies and non-convex thermal network equations. To maintain model linearity, energy conversion technologies, and storage technologies in the systems are extrapolated as simplified linear mathematical models from physical models to represent the off-design characteristics of each technology. Similarly, the operational performance of DHCN, such as heat losses and pumping energy, can be formulated as linear functions depending on the heat delivered and distance between the consumers and the supply systems [7]. For instance, Högnabba et al. [37] developed an hourly MILP optimization model over the 2023 year horizon to determine the cost optimal integration of P2H technologies within the district heating network of Turku, Finland. Similarly, Terlouw et al. [38] formulated a multi-objective MILP framework to optimize the structural design and life-cycle environmental impact of a MES, evaluating a broad technology portfolio across residential, mobility, and industrial sectors to minimize system costs and GHG emissions. Delangle et al. [39] likewise applied a MILP framework to identify optimal district heating network expansion strategies, selecting the ideal technology mix to maximize financial savings or minimize GHG emissions.

To address the inherent non-linearities of physical energy systems, Mixed-Integer Non-Linear Programming (MINLP) is occasionally employed to capture precise off-design performance curves and detailed operational constraints. For example, Mertz et al. [40] utilized an MINLP model to optimize the topological design of district heating networks, evaluating parallel versus cascade consumer connections across varying supply temperatures and balancing the trade-offs between pumping energy and network heat losses. However, despite their ability to reflect detailed physical behavior, MINLP formulations remain far less common than LP and MILP approaches due to their significantly higher computational complexity and convergence challenges [7].

Beyond the specific choice of mathematical formulation, existing literature strongly underscores the imperative need for integrated, co-optimized design and operational modeling when evaluating candidate technologies and multi-energy input carriers within transitioning 4th generation DHCNs. Figure 2.2 is representative of these networks transition by [12]. Sizing and dispatching system components in isolation risks oversimplifying physical operational dynamics or miscalculating the viability of investments. In this context, Cheng et al. [26] emphasized that simultaneous design and operational optimization is essential for yielding robust investment recommendations when expanding district heating networks in China to satisfy growing thermal loads while adhering to decarbonization mandates. Similarly, Sadeghi et al. [41] applied a joint design-and-operation framework to high-temperature HPs supplying both industrial and space heating loads, demonstrating that accounting for dynamic dispatch during the sizing phase yields significantly higher system efficiency and cost performance. In the context of system retrofits, Kontu et al. [28] utilized a combined design and operational approach to evaluate the techno-economic viability and carbon footprint reduction of integrating LSHPs into existing network infrastructures. Furthermore, Zhang et al. [42] established that joint capacity sizing and operational planning are critical to identifying economically viable, low-carbon hybrid heating strategies capable of achieving national emissions targets in the UK. As such, current literature demonstrates that evaluating the actual economic and environmental benefits of a supposed MES coupled with DHCNs requires a unified design-and-operation optimization framework, as isolated approaches fail to capture the critical interplay between capital investment decisions and real time optimal dispatch scheduling.

In addition to joint design and operational optimization, a rapidly growing body of literature highlights the operational value of coupling HPs and TES within DHCNs to provide demand-side flexibility services to the power grid, such as load shifting, peak shaving, and ancillary market participation. In particular, LSHPs are exceptionally well-suited for supplying rapid power grid balancing services, such as automatic Frequency Restoration Reserves (aFRR) and Frequency Containment Reserves (FCR) [29,30], with LSHPs of MW-size particularly suitable for aFRR balancing services [29]. Because HP compressors can modulate electrical consumption within seconds or minutes [29,30], LSHPs are ideal candidates for these sector-coupling operations; the thermal inertia of district heating networks and storage buffers these short-term fluctuations to ensure an uninterrupted thermal supply to end-users [43]

In this context, Javanshir et al. [29] validated that P2H units paired with thermal storage can reliably supply balancing services, using a multi-market operational strategy across DAM and BM. Similarly, Zhang et al. [44] demonstrated the techno-economic feasibility of reserve provision from LSHPs through a stochastic optimization model. Furthermore, Pesola [30] confirmed that capturing the complete value proposition of hybrid heating systems requires actively deploying HPs for ancillary services to monetize grid flexibility revenues.

However, participating in reserve and BM introduces significant exposure to volatile electricity prices and activation probabilities. While deterministic frameworks effectively

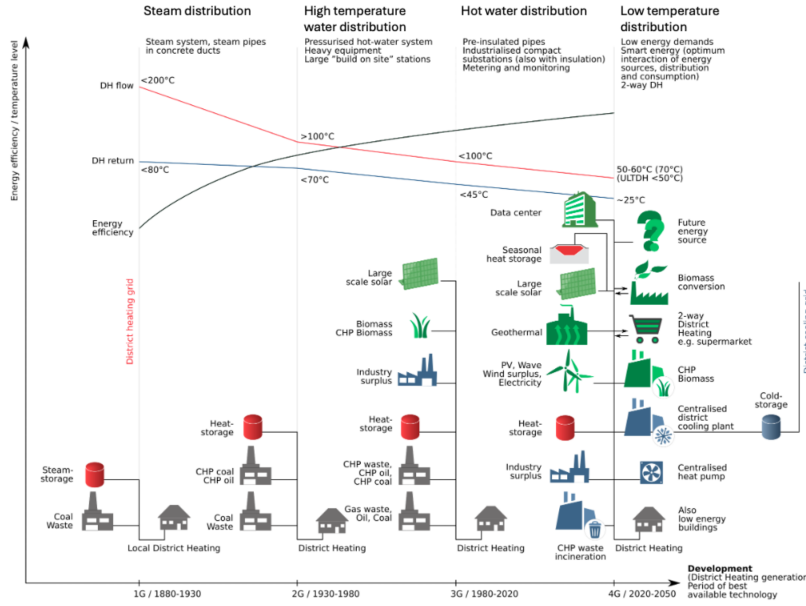


Figure 2.2: The historic transition towards 4GDH networks adopted by [12] licensed by CC BY-NC 4.0.

establish baseline system capacity and static operational dispatch, they fail to account for these stochastic market dynamics. Consequently, synthesized insights from the literature [29, 30, 44] demonstrate that to fully exploit the flexibility of P2H assets, operational optimization must incorporate a stochastic framework, capturing the high degree of market uncertainty to realistically evaluate the dynamic constraints regarding satisfying local thermal demand and providing electricity grid services.

Building on the above literature and to ensure a reproducible, structured literature review, a systematic review is conducted. This process is designed to identify the most relevant scholarly contributions aligned with the core scope of this thesis: the integration and optimization of LSHPs within DHCNs.

The following Boolean string is used in the Scopus search in order to export Tables 8.1, 8.2, 8.3, 8.4 and 8.5, visible in Appendix:

TITLE-ABS-KEY (("heat pump") AND ("district heat*" OR "district cooling" OR "district energy") AND ("optimization" OR "system integration"))*

To ensure thematic and technical focus, the literature search is refined using the following exclusionary criteria:

1. *Technological Boundary:* Studies regarding borehole heat exchangers and geothermal hybrid systems are omitted. The reason that justifies this exclusion is that those technologies are ideal for single buildings or small clusters, not entire districts

of cities because of their low heating temperature. The above deviates from the scope of this thesis which is strictly limited to a system level optimization including heating and cooling networks of districts of cities.

2. *Spatial and Sectoral Boundary*: Research is restricted to urban and residential energy demands. Consequently, papers focused on industrial-specific applications or isolated rural systems or single and complex of buildings are excluded, since they do not entail the systemic approach and scope of this thesis on modeling bigger DHC networks (i.e., networks of entire districts of cities).

As a result, the Tables 8.1, 8.2, 8.3, 8.4 and 8.5 visible in Appendix present the papers under the above search string that are also important to identify the research gaps of Chapter 1. These tables indicate that MILP is the dominant approach in the literature, typically applied with an hourly time resolution over a yearly horizon. The technologies considered vary significantly across studies with renewables and HPs being the main technologies, while the optimization objective is most often the minimization of system costs.

Chapter 3

Optimization Methodology

This section outlines the methodology behind the proposed design and annual operational routine for DHCN operators and energy investors in the district heating and cooling industry. The primary objective is to select and operate an optimal portfolio of technologies that provides the heating and cooling demand of a district network at the minimum TAC. Additionally, this framework assesses the potential for further revenues making/cost reduction by offering grid flexibility and contributing to electrical grid stability through participation in electricity BM auctions, specifically regarding aFRR provisions.

The proposed methodology is structured into two main phases. First, a first-stage optimization is performed to determine the minimum heating and cooling operation costs by considering also relevant technology investments. Within this first stage, different experiments are conducted by alternating the presence and absence of specific technologies. This allows evaluation of each technology's influence on the TAC and baseline operation, with a particular focus on the integration of an LSHP and a TES system. This stage establishes the design capacity of the examined technologies required to meet the demand, alongside their baseline operation under a scenario where BM participation is excluded. The first-stage optimization operates at an hourly resolution to comply with electricity DA market regulations, evaluating various system configurations.

Subsequently, a second-stage optimization is formulated to determine to which extent the operational costs can be further reduced through active participation in the BM. In this stage, all optimally selected technologies from the first stage are considered to be fully operational. To strictly adhere to BM rules, the second-stage optimization is modeled at a 15-minute resolution. Sections 3.1 and 3.2 present the comprehensive mathematical models and underlying equations governing the energy and economic analysis of each respective optimization stage. The case studies examined in this thesis incorporate real optimization parameters from the examined district networks and utilize simulations across all available asset technologies.

The case studies for Zurich and Amsterdam are simulated using the year 2025 as the optimization baseline, representing the most recent available year with complete seasonal input datasets. For the first-stage optimization, a deterministic approach is

adopted. This approach is highly suitable for estimating design capacities and baseline operations, given the availability of complete and materialized information regarding the required parameters. Furthermore, it reflects the coupling between day-ahead market dynamics, the evolution of annual climatic conditions, and the subsequent variance in heating and cooling demand load curves.

In addition, for the second-stage optimization, a scenario-based stochastic approach is introduced to model balancing prices, incorporating five representative annual scenarios along with their historic frequency of occurrence. This stochastic formulation provides deeper insights into the operational behavior and economic performance of the DHCN, as BM revenues exhibit high volatility and represent an uncertain revenue stream for flexible load providers, such as the configuration examined in this thesis. The simulations are executed using historical 2025 inputs, which include DAM electricity prices, fuel prices, EU ETS carbon prices, meteorological data, and real-time aFRR balancing energy remuneration prices spanning the 2016–2025 period. The following assumptions are considered in this analysis:

- **Security of Supply:** The primary objective of the DHC network is to guarantee an uninterrupted heat and cooling supply to all consumers; consequently, no unserved demand is permitted within the model.
- **Balancing Market Eligibility:** It is assumed that only the large-scale HP (LSHP) coupled with the TES unit can participate in the electricity balancing markets in the proposed system configuration, since the fast dynamic response of the LSHP combined with the coupled capability of the TES allows the system to rapidly adjust its electrical consumption to satisfy (Transmission System Operator) TSO bidding requirements without disrupting the network's thermal delivery constraints.
- **aFRR Participation and Regulatory Framework:** The large-scale HP exclusively participates in the aFRR balancing market, which is characterized by a longer activation time (5 min) [29] and only during its heating function mode because of the existence of the thermal heat storage. Participation is restricted solely to balancing energy auctions, without prior inclusion in the capacity availability merit order. Under the most recent European PICASSO regulations, the unit is thus classified as a flexible load provider, similar to electric vehicles (EVs) and data centers ([25] - Articles 7, 16 and 18) and its flexibility services are paid on the pay-as-clear basis with a common price for all awarded energy bids.
- **CHP Revenue Hedging:** The electricity generated by the CHP unit is assumed to be hedged via a yearly forward contract fixed at ($\Lambda_{\text{revenue}} = 100 \text{ €/MWh}$), reflecting typical industry long-term hedging practices for electrical sales executed by CHP operators [29,45], and is kept identical across both examined case studies. This assumption is justified on the basis that power-sector sales dynamics fall outside the scope of this study; hence, stabilizing revenues via a standardized, representative industry benchmark isolates the thermal and cooling optimization of this study.

- **Market Scope Exclusions:** To reflect the perspective of a risk-averse network operator and energy investor, this study excludes participation in alternative balancing markets as well as intraday arbitrage trading strategies due to their highly complex market dynamics.
- **Greenfield and Brownfield Approach:** This study incorporates both greenfield and brownfield optimization approaches. Under the greenfield approach, all candidate technologies are assumed to be constructed from scratch, thereby incurring full initial capital expenditures. Conversely, when the brownfield approach is examined, a baseline infrastructure is assumed to exist for the conventional district heating units (i.e., the BB and CHP). Consequently, these technologies are modeled with zero capital expenditures, meaning their costs derive solely from fixed operating expenses and fuel inputs. For all other newly integrated technologies across both approaches, investment costs are treated as optimization parameters.
- **Network Losses:** This study assumes a single-node spatial configuration, meaning that all demand is aggregated into a single node. Consequently, physical network and pipeline dynamics are disregarded. To account for this limitation, a scaling factor of 1.1 is applied to the total demand to model network losses [46, 47].
- **WtE Power Plants Heat Production:** This study excludes heat produced by WtE power plants from the optimization framework; consequently, their output is subtracted from the total demand in the implemented case studies. Because WtE heat production is a by-product of the waste treatment process, varying with both waste quality and quantity, it is realistically modeled as a baseload contribution that reduces net district heating demand. Additionally, thermal energy storage is assumed to be installed primarily to provide flexibility in combination with the LSHP, rather than to store excess heat from the WtE plants.

Figure 3.1 illustrates the schematic of the proposed configuration of the examined MES for this study along with the input and output operational decision variables of the first stage optimization as they have been defined in the Nomenclature.

The technologies included in Figure’s 3.1 MES configuration stem from those predominantly utilized in the examined case study cities. In Zurich, the district heating system is primarily supplied by waste heat from the Hagenholz waste-to-energy plant (approximately 53%), followed by wood-fired power plants (approximately 16%), and gas-fired boilers accounting for around 29% [48]. In Amsterdam, the primary energy sources for district heating include heat recovered from the municipal waste incineration facility operated by AEB, alongside Vattenfall’s Diemercentrale plant, a CHP plant fueled mainly by natural gas and complemented by a minor contribution from biomass [49].

Based on these configurations, a representative DHC network can be defined as shown in Figure 3.1, with primary energy inputs consisting of natural gas, biomass, and electricity. Within the MES framework developed in this study, these conventional technologies are initially considered in the baseline scenario. A LSHP and a coupled TES are subsequently introduced as one of the most prominent alternative technologies, as seen in the

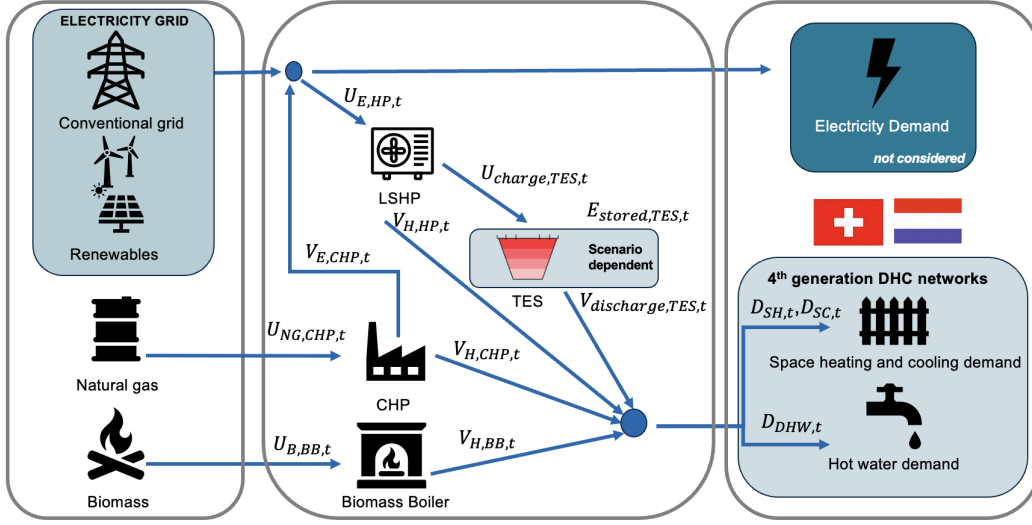


Figure 3.1: A schematic of the proposed MES system configuration assumed for this study with the following energy carriers: Biomass (B), Natural Gas (NG), Electricity (E), Heat (H) and Cooling (C).

literature and industry, to evaluate the central scope of this study: the potential and impact of HP integration within such networks. The LSHP operates by utilizing electricity sourced from both the electricity grid and its coupled renewable sources (wind and solar PV) to generate heat, which is either delivered directly to meet network demand or stored in the TES for later deployment.

Regarding the cooling demand, it is met by a large-scale electric chiller in the experimental configurations where the LSHP is absent from the MES. The selection of this specific standalone technology is driven by the fact that an electric chiller mirrors the exact physical cooling mechanism and performance characteristics of the LSHP when operating exclusively in cooling mode; consequently, utilizing it establishes a perfectly balanced, equivalent base that accurately isolates the benefits of the LSHP without introducing architectural discrepancies. When the LSHP is included, cooling is instead supplied via the hybrid operation of the HP, which is also powered by the electrical grid.

Evaluating these shifting system configurations, investment decisions, and operational modes requires a mathematical framework capable of handling both continuous physical processes and discrete choices. For instance, determining whether to invest in a specific technology, managing the on/off status of conversion assets, and enforcing capacity limits require binary variables. At the same time, modeling hourly energy flows, fuel consumption, and market dynamics requires continuous variables. Because these interactions can be captured through linear relationships and constraints, the optimization framework is structured as a MILP in accordance with the reviewed studies [37, 39, 50].

Overall, from a mathematical perspective, this MILP problem can be formulated in

its general form as follows:

$$\min_{\mathbf{x}, \mathbf{y}} (\mathbf{c}^T \mathbf{x} + \mathbf{d}^T \mathbf{y}), \quad (3.1)$$

s.t.

$$A\mathbf{x} + B\mathbf{y} = \mathbf{b}, \quad (3.2)$$

$$\mathbf{x} \geq 0 \in \mathbb{R}^{N_x}, \quad \mathbf{y} \in \{0, 1\}^{N_y}, \quad (3.3)$$

where $\mathbf{c}$ and $\mathbf{d}$ represent the cost vectors associated to continuous and binary decision variables, $\mathbf{x}$ and $\mathbf{y}$ respectively; A and B are the corresponding constraint matrices (parameters of the conversion technologies and of TES constraints) and $\mathbf{b}$ is the constraint known-term; N_x and N_y indicate the dimension of $\mathbf{x}$ and $\mathbf{y}$, respectively.

3.1 First Stage Optimization

This section outlines the general formulation of the economic optimization problem of the proposed MES **for the first stage optimization**. This initial stage focuses on the optimal capacity planning (design) and baseline unit commitment (operation) of the system assets as defined in Figure 3.1. The deterministic optimization problem is formulated at an hourly temporal resolution ($t \in \mathcal{T}$), where the optimization horizon encompasses a complete standard year ($\mathcal{T} = \{1, \dots, 8760\}$) parameterized against the parameter conditions of the year 2025 for the set of conversion technologies $k \in \mathcal{K}$ with $\mathcal{K} = \{BB, CHP, HP, CH\}$ and thermal storage (TES), which use the energy carriers $j \in \mathcal{J}$ with $\mathcal{J} = \{B, NG, E, H, C\}$. The mathematical formulation of the optimization model is structured systematically as follows. First, the core decision variables of the problem are defined. Subsequently, the economic objective function is established. Finally, the governing constraints of the system are detailed, starting with the global network-level constraints, followed by the technology specific constraints. This stage optimization relies on a set of input parameters, which are characterized also in detail in the Nomenclature but include: the capital recovery factor (CRF) (α), technology-specific investment and fixed maintenance costs (CAPEX_k , $\text{CAPEX}_{\text{TES}}$, OPEX_k), commodity fuel prices (π_{biomass} , $\pi_{\text{gas},t}$), DAM electricity spot prices ($\pi_{\text{elec},t}$), and the fixed CHP electricity export tariff (Λ_{revenue}). Thermodynamically the model requires the hourly district heating and cooling demand time series ($D_{H,t}$, $D_{C,t}$), the peak network's heating requirement ($P_{\text{demand,peak}}$), the time-dependent COP ($\text{COP}_{H,t}$, $\text{COP}_{C,t}$), the steady-state conversion efficiencies (η_{BB} , $\eta_{\text{el,CHP}}$, $\eta_{\text{th,CHP}}$), charging/discharging and self discharge storage parameters ($\eta_{c,\text{TES}}$, $\eta_{d,\text{TES}}$, λ_{TES}), minimum load fractions (δ_k), and minimum/maximum unit sizing boundaries ($P_{\min,k}$, $P_{\max,k}$, $P_{\min,\text{TES}}$, $P_{\max,\text{TES}}$).

3.1.1 Decision Variables

The following decision variables are returned by the MILP.

Design decision variables

These variables govern the investment, installation, and sizing decisions of the candidate MES technologies, representing the first-stage here and now decisions:

- **Installed capacity sizing** ($P_k \in \mathbb{R}^+$, $P_{TES} \in \mathbb{R}^+$, $\forall k \in \mathcal{K}$): Represents the continuous optimal design capacity to be constructed for each candidate technology k within the set of technologies $\mathcal{K}$ in MW. For the storage technology, this represents the nominal energy storage capacity in MWh.
- **Technology installation status** ($b_k \in \{0, 1\}$, $\forall k \in \mathcal{K}$): A binary decision variable introduced to indicate whether candidate technology k is selected for installation ($b_k = 1$) or rejected ($b_k = 0$).

Baseline operational decision variables

These continuous variables determine the high-resolution energy dispatch profiles throughout the annualized operational horizon $t \in \mathcal{T}$:

- **Input energy consumption rate** ($U_{j,k,t} \in \mathbb{R}^+$, $\forall j \in \mathcal{J}, \forall k \in \mathcal{K}, \forall t \in \mathcal{T}$): Specifies the operational energy input consumption rate of energy carrier j by technology k at time interval t in MW.
- **Output energy generation rate** ($V_{j,k,t} \in \mathbb{R}^+$, $\forall j \in \mathcal{J}, \forall k \in \mathcal{K}, \forall t \in \mathcal{T}$): Represents the operational energy generation rate of carrier j produced by technology k at time interval t in MW.
- **TES charging rate** ($U_{\text{charge},\text{TES},t} \in \mathbb{R}^+$, $\forall t \in \mathcal{T}$): Quantifies the operational thermal energy rate being actively charged into the TES during time interval t in MW.
- **TES discharging rate** ($V_{\text{discharge},\text{TES},t} \in \mathbb{R}^+$, $\forall t \in \mathcal{T}$): Quantifies the operational thermal energy rate being actively discharged from the TES during time interval t in MW.
- **TES thermal inventory level** ($E_{\text{state},\text{TES},t} \in [0, P_{\text{TES}}]$, $\forall t \in \mathcal{T}$): Represents the physical state of charge (stored thermal energy inventory) retained inside the TES boundaries at the end of time interval t , bounded by the maximum storage capacity P_{TES} in MWh.
- **On/off technology status** ($y_{k,t} \in \{0, 1\}$, $\forall k \in \mathcal{K}, \forall t \in \mathcal{T}$): A scheduling binary variable indicating whether technology k is actively online and operating ($y_{k,t} = 1$) or inactive ($y_{k,t} = 0$) at each time interval t . This status restriction is specifically applied only to the energy conversion technologies, namely the BB ($y_{\text{BB},t}$), the CHP ($y_{\text{CHP},t}$), and the LSHP ($y_{\text{HP},t}$).
- **Heat pump mode exclusivity** ($y_{\text{H},\text{HP},t}, y_{\text{C},\text{HP},t} \in \{0, 1\}$, $\forall t \in \mathcal{T}$): Interlocked binary variables introduced to physically prevent the LSHP from simultaneously operating in heating and cooling modes within the same time interval t . Here, $y_{\text{H},\text{HP},t}$ represents active heating operation, while $y_{\text{C},\text{HP},t}$ enforces active cooling operation.

- **TES charge/discharge exclusivity** ($c_{\text{TES},t} \in \{0,1\}$, $\forall t \in \mathcal{T}$): A binary decision variable introduced to physically prevent the TES from simultaneously charging and discharging within the same time interval t .

3.1.2 Objective Function

The system optimization targets the minimization of the Total Annualized Cost (TAC), which incorporates investments scaled by the CRF (α), annual operational fixed costs, dynamic fuel expenditures, and co-generation grid sales revenues. The individual financial and energy streams are evaluated over the hourly-resolution temporal domain $t \in \mathcal{T}$ with an operational time step of $\Delta t = 1$ h.

The optimization objective function is formulated mathematically as follows:

$$\min \quad \text{TAC} = C_{\text{inv}} + C_{\text{fuel}} - R_{\text{elec}}. \quad (3.4)$$

Annualized investment and fixed O&M costs (C_{inv}): This component aggregates the capacity-dependent investment and fixed maintenance and operational fees per capacity installed, where the capital expenditure initial parameters are annualized using the CRF (α):

$$C_{\text{inv}} = \sum_{k \in K} (\alpha \cdot \text{CAPEX}_k + \text{OPEX}_k) \cdot P_k + (\alpha \cdot \text{CAPEX}_{\text{TES}} \cdot P_{\text{TES}}). \quad (3.5)$$

Dynamic network operational fuel and electricity expenditures (C_{fuel}): This factor models the integrated sum of variable hourly commodity resource consumption costs (natural gas and biomass) spanning fuel inputs and heat pump grid electricity DAM prices:

$$C_{\text{fuel}} = \sum_{t \in T} \Delta t \cdot (U_{B,\text{BB},t} \cdot \pi_{\text{biomass}} + U_{\text{NG},\text{CHP},t} \cdot \pi_{\text{gas},t} + U_{E,\text{HP},t} \cdot \pi_{\text{elec},t}). \quad (3.6)$$

In the configuration where the LSHP is inactive the ($U_{E,\text{HP},t}$) is substituted with the electricity consumption rate of the Chiller ($U_{E,\text{CH},t}$).

Co-generation electricity export revenue (R_{elec}): The system total economic expenditure is offset through revenues earned by selling co-generated electricity from the active operation of the CHP system to the grid:

$$R_{\text{elec}} = \sum_{t \in T} \Delta t \cdot (V_{E,\text{CHP},t} \cdot \Lambda_{\text{revenue}}). \quad (3.7)$$

3.1.3 Global Energy Network Balance Constraints

The optimization framework must guarantee that the integrated energy generation fleet simultaneously and continuously satisfies the localized consumer loads across every operational hour $t \in T$.

Global district heating balance: The total heat energy generated across all active technical units combined with the net discharging yield ($V_{\text{discharge, TES},t}$) from the thermal energy storage must balance the hourly structural space heating and domestic hot water network heat demands ($D_{H,t}$) and storage charging extractions ($U_{\text{charge, TES},t}$):

$$\sum_{k \in K} \Delta t \cdot (V_{H,k,t} + V_{\text{discharge, TES},t} - U_{\text{charge, TES},t}) = D_{H,t} \quad \forall t \in T. \quad (3.8)$$

Global district cooling balance: The cooling network requirements ($D_{C,t}$) must match the aggregate cooling yields produced across the system's eligible cooling-ready technologies which steams exclusively from the the LSHP, or from the Large Scale Electric Chiller during the experiment when the LSHP is inactive:

$$\sum_{k \in K | \exists V_{C,k}} \Delta t \cdot V_{C,k,t} = D_{C,t} \quad \forall t \in T. \quad (3.9)$$

Which simplifies explicitly under this framework configuration to:

$$\begin{cases} \Delta t \cdot V_{C,\text{HP},t} = D_{C,t} & \forall t \in \mathcal{T} \quad (\text{if LSHP is active}). \\ \Delta t \cdot V_{C,\text{CH},t} = D_{C,t} & \forall t \in \mathcal{T} \quad (\text{if LSHP is inactive}). \end{cases} \quad (3.10)$$

3.1.4 Technology Specific Constraints

Market capacity design limits: The designed capacity of energy conversion technologies must stay within market size boundaries if selected ($b_k = 1$), and is forced to 0 if not selected ($b_k = 0$) and the TES capacity size must stay within the market size boundaries defined before:

$$b_k \cdot P_{\min,k} \leq P_k \leq b_k \cdot P_{\max,k} \quad \forall k \in \mathcal{K}, \quad (3.11)$$

$$P_{\min,\text{TES}} \leq P_{\text{TES}} \leq P_{\max,\text{TES}}. \quad (3.12)$$

Energy conversion performance: The mathematical relationship mapping input energy carrier consumption to thermal, electrical and cooling generation for each energy conversion technology is formulated as follows:

$$V_{H,\text{BB},t} = U_{B,\text{BB},t} \cdot \eta_{\text{BB}} \quad \forall t \in \mathcal{T}, \quad (3.13)$$

$$V_{E,\text{CHP},t} = U_{\text{NG},\text{CHP},t} \cdot \eta_{\text{el},\text{CHP}} \quad \forall t \in \mathcal{T}, \quad (3.14)$$

$$V_{H,\text{CHP},t} = U_{\text{NG},\text{CHP},t} \cdot \eta_{\text{th},\text{CHP}} \quad \forall t \in \mathcal{T}, \quad (3.15)$$

$$U_{E,\text{HP},t} = \frac{V_{H,\text{HP},t}}{\text{COP}_{H,t}} + \frac{V_{C,\text{HP},t}}{\text{COP}_{C,t}} \quad \forall t \in \mathcal{T}, \quad (3.16)$$

$$U_{E,\text{CH},t} = \frac{V_{C,\text{CH},t}}{\text{COP}_{C,t}} \quad \forall t \in \mathcal{T}. \quad (3.17)$$

Minimum and maximum heating output of the BB and the CHP: The mathematical relationship constraining heat production of the CHP and BB within its capacity sizes boundaries and minimum load fraction is modeled as follows. However, to avoid nonlinear bilinear terms ($y_{k,t} \cdot P_b$), the operational constraints dictate that if $y_{k,t} = 1$, the BB or the CHP operates between its minimum load fraction (δ_k) and its designed capacity (P_k). If $y_{k,t} = 0$, output is forced to 0. This is modeled via Big- M formulations using the upper market limit $P_{k,\max}$, which is a typical way to handle bilinearity in optimization theory [51]:

$$V_{H,k,t} \leq P_k \quad \forall t \in T, \quad (3.18)$$

$$V_{H,k,t} \leq y_{k,t} \cdot P_{k,\max} \quad \forall t \in T, \quad (3.19)$$

$$V_{H,k,t} \geq \delta_k \cdot P_k - (1 - y_{k,t}) \cdot P_{k,\max} \quad \forall t \in T. \quad (3.20)$$

Minimum and maximum energy output of the LSHP and Chiller: Because the LSHP features a dual-mode operational design capable of providing both heating and cooling services, mutual exclusivity must be strictly enforced. To mathematically prevent the simultaneous operation of these conflicting modes within any single dispatch interval, the system scheduling is governed by the following interlocking constraint:

$$y_{H,HP,t} + y_{C,HP,t} \leq y_{HP,t} \quad \forall t \in \mathcal{T}. \quad (3.21)$$

Analogous to the operational capacity boundary formulations previously defined for the BB and CHP conversion technologies, the operational limits of the LSHP are dynamically mapped to these mode-specific binary variables. Consequently, the heating output constraints are formulated as follows:

$$V_{H,HP,t} \leq P_{HP} \quad \forall t \in T, \quad (3.22)$$

$$V_{H,HP,t} \leq y_{H,HP,t} \cdot P_{HP,\max} \quad \forall t \in T, \quad (3.23)$$

$$V_{H,HP,t} \geq \delta_{HP} \cdot P_{HP} - (1 - y_{H,HP,t}) \cdot P_{HP,\max} \quad \forall t \in T. \quad (3.24)$$

And the cooling output constraints for the LSHP are formulated as follows and by taking into account the absence of a minimum load fraction for providing cooling services.

$$V_{C,HP,t} \leq P_{HP} \quad \forall t \in T, \quad (3.25)$$

$$V_{C,HP,t} \leq y_{C,HP,t} \cdot P_{HP,\max} \quad \forall t \in T. \quad (3.26)$$

And similarly for the large scale electric chiller during the experiment where the LSHP remains inactive.

$$V_{C,CH,t} \leq P_{CH} \quad \forall t \in T, \quad (3.27)$$

$$V_{C,CH,t} \leq y_{C,CH,t} \cdot P_{CH,\max} \quad \forall t \in T. \quad (3.28)$$

TES inventory bounds restriction: During any operational timestep t , the stored thermal energy level within the TES cannot exceed the structurally designed investment capacity sizing:

$$0 \leq E_{\text{state, TES}, t} \leq P_{\text{TES}} \quad \forall t \in T. \quad (3.29)$$

TES operational charging and discharging limits: Regarding the TES charging operation, the thermal input rate is physically bounded by the heat output of the LSHP ($V_{H, \text{HP}, t}$), as the Pit storage is exclusively charged by the LSHP within the MES framework. To avoid the bilinear term ($C_{\text{TES}, t} \cdot V_{H, \text{HP}, t}$), a linear Big- M formulation enforces a zero-charging constraint whenever the charging flag is inactive ($C_{\text{TES}, t} = 0$), utilizing the maximum heat pump capacity ($P_{\text{HP}, \text{max}}$) as the upper bound:

$$U_{\text{charge, TES}, t} \leq V_{H, \text{HP}, t} \quad \forall t \in T, \quad (3.30)$$

$$U_{\text{charge, TES}, t} \leq C_{\text{TES}, t} \cdot P_{\text{HP}, \text{max}} \quad \forall t \in T. \quad (3.31)$$

Regarding the discharging operation of the TES, the storage discharging rate cannot exceed the network's maximum absolute system peak heating requirement ($P_{\text{demand, peak}}$), and is restricted to zero if the charging flag is active ($C_{\text{TES}, t} = 1$):

$$V_{\text{discharge, TES}, t} \leq (1 - C_{\text{TES}, t}) \cdot P_{\text{demand, peak}} \quad \forall t \in T. \quad (3.32)$$

TES dynamic energy balance & horizon periodicity: The state of charge energy profile models hourly storage level variations using the predefined TES efficiencies. For seasonal consistency, a closed-loop boundary condition locks the starting timestep ($t = 1$) directly to the last timestep ($t = 8760$):

$$E_{\text{state, TES}, t} = \begin{cases} E_{\text{state, TES}, 8760} & \text{if } t = 1 \\ (1 - \lambda_{\text{TES}}) \cdot E_{\text{state, TES}, t-1} + \Delta t \cdot \left(\eta_{c, \text{TES}} \cdot U_{\text{charge, TES}, t} - \frac{V_{\text{discharge, TES}, t}}{\eta_{d, \text{TES}}} \right) & \text{if } t > 1 \end{cases} \quad (3.33)$$

3.2 Second Stage Optimization

This section outlines the mathematical formulation of the stochastic economic operational optimization problem for the MES within the **second stage scheduling** horizon. Building upon the design capacities established in the first stage design phase, this operational stage incorporates high resolution flexibility by also evaluating the system's participation in aFRR activation energy provisions.

To follow BM rules, this stochastic optimization stage is formulated at a 15-minute temporal resolution ($t \in \mathcal{T}_{15}$) over the same time horizon as in the first stage ($\mathcal{T} = \{1, \dots, 35040\}$). The framework is parameterized against the parameter conditions of the year 2025 to maintain boundary consistency with the first stage. Furthermore, balancing price uncertainty is captured through a multi-scenario tree comprising five discrete upward and downward balancing energy price scenarios, derived from historical market data spanning the last decade (2016–2025). This formulation evaluates the identical set of conversion technologies ($k \in \mathcal{K}$) and energy storage (TES), utilizing the same set of

energy carriers ($j \in \mathcal{J}$) defined previously under five different balancing prices scenario ($s \in \mathcal{S}$).

Consistent with the structural paradigm of the first stage, the mathematical formulation of the second stage optimization problem is systematically organized as follows. First, the core operational decision variables are defined. Subsequently, the stochastic economic objective function is established to minimize expected TAC. Finally, the constraints of the system are explained, starting with the global network-level constraints, followed by the technology-specific constraints and the subsequent definition of the flexibility offered metrics. Similar to the first-stage optimization, the same underlying set of input parameters applies, with the additional inclusion of the scenario-dependent upward and downward BM prices ($\pi_{\text{bal,up},t,s}$, $\pi_{\text{bal,down},t,s}$), their scenario occurrence probabilities (ρ_s) and the new 15-min self-discharge loss rate of the TES ($\lambda_{15\text{min}}$).

3.2.1 Decision Variables

To formulate this second-stage optimization, a *co-optimization* modeling approach is adopted, wherein all the operational decision variables are structured as scenario-dependent. This methodology is selected to enable the optimizer to evaluate deterministic DAM clearing prices and stochastic BM prices scenarios at the same time. Consequently, the model determines an optimal electricity procurement schedule for the LSHP that mathematically guarantees sufficient physical storage capacity of the TES in order to capitalize on lucrative real-time balancing opportunities as they materialize.

While this formulation differs from the traditional sequential optimization approaches dominant in the literature (e.g., [29,30]), where DAM procurement decisions strictly precede and fix the baseline for subsequent BM bidding, it reflects a globally optimal strategy from the perspective of the MES network operator. By internalizing cross-market synergies, the co-optimization approach fully exploits the flexibility potential inherently offered by the demand-response capabilities under the PICASSO market clearing mechanism [25].

Accordingly, the following decision variables are returned by the stochastic MILP for this second stage optimization.

- **Input and output energy rates** ($U_{j,k,t,s}, V_{j,k,t,s} \in \mathbb{R}^+, \forall j \in \mathcal{J}, \forall k \in \mathcal{K}, \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}$): As defined in the first stage optimization but now for every $s \in \mathcal{S}$.
- **TES charge and discharge rates** ($U_{\text{charge, TES}, t, s}, V_{\text{discharge, TES}, t, s} \in \mathbb{R}^+, \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}$): As defined in the first stage optimization but now under $s \in \mathcal{S}$.
- **TES thermal inventory level** ($E_{\text{state, TES}, t, s} \in [0, P_{\text{TES}}], \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}$): As defined in the first stage optimization but now under $s \in \mathcal{S}$.
- **Technology commitment and LSHP mode exclusivity** ($y_{k,t,s}, y_{\text{H,HP}, t, s}, y_{\text{C,HP}, t, s} \in \{0, 1\}, \forall k \in \mathcal{K}, \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}$): As defined in the first stage optimization but now under $s \in \mathcal{S}$.

- **TES charge/discharge exclusivity indicator** ($c_{\text{TES},t,s} \in \{0, 1\}, \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}$): As defined in the first stage but now under $s \in \mathcal{S}$.
- **LSHP DA and BM thermal generation rates** ($V_{\text{H,DA,HP},t,s}, V_{\text{H,bal,down,HP},t,s} \in \mathbb{R}^+, \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}$): Represents the newly introduced scenario-dependent heat production rates partitioned by market clearing mechanism in MW:
 - $V_{\text{H,DA,HP},t,s}$: Represents the operational energy generation rate of heat produced by the LSHP attributable to the DAM electricity schedule after real-time upward curtailment under scenario s at time interval t .
 - $V_{\text{H,bal,down,HP},t,s}$: Represents the operational energy generation rate of heat produced by the LSHP through electricity procured exclusively from the BM when providing downward flexibility under scenario s at time interval t .
- **LSHP balancing market electricity flexibility rates** ($V_{\text{balancing,up},t,s}, V_{\text{balancing,down},t,s} \in \mathbb{R}^+, \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}$): Represents the newly introduced scenario-dependent BM operational electricity consumption adjustment rates utilized for the provision of upward and downward energy grid flexibility services in MW. According to the PICASSO market clearing regulations for demand-response agents, upward flexibility dictates a strategic reduction in baseline electricity consumption, whereas downward flexibility mandates an incremental increase in consumption to support grid frequency stabilization [25]:
 - $V_{\text{balancing,up},t,s}$: Represents the upward power flexibility rate provided by the LSHP, quantified as the rate of operational electricity consumption reduced under scenario s at time interval t .
 - $V_{\text{balancing,down},t,s}$: Represents the downward power flexibility rate provided by the LSHP, quantified as the rate of additional operational electricity consumed under scenario s at time interval t .
- **LSHP balancing market flexibility exclusivity** ($b_{\text{market,dir},t,s} \in \{0, 1\}, \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}$): A binary decision variable introduced to physically prevent the LSHP from simultaneously providing upward and downward flexibility services within the same time interval t under scenario s .

3.2.2 Objective Function

The second-stage operational optimization framework minimizes the Total Expected Annualized Cost (TEAC). This objective aggregates the deterministic, pre-allocated asset capacity expenditures carried over from the first stage alongside the probability-weighted operational expenses across all evaluated stochastic scenarios:

$$\min \quad \text{TEAC} = C_{\text{fixed}} + \mathbb{E}[C_{\text{ops}}]. \quad (3.34)$$

Fixed annualized investment and O&M costs (C_{inv}): This component captures the annualized investment and fixed O&M costs corresponding to the optimal configuration determined in the first-stage design phase. Because the asset capacities are fixed, these terms function as constant parameters within the second-stage optimization:

$$C_{\text{inv}} = \sum_{k \in K} (\alpha \cdot \text{CAPEX}_k + \text{OPEX}_k) \cdot P_k + (\alpha \cdot \text{CAPEX}_{\text{TES}} \cdot P_{\text{TES}}). \quad (3.35)$$

Expected scenario operational cost ($\mathbb{E}[C_{\text{Ops}}]$): The expected variable operational expenditures capture multi-market dynamics by scaling the net operational cash flows of each scenario $s \in S$ by its respective historical probability of occurrence (ρ_s). This formulation also introduces the participation scaling factors μ and ν . These continuous parameters, bounded in the range $[0, 1]$, denote the fraction of submitted energy activation bids by the LSHP that are successfully cleared in the BM for upward (μ) and downward (ν) flexibility under every single timestep and scenario, respectively.

When both factors are set to zero ($\mu = \nu = 0$) and the optimization decision variables involving downward flexibility provisions are erased from the optimization problem, the equation collapses back to the first-stage deterministic optimization formulation, representing a baseline scenario with no BM participation. Conversely, when both factors equal universal unity ($\mu = \nu = 1$), it models a perfect market clearing scenario where all bids submitted by the LSHP are fully accepted by the TSO, maximizing balancing energy provision. Although these activation rates fluctuate unpredictably in real-world market operations, the primary scope of this work is to establish the theoretical bounds of TEAC reductions between complete non-participation and maximum market integration. Consequently, these factors are assigned unit values ($\mu = \nu = 1$) for the optimal results analysis:

$$\mathbb{E}[C_{\text{Ops}}] = \sum_{s \in S} \rho_s \cdot (C_{\text{base},s} - R_{\text{base},s} - \mu \cdot R_{\text{bal,up},s} + \nu \cdot C_{\text{bal,down},s}). \quad (3.36)$$

The individual financial and energy streams within each scenario are evaluated over the high-resolution temporal domain $t \in \mathcal{T}_{15}$ with an operational time step of $\Delta t = 0.25$ h, defined as follows:

1. **Baseline fuel costs spending ($C_{\text{base},s}$):** Accumulates scenario-dependent input energy fuel costs for each conventional technology and DAM electricity procurement costs:

$$C_{\text{base},s} = \sum_{t \in \mathcal{T}_{15}} \Delta t \cdot (U_{B,\text{BB},t,s} \cdot \pi_{\text{biomass}} + U_{\text{NG},\text{CHP},t,s} \cdot \pi_{\text{gas},t} + U_{E,\text{HP},t,s} \cdot \pi_{\text{elec},t}). \quad (3.37)$$

2. **Baseline CHP electricity revenue ($R_{\text{base},s}$):** Quantifies revenues earned via fixed-tariff feed-in electricity sales generated by the co-generation CHP system:

$$R_{\text{base},s} = \sum_{t \in \mathcal{T}_{15}} \Delta t \cdot (V_{E,\text{CHP},t,s} \cdot \Lambda_{\text{revenue}}). \quad (3.38)$$

3. **Upward flexibility revenue ($R_{\text{bal,up},s}$):** Represents payments received from the TSO for strategic demand-response load reductions below baseline clearing values to stabilize grid scarcity:

$$R_{\text{bal,up},s} = \sum_{t \in \mathcal{T}_{15}} \Delta t \cdot (V_{\text{balancing,up},t,s} \cdot \pi_{\text{bal,up},t,s}). \quad (3.39)$$

4. **Downward flexibility cash flow ($C_{\text{bal,down},s}$):** Captures cash flows resulting from incremental electrical consumption increase over baselines to absorb grid over-generation. The directionality of the financial flow relies on the algebraic sign of the market clearance price $\pi_{\text{bal,down},t,s}$:

$$C_{\text{bal,down},s} = \sum_{t \in \mathcal{T}_{15}} \Delta t \cdot (V_{\text{balancing,down},t,s} \cdot \pi_{\text{bal,down},t,s}). \quad (3.40)$$

3.2.3 Global Energy Network Balance Constraints

As in the first stage, the optimization framework must continuously satisfy localized energy demands across all operational time steps $t \in \mathcal{T}_{15}$.

Global district heating balance: As defined in the first stage, the total thermal energy generated across all active units and net storage yields must strictly balance the dynamic network heat demands. Crucially, to reflect that heat demand is satisfied exclusively by electricity procurement determined prior to BM participation, thereby capturing the true operational sequencing where DAM commitments precede any real time balancing revenue streams, the heat production rate allocated from the LSHP is separated into its DA ($V_{\text{H,DA,HP},t,s}$) and resulting downward flexibility ($V_{\text{H,bal,down,HP},t,s}$) cleared volumes. These variables are both scenario dependent since under the co-optimization framework adopted in this thesis the scheduling commitment in the DAM depends on the prevailing BM prices of each scenario. Furthermore, because this second stage optimization captures sub-hourly market interactions where the TES plays a pivotal operational role in it, two additional constraints are introduced for the BB and CHP units. Because neither the BB nor the CHP is physically linked to thermal storage or participating in the BM, their individual thermal outputs are strictly capped by the instantaneous district heat demand power:

$$\Delta t \cdot (V_{\text{H,BB},t,s} + V_{\text{H,CHP},t,s} + V_{\text{H,DA,HP},t,s} + V_{\text{H,bal,down,HP},t,s} + V_{\text{discharge,TES},t,s} - U_{\text{charge,TES},t,s}) = D_{\text{H},t} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}, \quad (3.41)$$

$$\Delta t \cdot V_{\text{H,BB},t,s} \leq D_{\text{H},t} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}, \quad (3.42)$$

$$\Delta t \cdot V_{\text{H,CHP},t,s} \leq D_{\text{H},t} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}. \quad (3.43)$$

Global district cooling balance: As defined in the first stage, the total scenario-dependent cooling yields, generated exclusively from the LSHP under this framework stage configuration, must strictly match the localized dynamic network cooling demands:

$$\Delta t \cdot V_{\text{C,HP},t,s} = D_{\text{C},t} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}. \quad (3.44)$$

3.2.4 Technology Specific Operational Constraints

Since the asset capacities (P_k , P_{TES}) are fixed constants determined in the first-stage design phase, this operational stage governs the scenario-dependent dispatch limits, multi-market interactions, and storage dynamics across all time steps $t \in \mathcal{T}_{15}$ and scenarios $s \in \mathcal{S}$.

Energy conversion performance (BB and CHP): As formulated in the first stage, the mathematical mapping between input energy carrier consumption and technology specific generation is updated to incorporate scenario dependencies as follows:

$$V_{\text{H,BB},t,s} = U_{\text{B,BB},t,s} \cdot \eta_{\text{BB}} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}, \quad (3.45)$$

$$V_{\text{E,CHP},t,s} = U_{\text{NG,CHP},t,s} \cdot \eta_{\text{el,CHP}} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}, \quad (3.46)$$

$$V_{\text{H,CHP},t,s} = U_{\text{NG,CHP},t,s} \cdot \eta_{\text{th,CHP}} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}. \quad (3.47)$$

Operational capacity limits (BB and CHP): As formulated in the first stage, the operational constraints bounding the thermal generation of the BB and the CHP unit within their technical capacity limits (δ_k and P_k) are adjusted for scenario dependencies as follows:

$$V_{\text{H},k,t,s} \leq P_k \cdot y_{k,t,s} \quad \forall k \in \{\text{BB, CHP}\}, \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}, \quad (3.48)$$

$$V_{\text{H},k,t,s} \geq \delta_k \cdot P_k \cdot y_{k,t,s} \quad \forall k \in \{\text{BB, CHP}\}, \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}. \quad (3.49)$$

LSHP-specific constraints for BM participation: This section establishes the operational rules regulating the LSHP as it simultaneously interacts with both the DAM and BM across all time steps $t \in \mathcal{T}_{15}$ and scenarios $s \in \mathcal{S}$.

Mode exclusivity and power limits: As formulated in the first stage, the LSHP is strictly prohibited from simultaneously operating in heating and cooling modes within the same time interval and the operational limits of the LSHP are dynamically mapped to these mode-specific binary variables:

$$y_{\text{H,HP},t,s} + y_{\text{C,HP},t,s} \leq y_{\text{HP},t,s} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}, \quad (3.50)$$

$$V_{\text{H,HP},t,s} \leq P_{\text{HP}} \cdot y_{\text{H,HP},t,s} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}, \quad (3.51)$$

$$V_{\text{H,HP},t,s} \geq \delta_{\text{HP}} \cdot P_{\text{HP}} \cdot y_{\text{H,HP},t,s} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}, \quad (3.52)$$

$$V_{\text{C,HP},t,s} \leq P_{\text{HP}} \cdot y_{\text{C,HP},t,s} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}. \quad (3.53)$$

Net electrical-thermal conversion performance: The conversion of the net real-time electrical power input of the LSHP to heating and cooling generation is defined by its scenario-specific DA schedule, optimized jointly with each BM scenario ($U_{\text{E,HP},t,s}$) adjusted for upward balancing load shedding ($V_{\text{balancing,up},t,s}$) and downward balancing load increases ($V_{\text{balancing,down},t,s}$). To isolate scenario-specific baseline DAM procurement

from balancing adjustments, the total physical heat generation is explicitly separated into market-cleared components and capped by their electrical procurement inputs:

$$\frac{V_{H,HP,t,s}}{COP_{H,t}} + \frac{V_{C,HP,t,s}}{COP_{C,t}} = U_{E,HP,t,s} + V_{\text{balancing,down},t,s} - V_{\text{balancing,up},t,s} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}, \quad (3.54)$$

$$V_{H,HP,t,s} = V_{H,DA,HP,t,s} + V_{H,bal,down,HP,t,s} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}, \quad (3.55)$$

$$V_{H,DA,HP,t,s} \leq U_{E,HP,t,s} \cdot COP_{H,t} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}, \quad (3.56)$$

$$V_{H,bal,down,HP,t,s} \leq V_{\text{balancing,down},t,s} \cdot COP_{H,t} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}. \quad (3.57)$$

Multi-Market energy headroom and bidding boundaries: To ensure that the scenario-specific baseline DAM schedule retains sufficient physical capacity margins to execute sub-hourly, real-time balancing actions, the flexibility bids are strictly bounded. The first two constraints dictate that the provision of upward flexibility, achieved by diminishing electrical intake, is physically constrained by the baseline market clearing conditions; it can neither surpass the DA imported electricity ($U_{E,HP,t,s}$) nor exceed the equivalent DA procured physical heat generation. Conversely, downward balancing bids must satisfy both absolute asset capacity limits and realistic market bounds. The third constraint guarantees that the cumulative electrical load (i.e., the sum of the scenario-specific DAM baseline and the incremental downward flexibility) respects the maximum physical electrical capacity of the LSHP under active commitment status. Finally, the fourth constraint enforces a proportional scaling threshold introduced as a realistic modeling assumption. By capping downward flexibility bids at twice the level of the current scenario-specific DAM electrical baseline, the optimization framework guarantees that real-time load increases remain structurally bounded to a feasible operating envelope. This prevents the LSHP from submitting overly aggressive bids to the TSO, minimizing the severe penalty risks of non delivery and ensuring that the asset retains a realistically sound baseline of grid electricity before committing to intensive load-addition services motivated from a similar approach performed by [50]. This twice factor fits within the accepted optimization results of the studies performed by [29,30] for downward flexibility compared to the committed electricity procurement in the DAM:

$$V_{\text{balancing,up},t,s} \leq U_{E,HP,t,s} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}, \quad (3.58)$$

$$V_{\text{balancing,up},t,s} \leq \frac{V_{H,DA,HP,t,s}}{COP_{H,t}} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}, \quad (3.59)$$

$$U_{E,HP,t,s} + V_{\text{balancing,down},t,s} \leq \left(\frac{P_{HP}}{COP_{H,t}} \right) \cdot y_{HP,t,s} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}, \quad (3.60)$$

$$V_{\text{balancing,down},t,s} \leq 2 \cdot U_{E,HP,t,s} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}. \quad (3.61)$$

Grid flexibility directional exclusivity: To ensure realistic physical bidding behaviors within the dynamic PICASSO market clearing mechanism, directional binary interlocking gates are enforced via $b_{\text{market,dir},t,s}$, strictly preventing the LSHP from simultaneously executing upward and downward flexibility within the same dispatch interval:

$$V_{\text{balancing,down},t,s} \leq \left(\frac{P_{\text{HP}}}{\text{COP}_{\text{H},t}} \right) \cdot b_{\text{market,dir},t,s} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}, \quad (3.62)$$

$$V_{\text{balancing,up},t,s} \leq \left(\frac{P_{\text{HP}}}{\text{COP}_{\text{H},t}} \right) \cdot (1 - b_{\text{market,dir},t,s}) \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}. \quad (3.63)$$

TES inventory and operational dynamic limits: The thermal inventory ($E_{\text{state, TES},t,s}$) is strictly bounded by the fixed capacity P_{TES} , determined in the first stage optimization, for every timestep and scenario:

$$0 \leq E_{\text{state, TES},t,s} \leq P_{\text{TES}} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}. \quad (3.64)$$

Similar to the first stage, the charging and discharging limits enforce mutual exclusivity and bound operational rates to resolve bilinear tracking configurations. For the charging operation, the thermal energy input rate ($U_{\text{charge, TES},t,s}$) is physically constrained by the aggregate scenario-dependent real-time heat production allocated from both scenario-specific DAM baseline and downward BM load increments, while remaining locked to the charging binary flag ($C_{\text{TES},t,s}$):

$$U_{\text{charge, TES},t,s} \leq V_{\text{H,DA,HP},t,s} + V_{\text{H,bal,down,HP},t,s} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}, \quad (3.65)$$

$$U_{\text{charge, TES},t,s} \leq C_{\text{TES},t,s} \cdot P_{\text{TES}} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}. \quad (3.66)$$

Conversely, the storage discharging rate ($V_{\text{discharge, TES},t,s}$) cannot exceed the district network's absolute system peak heating requirements ($P_{\text{demand,peak}}$), and is restricted to zero when active charging occurs ($C_{\text{TES},t,s} = 1$):

$$V_{\text{discharge, TES},t,s} \leq (1 - C_{\text{TES},t,s}) \cdot P_{\text{demand,peak}} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}. \quad (3.67)$$

TES charging limit for BM participation: To ensure that the TES directly facilitates the BM participation of the LSHP, and to guarantee that district heat demand is supplied exclusively by the conventional units (BB and CHP) or by DA electricity commitments and discharging energy from the TES, an operational coupling constraint is formulated. This structure strictly reflects the realistic sequential market clearing process, where DA procurement precedes real-time balancing execution.

First, the storage charging rate ($U_{\text{charge, TES},t,s}$) must be greater than or equal to the heat rate generated from downward balancing electricity procurement. This forces all real-time downward balancing heat output rate to be directed entirely into thermal storage rather than serving end-user heat demand:

$$U_{\text{charge, TES},t,s} \geq V_{\text{H,bal,down,HP},t,s} \quad \forall t \in \mathcal{T}_{15}, \forall s \in \mathcal{S}. \quad (3.68)$$

Conversely, when the LSHP offers upward flexibility by curtailing its electrical consumption, the resulting thermal deficit can directly be substituted by either thermal energy discharged from the TES or by dispatching more heat from the BB or CHP and as a result there is no need for imposing a similar constraint regarding the upward flexibility provisions.

Although this constraint restricts complex operational strategies where real-time balancing procurement can directly supply district heating, they accurately capture the perspective of a DHCN operator, mainly seeking to provide heating and cooling services to consumers. Under this paradigm, the SoC of the TES is deployed as the dedicated flexibility provider, maintaining the integrity and sequence of the DA schedule co-optimized against stochastic BM price clusters.

TES dynamic energy balance & horizon periodicity: As formulated in the first stage, the state of charge profile of the TES maps chronological inventory variations using the predefined TES efficiencies, updated here to capture 15-minute resolution ($\Delta t = 0.25$ h) across each scenario path. For seasonal consistency, a closed-loop boundary condition locks the starting timestep ($t = 1$) directly to the last timestep ($t = 35040$) independently for every scenario $s \in \mathcal{S}$:

$$E_{\text{state, TES}, t, s} = \begin{cases} E_{\text{state, TES}, t_{35040}, s} & \text{if } t = 1, \\ (1 - \lambda_{15\text{min}}) \cdot E_{\text{state, TES}, t-1, s} + \Delta t \cdot \left(\eta_{c, \text{TES}} \cdot U_{\text{charge, TES}, t, s} - \frac{V_{\text{discharge, TES}, t, s}}{\eta_{d, \text{TES}}} \right) & \text{if } t > 1. \end{cases} \quad (3.69)$$

3.2.5 Expected BM Energy Volumes Metrics

In order to quantify the upper bound of flexibility that the LSHP can offer to the power grid over the optimization horizon, the maximum expected balancing energy volumes are evaluated under complete bid acceptance ($\mu = 1$ and $\nu = 1$). The expected upward flexibility captures the cumulative electricity draw reductions achieved by the LSHP when providing upward grid support, scaled across all scenarios $s \in S$ by their historical probabilities of occurrence (ρ_s) in *MWh*. Conversely, the expected downward flexibility measures the additional power absorbed by the asset to mitigate grid oversupply in *MWh*.

These expected annual energy volumes are calculated over the set of discrete time steps $t \in T_{15}$, representing the operational resolution of the BM. The upward ($E_{\text{flex, up}}$), downward ($E_{\text{flex, down}}$), and total expected flexibility services ($E_{\text{flex, total}}$) are formally defined as follows:

$$E_{\text{flex, up}} = \sum_{s \in S} \rho_s \cdot \left(\sum_{t \in T_{15}} \Delta t \cdot V_{\text{balancing, up}, t, s} \right), \quad (3.70)$$

$$E_{\text{flex, down}} = \sum_{s \in S} \rho_s \cdot \left(\sum_{t \in T_{15}} \Delta t \cdot V_{\text{balancing, down}, t, s} \right), \quad (3.71)$$

$$E_{\text{flex, total}} = E_{\text{flex, up}} + E_{\text{flex, down}}. \quad (3.72)$$

Chapter 4

Input Data & Post-Optimization Analysis

Following the mathematical formulation of the two-stage optimization problem established in Chapter 3, this chapter presents the techno-economic input data, operational parameters, and case study exogenous parameters for the district networks examined of Zurich and Amsterdam. It mainly details the localized thermal and cooling demand profiles synthesis, energy performance characteristics, fuel and electricity market pricing structures, BM price scenario clusters and other considered optimization input costs and data. Furthermore, this chapter establishes the environmental analysis utilized to quantify both operational and embedded GHG emissions of the optimization results, and defines the examined MES configurations and levelized cost metric evaluated across both greenfield and brownfield investment pathways.

4.1 Input Data and Parameters

The most important inputs and parameters to the optimization problem are described in the following:

4.1.1 Heating and cooling yearly demand profiles ($D_{H,t}$ and $D_{C,t}$)

To construct the hourly demand profiles for the 2025 optimization horizon in both case study networks, a high-resolution synthesizing procedure is implemented to capture both residential and tertiary requirements for space heating, cooling, and domestic hot water. This approach utilizes generic, unitless demand profiles derived from the synthetic load profiles (SLPs) for heat pumps provided by the Municipal Utility of Munich (*Stadtwerke München*) [52] and the German Federal Association of Energy and Water Management (BDEW) [53] both retrieved from the European Open Source Data Repositories project, named “Hotmaps project - The open source mapping and planning tool for heating and cooling demand in the EU” [54]. These generic profiles establish a functional relationship between the hour of the day, the day type, ambient temperature levels, and the

corresponding relative energy demand.

The synthesis of these continuous 8,760-hour profiles for Zurich (CH04) and Amsterdam (NL32) is executed through a three-stage mapping, normalization, and scaling framework which can be summarized below:

1. **Multi-Dimensional Attribute Matching:** For every individual hour t ($t \in \{1, 2, \dots, 8760\}$) of the year 2025, a multi-dimensional lookup is performed to map local exogenous data to the generic SLP matrix. Specifically, each hour is matched based on four primary criteria:
 - **Day-Type Classification (d):** Differentiating between weekdays ($d = 0$), Saturdays or days preceding a public holiday ($d = 1$), and Sundays or public holidays ($d = 2$), tailored strictly to the specific 2025 calendar and holiday schedule of Zurich and Amsterdam (relevant for both DHW and SH and SC).
 - **Diurnal Hour (h):** The specific hour of the day ($h \in \{1, \dots, 24\}$) (relevant for both DHW and SH and SC).
 - **Meteorological Temperature (T_t):** The localized, real-time hourly ambient temperature recorded for that specific hour in 2025 in both cities. These hourly temperature data for 2025 are retrieved for Zurich from the Fluntern meteorological station and for Amsterdam from the Schiphol meteorological station (relevant only for SH and SC).
 - **Seasonal Influence (s):** Accounting for macro-seasonal variations in hot water demand by mapping each hour to a specific period of the year, categorized into summer ($s = 0$, spanning 15/05–14/09), winter ($s = 1$, spanning 01/11–20/03), and transition periods ($s = 2$, spanning 21/03–14/05 and 15/09–31/10) (relevant only for DHW).

Through this matching process, a raw, unitless load value $\hat{L}_t^c$ is assigned to each hour t for each city c for domestic hot water and space heating and cooling both residential and tertiary.

2. **Mathematical Normalization:** To eliminate any residual volumetric bias from the original German reference dataset and isolate the localized temporal distribution, the raw hourly profile is normalized. The unitless normalized load factor for each hour, $\bar{L}_t^c$, representing its fractional share of the total annual consumption, is calculated as follows:

$$\bar{L}_t^c = \frac{\hat{L}_t^c}{\sum_{k=1}^{8760} \hat{L}_k^c}$$

By definition, this ensures that the integral of the annual normalized profile equals unity ($\sum_{t=1}^{8760} \bar{L}_t^c = 1$).

3. **Volumetric Demand Scaling:** Finally, the hourly load profiles are converted into actual energy demand vectors (D_t , in MWh). The normalized hourly shares are multiplied by the actual, annualized total historical heating and cooling demand

along with the shares of residential and tertiary consumption of each city as explained further in detail.

Annual Volume Determination and Consumer Sector Shares

Through this multi-attribute matching process, a raw, unitless load value $\bar{L}_t^c$ is assigned to each hour t for each city c , differentiating by application (space heating, domestic hot water, and cooling) as well as consumer sector (residential and tertiary).

To scale these generic, unitless profiles into absolute hourly energy flows (MWh), the annual consumer demand volumes for each city are systematically derived based on public available reports and realistic assumptions.

For Zurich, the baseline annual heat demand (D_H^{ZH}), which encompasses both space heating (SH) and domestic hot water (DHW), is established at 434,340 MWh based on the energy system modeling of the ERZ district heating network [48] after subtracting heat generation of WtE power plants from the total heat demand value. To ensure strict compatibility and eliminate volumetric bias when comparing the two geographic locations since the comparison's scope is technology-focused rather than city-scale forecasting, the same baseline annual heat demand is applied to Amsterdam ($D_H^{AM} = 434,340$ MWh), matching the approximate scale of the Amsterdam-Zuid district network. To account for transmission and distribution losses before sector allocation, the network thermal demand are scaled across all operational hours using the predefined network loss factor of 1.1.

Because localized historical cooling data are often less transparent and publicly unavailable due to limited developed district cooling networks in the examined cities, a climate-driven threshold methodology is developed to approximate the annual cooling energy demand (D_C) of each city. First, a critical ambient temperature threshold is defined at 25.5°C to identify hours requiring active space cooling, which is motivated by the upper cooling temperature thresholds of study [55] and the method of Cooling Degree Days (CDD). This temperature threshold is defined since it lays within the boundaries of the ambient temperature where most buildings activate machines which provide cooling services like air conditioning [56]. The annual cooling demand is then derived as a function of the total heating demand using a structural seasonal ratio:

$$D_C = D_H^c \times \left(\frac{\mathcal{H}^{c>25.5}}{8760 - \mathcal{H}^{c>25.5}} \right) \quad (4.1)$$

where $\mathcal{H}^{c>25.5}$ represents the number of hours where the ambient temperature exceeded 25.5°C in the baseline year 2025. For Zurich, 296 hours exceeded this threshold, yielding an annual cooling requirement of 15,202 MWh. For Amsterdam, 131 hours exceeded the threshold, resulting in an annual cooling requirement of 6,515 MWh.

The fundamental parameters governing end-use and sectoral demand distribution are established using a combination of international energy benchmarks and structural assumptions:

- **Domestic Hot Water Share:** The fraction of total thermal heating demand attributed to domestic hot water (DHW) is defined as 15% ($\alpha^c = 0.15$) in France according to the work performed by Hoffman et al. [57] and as a result it is assumed the same value for the examined cities of this thesis, since their residential/tertiary characteristics are closely linked with the climatic conditions prevailing across Central and West European countries.
- **Sectoral Thermal Distribution:** The shares of residential and tertiary sector contributions to the total heating grid demand are set to 70% (β_{res}^c) and 30% (β_{ter}^c), respectively, in alignment with the latest energy efficiency report by the IEA [58].
- **Sectoral Cooling Distribution:** Because a fully mature urban district cooling infrastructure has not yet been deployed in either city, a pragmatic engineering assumption is adopted for the cooling grid. Specifically, it is assumed that 90% (ω_{ter}^c) of the centralized cooling demand originates from the tertiary sector (e.g., shopping malls and corporate office spaces), while the remaining 10% (ω_{res}^c) is allocated to the limited residential units connected to this multi-energy network.

As a result of these design conditions, the comprehensive breakdown of structural network parameters, sector shares, and volumetric allocations used to extract the final profile assembly of the MES is summarized in Table 4.1.

Table 4.1: Structural energy demand and sector breakdown parameters

Parameter	Zurich (ZH)	Amsterdam (AM)
Annual Heat Demand, D_H^c (MWh)	434,340	434,340
Annual Cooling Demand, D_C^c (MWh)	15,202	6,515
DHW Share of Heat Demand (α^c)	0.15	0.15
Residential Heating Share (β_{res}^c)	0.7	0.7
Tertiary Heating Share (β_{ter}^c)	0.3	0.3
Residential Cooling Share (ω_{res}^c)	0.1	0.1
Tertiary Cooling Share (ω_{ter}^c)	0.9	0.9

Sectoral Disaggregation and Final Profile Assembly

To transform the generalized unitless load vectors into a single, cohesive network demand profile, the energy volumes are segregated into specific energy demands (i.e., SH, SC and DHW) and multiplied by their respective sectoral grid shares as indicated in Table 4.1.

For each hour t , the space heating and domestic hot water components are separated using the DHW share parameter α^c , whereby the discrete annual energy volumes for DHW (D_{DHW}^c) and SH (D_{SH}^c) for each city c are isolated via:

$$D_{DHW}^c = D_H^c \times \alpha^c, \quad (4.2)$$

$$D_{SH}^c = D_H^c \times (1 - \alpha^c). \quad (4.3)$$

Let $\bar{L}_{res,sh,t}^c$ and $\bar{L}_{ter,sh,t}^c$ represent the normalized, unitless hourly profile values for residential and tertiary space heating, respectively. Similarly, let $\bar{L}_{res,dhw,t}^c$ and $\bar{L}_{ter,dhw,t}^c$ denote the corresponding domestic hot water factors. The final aggregate hourly thermal grid demand for heating ($D_{H,t}^c$) of each city is constructed by compounding the sectoral profiles with their corresponding network distribution shares (β_{res}^c and β_{ter}^c):

$$D_{H,t}^c = [(\bar{L}_{res,sh,t}^c \cdot \beta_{res}^c + \bar{L}_{ter,sh,t}^c \cdot \beta_{ter}^c) \cdot D_{SH}^c] + [(\bar{L}_{res,dhw,t}^c \cdot \beta_{res}^c + \bar{L}_{ter,dhw,t}^c \cdot \beta_{ter}^c) \cdot D_{DHW}^c] \quad (4.4)$$

An analogous formulation is applied to resolve the network's aggregate hourly cooling demand profile ($D_{C,t}^c$) of each city. Using the global residential and tertiary cooling distribution constants (ω_{res}^c and ω_{ter}^c), the final cooling profile is calculated as:

$$D_{C,t}^c = (\bar{L}_{res,c,t}^c \cdot \omega_{res}^c + \bar{L}_{ter,c,t}^c \cdot \omega_{ter}^c) \cdot D_C^c \quad (4.5)$$

Consequently, the extracted hourly heating load profile ($D_{H,t}$ in MWh per timestep) and hourly cooling load profile ($D_{C,t}$ in MWh per timestep) for each city serve as deterministic exogenous parameters in the first-stage optimization. These vectors dictate the continuous energy demand that must be satisfied uninterrupted at each hourly timestep across the annual optimization horizon.

For the second-stage optimization, which operates at a finer 15-minute temporal resolution, the same underlying energy load profiles are utilized. To adapt the hourly data to the quarter-hourly framework, a pragmatic engineering assumption is adopted whereby the total energy volume of each discrete hour is equally distributed across four uniform 15-minute intervals. This constant-power allocation ensures total volumetric consistency between both optimization stages while facilitating the high-resolution operational simulation required for BM participation.

Overall, this downscaling methodology preserves the underlying national consumer habits embedded in the SLPs while ensuring strict alignment with the specific 2025 calendar structure, local meteorological conditions, and actual annual energy volumes of the investigated district networks.

4.1.2 District network's temperature levels and LSHP's heating and cooling coefficient of performance (COP_H and COP_C)

The modeling of the LSHP is predicated on high-capacity, industrial-scale configurations, reflecting the state-of-the-art architectures found in modern district energy systems [26], [10]. The 4th generation DHCNs investigated in this thesis and assumed for both examined district network cities, integrates a centralized heat pump unit to fulfill thermal demand dynamically, moving away from decentralized, building-level generation assets. Given the geographical and environmental characteristics of the selected case study locations, hydrothermal (water-sourced) systems represent the most viable configuration. Thermodynamically, these LSHP units extract low-grade heat from an aquatic source (e.g., a lake, sea, or canal) and elevate it to the required network grid

temperature, utilizing the district network as the primary thermal sink. For that reason, the following temperature levels are assumed for the calculation of the heating and cooling COP:

- **Heat Source Temperature for Heating:** The monthly temperature variations of the lake or seawater body serving as the hydrothermal source are derived from official datasets provided by the report of the Danish Energy Agency on the “Generation of Electricity and District heating” [59] for the seawater temperatures in Copenhagen across last 5 years. For modeling consistency, this identical monthly water temperature profile is applied to both examined case study locations and is given in monthly values in the Appendix Table 8.6.
- **Heat Sink Temperature for Heating:** The supply temperature delivered to the district heating network is assumed to be 70°C. This operational profile falls within the established boundaries of a system transitioning towards a 4th generation district network [12,60].
- **Network Return Temperature:** The return temperature of the district heating grid is assumed to be 30°C, which aligns with the low-temperature return parameters characteristic of a system transitioning towards a 4th generation infrastructure [12,60].
- **Cooling Supply Temperature:** The chilled water supply temperature for district cooling services is assumed to be 6°C, reflecting standard industrial boundary conditions for centralized cooling distribution according to Eurostat [61].

In Zurich, the energy company EWZ is actively developing the *CoolCity* project, which leverages Lake Zurich as a renewable hydrothermal source for both district heating and district cooling services [62]. Similarly, Amsterdam utilizes its network of seawater canals and open water bodies for decentralized thermal energy extraction.

Regarding the thermodynamic cycle of the central unit, industrial-scale heat pumps predominantly utilize R717 (ammonia) [63] or R744 (carbon dioxide) [32] as working fluids. Consequently, it is a robust engineering assumption within the scope of this thesis to model a water-sourced LSHP operating with either ammonia or carbon dioxide.

Calculation of heating COP vector ($COP_{H,t}$)

To dynamically model the heating operational efficiency of the LSHP, the monthly $COP_{H,t}$ should be calculated as a function of the temperature lift between the heat source and sink temperature levels. Rather than assuming a constant second-law Carnot efficiency, a deterministic empirical regression model is used, which was developed by Jesper et al. [63] for MW-installed industrial mechanical compression heat pumps of high temperature lifts utilizing R717 (ammonia) as a working fluid. This quadratic function represents the highest-accuracy regression formulation presented in [63] for characterizing the performance of this class of LSHPs.

The temperature lift ΔT_t at any given hour t represents the difference between the constant network distribution heat sink delivery temperature (T_{SINK}) and the localized, monthly fluctuating water heat source temperature ($T_{s,t}$):

$$\Delta T_t = T_{SINK} - T_{s,t} \quad (4.6)$$

where the district heating supply temperature (T_{SINK}) is held constant at 70°C. Using the derived empirical quadratic regression model for ammonia-based refrigeration cycles, the monthly baseline heating efficiency is formulated as:

$$\text{COP}_{H,t} = 0.0014515 \cdot (\Delta T_t)^2 - 0.23104 \cdot \Delta T_t + 11.684 \quad (4.7)$$

This continuous 8,760-hour array of $\text{COP}_{H,t}$ values serves as a direct technical parameter vector within the MILP optimization matrix of the first stage problem, dictating the exact hourly electricity input requirements necessary to produce heat by the LSHP.

For the second-stage optimization framework, where the temporal resolution shifts from an hourly to a quarter-hourly (15-minute) frequency, the monthly heating COP values are held constant across all four constituent quarters of each respective hour.

Calculation of cooling COP vector ($\text{COP}_{C,t}$)

To model the operational efficiency of the centralized cooling hybrid function of the LSHP and of the electric chiller when the MES operates in the absence of the LSHP, a dynamic monthly $\text{COP}_{C,t}$ is calculated. Rather than assuming a constant efficiency, the electric motor-driven water chiller and the LSHP are both characterized by a temperature-dependent linear scaling model calibrated against the standard efficiency requirements for central cooling outlined in ASHRAE report for HVAC equipment minimum efficiency requirements [64].

The cooling lift $\Delta T_{C,t}$ at any given hour t represents the gradient between the monthly fluctuating lake/seawater temperature ($T_{s,t}$) which is identical to the one used to calculate the heating $\text{COP}_{H,t}$ and the constant district cooling supply temperature ($T_{COOLING}$), which is held at 6.0°C:

$$\Delta T_{C,t} = T_{s,t} - T_{COOLING} \quad (4.8)$$

The cooling COP is calculated by taking a baseline efficiency factor equal to 4.7 and a temperature sensitivity factor equal to 0.0045 and reducing it linearly based on the summer occurred temperature lift. This relationship is formulated as follows:

$$\text{COP}_{C,t} = 4.70 \cdot (1.0 - 0.0045 \cdot \Delta T_{C,t}) \quad (4.9)$$

The empirical parameters (4.70 and 0.0045) are explicitly calibrated based on ASHRAE's report [64] to ensure the LSHP or the chiller maintains a representative nominal efficiency around a standard COP of 4.5 under realistic summer lake or seawater temperature profiles.

Analogous to the heating framework, for the second-stage optimization, the cooling COP time interval values are held constant across all four constituent quarters of each respective hour. This 8,760-hour parameter array ($\text{COP}_{C,t}$) is directly ingested by the MILP model to determine the interval electricity required to produce cooling either from the LSHP or the chiller.

4.1.3 Energy efficiencies of BB, CHP and TES

Apart from the modeled energy efficiencies (COP) of the LSHP and the Chiller the rest of the candidate energy conversion technologies and of the TES are dictated by steady-state efficiency parameters and thermodynamic coefficients. These parameters are assumed constant throughout the annualized optimization horizon for both stages of the optimization problem and identical for both examined district networks.

Thermal and electrical efficiencies of the BB and CHP

The static thermal efficiency of the BB is defined as $\eta_{\text{BB}} = 0.86$, a value that aligns with the empirical performance boundaries established in the literature by Doninelli et al. [65]. The CHP unit is parameterized with a constant electrical efficiency of $\eta_{\text{el,CHP}} = 0.35$ and a concurrent thermal co-generation efficiency of $\eta_{\text{th,CHP}} = 0.65$. These performance values are representative of state-of-the-art commercial units and are sourced from the “Future Markets Insights” regarding the CHP’s 2035 outlook [66].

Thermal efficiencies of the TES

To accurately capture the time-varying state of charge profile of the TES system, the model uses self-discharge standing loss rate (λ_{TES}), thermal input charging efficiency ($\eta_{c,\text{TES}}$), and thermal output discharging efficiency ($\eta_{d,\text{TES}}$).

To parameterize the TES efficiencies, reference is made to the findings of [33, 67], which provides representative operational ranges for Pit TES systems. For the specific applications of the proposed MES, the round-trip efficiency can be established at $\eta_{\text{TES}} = 81\%$ according to [67]. Consequently, a pragmatic engineering assumption is adopted by defining the round-trip efficiency as the product of the charging and discharging efficiencies, such that:

$$\eta_{\text{TES}} = \eta_{c,\text{TES}} \cdot \eta_{d,\text{TES}} \quad (4.10)$$

Assuming symmetrical energy conversion losses during cyclic charging and discharging phases, these parameters are defined equally as $\eta_{c,\text{TES}} = \eta_{d,\text{TES}} = 0.9$ from equation (4.10).

Regarding the thermal degradation over time, the hourly continuous self-discharge standing loss rate is assumed to be $\lambda_{\text{TES}} = 0.0005$, meaning that the storage loses 0.05% of its currently stored thermal energy inventory every single hour. The above serves as a robust engineering approach for water-Pit thermal storage configurations [33, 67, 68]. Crucially, to preserve thermodynamic consistency during the high-resolution second-stage optimization, where the temporal resolution scales down to a 15-minute operational

interval ($\Delta t = 15$ min), the hourly loss rate must be nonlinearly scaled. This non-linear scaling stems from the fact that standing thermal self-discharge is fundamentally governed by an exponential decay process rather than a linear approximation. Over an arbitrary elapsed duration t , the remaining energy level $E(t)$ subject to standing losses follows a first-order differential decay profile which can be mathematically expressed by the following relationship:

$$1 - \lambda_{15\min} = (1 - \lambda_{\text{TES}})^{\frac{1}{4}} \quad (4.11)$$

Rearranging Equation (4.11) yields the explicit parameterization for the second-stage self-discharge efficiency:

$$\lambda_{15\min} = 1 - (1 - \lambda_{\text{TES}})^{0.25} \quad (4.12)$$

Overall, the steady-state efficiency parameters of the BB, CHP and TES are summarized in Table 4.2.

Table 4.2: Summary of steady-state energy efficiencies

Technology	Efficiency Parameter	Symbol	Value
BB	Thermal Efficiency	η_{BB}	0.86
CHP	Electrical Efficiency	$\eta_{\text{el,CHP}}$	0.35
	Thermal Efficiency	$\eta_{\text{th,CHP}}$	0.65
TES	Charging Efficiency	$\eta_{c,\text{TES}}$	0.9
	Discharging Efficiency	$\eta_{d,\text{TES}}$	0.9
	Hourly Loss Rate	λ_{TES}	0.0005

4.1.4 Deterministic fuel prices parameters

CHP input fuel prices

To accurately characterize the operational constraints and economic dispatch of the CHP plant, the total input fuel price must be defined through its underlying cost components. The total effective natural gas procurement cost is modeled as an exogenous time-series parameter comprising two primary mechanisms: the wholesale natural gas commodity price and the corresponding carbon emission costs regulated under the European Union Emissions Trading System (EU ETS).

The total hourly input fuel price $\pi_{\text{gas},t}$ [€/MWh_{gas}] for both examined district network cities is mathematically formulated as follows:

$$\pi_{\text{gas},t} = \kappa_{\text{gas},t} \cdot \text{CF}_{\text{CO}_2/\text{gas}} + \Lambda_{\text{gas},t} \cdot f_{\text{delivery}} \quad (4.13)$$

Where the individual variables, parameters, and scaling coefficients remain identical for both cities and are explicitly defined below:

- $\pi_{\text{gas},t}$ represents the comprehensive hourly input fuel price for the CHP plant [€/MWh_{gas}], capturing the true combined operational cost of fuel procurement and climate liabilities at each timestep t .

- $\Lambda_{\text{gas},t}$ represents the wholesale commodity price of natural gas [$\text{€}/\text{MWh}_{\text{gas}}$]. For this analysis, the vector is indexed against the monthly price variations of the Title Transfer Facility (TTF), the Dutch natural gas wholesale marketplace, virtual trading point for the baseline year 2025 [69] and are given in monthly values in Appendix Table 8.7. This resolution aligns with standard European industrial hedging strategies, where generation assets typically secure fuel volumes based on forward-looking monthly TTF indices.
- f_{delivery} is a static premium multiplication factor established at 1.15. This dimensionless coefficient represents a pragmatic engineering assumption introduced to account for localized grid transmission tariffs, distribution network usage fees, and supplier volumetric delivery premiums [70].
- $\kappa_{\text{gas},t}$ represents the daily varying EU ETS carbon allowance prices for the examined year 2025 [$\text{€}/\text{tCO}_2$]. These 2025 historical daily spot market settlement values for European Emission Allowances are retrieved from [71].
- $\text{CF}_{\text{CO}_2/\text{gas}}$ is the specific carbon emission intensity conversion factor [$\text{tCO}_2/\text{MWh}_{\text{gas}}$] of the natural gas fuel and is equal to $0.201 \text{ tCO}_2/\text{MWh}_{\text{gas}}$ [72]. This parameter establishes dimensional compatibility within the equation, converting the unit of the carbon penalty from a mass basis ($\text{€}/\text{tCO}_2$) directly into an energy basis ($\text{€}/\text{MWh}_{\text{gas}}$).

By applying this methodology, the monthly wholesale TTF prices and the daily carbon emission cost allowances are mapped across a continuous 8,760-hour horizon through equation (4.13), capturing both the gas price seasonality and the emissions cost component fluctuations. As a result, a comprehensive, time-varying, hourly fuel input cost profile is yielded that is utilized within the optimization matrix of the first stage optimization problem as the natural gas fuel price parameter.

Analogous to the COP calculation framework, for the second-stage optimization, the CHP's fuel input cost time interval values ($\pi_{\text{gas},t}$) are held constant across all four constituent quarters of each respective hour.

Day Ahead Electricity prices

To accurately model the procurement cost of the LSHP, its driving energy cost must be quantified. Because the system is driven by electric motor compressors of MW-size, its primary operational expenditure is determined by electricity procurement from the wholesale market rules, augmented by regulated grid network charges. Consequently, the total hourly variable electricity procurement cost for 2025 is computed for each examined city c according to the following formulation:

$$\pi_{\text{elec},t}^c = P_{\text{DAM},t}^c + \text{GT}^c \quad (4.14)$$

Where the individual variables are explicitly defined below:

- $\pi_{\text{elec},t}^c$ represents the comprehensive 2025 hourly electricity procurement cost for the LSHP in city c at timestep t [€/MWh].
- $P_{\text{DAM},t}^c$ denotes the hourly wholesale DAM electricity spot price of the year 2025 for the respective bidding zone of city c at timestep t [€/MWh]. These historical price vectors are retrieved directly from the ENTSO-E Transparency Platform [73].
- GT^c represents the localized, static grid tariff applied to city c [€/MWh]. This parameter accounts for the regulated transmission and distribution network usage fees levied on high-capacity industrial consumers and are equal to 49 €/MWh in Zurich [74] and to 30 €/MWh in Amsterdam [75].

By applying this methodology, the 2025 hourly DAM spot market prices and the static localized grid tariffs are mapped across a continuous 8,760-hour horizon for each city, capturing both the electricity price volatility and the fixed network distribution costs. As a result, a comprehensive, time-varying, hourly electricity procurement cost profile is yielded that is utilized within the optimization matrix of the first-stage optimization problem as the LSHP driving energy price parameter.

Analogous to the rest parameter analysis before, for the second-stage optimization, the LSHP's electricity procurement cost time interval values ($\pi_{\text{elec},t}^c$) are held constant across all four constituent quarters of each respective hour.

Biomass price

To accurately model the operation of the Biomass Boiler, its primary fuel procurement cost must be quantified. For the purposes of this thesis, wood pellets are designated as the reference biomass fuel source. Because biomass procurement is highly dependent on geographical factors and typically secured via long-term forward contracts, the fuel price is assumed to be fixed on an annual basis. Consequently, this procurement parameter maintains a constant value across the entire optimization horizon and is considered identical for both examined case-study cities.

The biomass fuel price π_{biomass} adopted in this optimization problem, is mathematically defined as:

$$\pi_{\text{biomass}} = 50 \text{ €/MWh} \quad (13.9 \text{ €/GJ}) \quad (4.15)$$

The π_{biomass} represents the static procurement price for the Biomass Boiler [€/MWh]. Its nominal price is indexed against high-moisture wood chips from a study regarding the role of wood in the Polish energy sustainability [76] and is adjusted slightly upward to integrate localized transportation tariffs.

4.1.5 Stochastic energy balancing prices clustering

To accurately model the opportunities for BM participation during the second-stage optimization framework, where the LSHP-TES systems interact with the district heating network, the prices of the aFRR energy remuneration must first be characterized. Because aFRR balancing energy prices are notorious for high volatility and extreme pricing

spikes, a stochastic programming approach is implemented. The goal of this framework is to feed five representative, internally consistent price scenarios for each examined city into the model, capturing both upward and downward balancing energy remuneration prices.

Data Collection and Preprocessing

To perform the scenario reduction, historical quarterly (15-minute resolution) balancing energy prices spanning a 10-year horizon (2016–2025) are gathered according to this market’s rules and time resolution. The localized dataset source varies by geographic case study:

- **Switzerland (Zurich):** Historical 15-minute time-series data are retrieved directly from the SwissGrid national database labeled as (Average positive and negative secondary control energy prices) [77].
- **Netherlands (Amsterdam):** Likewise, historical price data are retrieved from TenneT transparency page labeled as (Balancing settlement prices - Price Dispatch Up and Down) [78].

Prior to clustering, the datasets are preprocessed to ensure structural uniformity. Leap years are standardized to non-leap year lengths by omitting February 29th, yielding exactly 35,040 values per year for both positive (upward) and negative (downward) secondary reserve prices.

K-Medoids Clustering

Because electricity BM data are heavily influenced by extreme outliers (such as pricing spikes), standard clustering methods like K-means can distort the feature space [79]. To address this, the features are normalized utilizing a robust scaling technique based on medians and Interquartile Ranges (IQR).

Scenario reduction is subsequently executed using the K-Medoids clustering algorithm (specifically, the Partitioning Around Medoids, or PAM, algorithm), whose mathematical formulation is provided in the Appendix (Section 8.4), as it is performed in the study of Sun et al. for “Uncovering residential electricity consumption patterns” [80] and the study of Anderson et al. for energy efficiency optimization [79] to yield five representative scenario clusters. K-Medoids is selected as the optimal clustering methodology for two fundamental reasons:

1. **Insensitivity to outliers:** *K*-Medoids is far less sensitive to extreme, non-representative annual pricing anomalies during cluster assignment compared to traditional *K*-means.
2. **Temporal validity:** Unlike *K*-means and other similar clustering methods, which computes virtual “average” centroids that smooth out physical volatility and create artificial price profiles, *K*-Medoids selects actual, real-world historical years

(medoids) from the input dataset as cluster representatives. This guarantees that the final 35,040 resolution balancing prices profiles fed to the second-stage optimization preserve historical correlations and volatility.

The probability of occurrence for each representative scenario is calculated based on the relative density of each cluster (the ratio of historical years assigned to a cluster relative to the total decade dataset). This methodology ultimately yields a representative, structurally complete, 35,040-interval time-series matrix of 5 scenario of upward ($\pi_{\text{bal,up},t,s}$) and downward ($\pi_{\text{bal,down},t,s}$) flexibility pricing in €/MWh along with its mathematically sound probability weights (ρ_s) for the stochastic optimization stage.

4.1.6 Capital Recovery Factor

To integrate the initial capital expenditures (CAPEX_k) of long-term assets into the single-year 2025 optimization horizon, the investments must be annualized. This is achieved by utilizing the CRF, also known as the annuity factor (α). The CRF converts a one-time capital investment into an equivalent uniform annual cost, formulated as follows:

$$\text{CRF} = \alpha = \frac{i(1+i)^n}{(1+i)^n - 1} \quad (4.16)$$

Where the economic parameters and structural lifetimes are defined as:

- i [%] represents the nominal interest (or discount) rate, which is established at 12%. This rate reflects the cost of capital and investment risks associated with large-scale municipal energy infrastructure, which are notorious for having higher perceived risk [81].
- n [years] denotes the operational and economic lifetime of the MES configuration, set at 20 years, which is particularly relevant for the HP technology according to [81].

This choice of economic lifetime ($n = 20$ years) and interest rate ($i = 12\%$) aligns with standard assumptions made in reviewed techno-economic optimization studies of Chapter 2. By multiplying the initial investment of the LSHP and TES by the calculated CRF, the annualized investments are directly integrated into the objective function of the optimization problem.

4.1.7 Technology specific costs, market sizes and minimum load fractions

These modeling parameters remain the same for both examined networks and during both stage optimizations and have as follows.

Technology specific CAPEX and fixed OPEX

The capital expenditure (CAPEX_k) and fixed operational expenditure (OPEX_k) of each candidate technology k constitute fundamental inputs to the optimization model, as they directly dictate the system's TAC. To maintain a realistic economic representation, these parameters are indexed against publicly available project data and pragmatic engineering assumptions and remain the same for both examined case studies for district network technologies. Also, it is assumed that capacities scale linearly in terms of investments and do not follow economies of scale approach.

For the primary generation assets, including the BB, the CHP plant, and the LSHP, cost factors are normalized per unit of thermal capacity [$\text{€}/\text{MW}_{\text{th}}$]. Conversely, the TES cost parameters are normalized per unit of energy storage capacity [$\text{€}/\text{MWh}_{\text{th}}$].

Then in order to model labor wages and regular scheduled maintenance a fixed annual operational expenditure (OPEX_k) is established. Following a common engineering economics convention, this parameter is assumed to be a fixed percentage of the asset's initial capital expenditure:

$$\text{OPEX}_k = 0.02 \cdot \text{CAPEX}_k \quad (4.17)$$

Equation (4.17) applies a baseline rate of 2% of the initial CAPEX_k for each respective technology k , except for the TES and the CHP which has been retrieved from [27] and [82], to model annual fixed operational overheads.

When a brownfield modeling approach is adopted for the BB and the CHP plant, these units are treated as pre-existing infrastructure with fully amortized investments. Consequently, their initial capital procurement cost is defined as zero ($\text{CAPEX}_{\text{BB}} = \text{CAPEX}_{\text{CHP}} = 0$), meaning only their respective fixed operational costs (OPEX_k) remain active during the economic dispatch optimization.

On the other hand, the investment parameters for the LSHP and the electric water chiller are derived from the empirical cost of Everllence's large-scale seawater heat pump installations in Denmark (e.g., the Esbjerg project). The investment parameters for the TES system are benchmarked against the seasonal water-Pit thermal energy storage (PTES) system implemented in Dronninglund, Denmark. In this project by Sifnaios et al. [27], the Pit storage construction and installation costs are typically reported on a volumetric basis $\dot{Q}_{\text{TES}}$ [$\text{€}/\text{m}^3$]. To integrate this into the energy-centric MILP model, this volumetric parameter must be converted into a standard energy capacity metric $\dot{P}_{\text{TES}}$ [$\text{€}/\text{MWh}$]. This thermodynamic conversion [83] is dictated by the specific heat capacity of water, the density of the water storage medium, and the operating thermal lift of the district heating network that is defined in the COP calculation section:

$$\Delta T_{\text{lift}} = T_{\text{SINK}} - T_{\text{RETURN}} = 70^\circ\text{C} - 30^\circ\text{C} = 40^\circ\text{C} \quad (4.18)$$

$$P_{\text{TES}} = \rho_{\text{H}_2\text{O}} \cdot Q_{\text{TES}} \cdot c_{p,\text{H}_2\text{O}} \cdot \Delta T_{\text{lift}} \cdot \text{CF}_{\text{Wh} \rightarrow \text{MWh}} \quad (4.19)$$

Where:

- $\rho_{\text{H}_2\text{O}}$ is the density of the storage water, assumed to be $997 \text{ kg}/\text{m}^3$ [83].

- c_{p,H_2O} is the specific heat capacity of water, established at $4182 \text{ J kg}^{-1} \text{ K}^{-1}$ or $1.162 \text{ Wh}/(\text{kg} \cdot ^\circ \text{C})$ [83].
- $CF_{Wh \rightarrow MWh}$ is a dimensional scale conversion factor equal to 10^{-6} MWh/Wh .

Using this formulation, the physical volume Q_{m^3} required to store 1 MWh of thermal energy under the network's operational constraints is directly determined:

$$Q_{TES} = \frac{P_{TES}}{1.158 \cdot 10^{-3} \cdot \Delta T_{lift}} \quad (4.20)$$

This mathematical relationship allows the volumetric capital costs from the Dronninglund project to be converted into energy-specific costs [€/MWh] for direct integration into the system-level optimization framework. In addition to these direct TES specific investments, supplementary expenditures are incorporated to account for land acquisition rights. Consequently, the comprehensive capital ($CAPEX_k$) and fixed operational ($OPEX_k$) cost parameters utilized for each candidate technology k during the Greenfield approach in this study are summarized in Table 4.3

Table 4.3: Investment economic parameters used for the examined technologies in (M€) during the Greenfield approach

Technology	Unit	CAPEX	Fixed OPEX	Source
Biomass Boiler	M€/MW _{th}	0.35	0.007	[84]
CHP	M€/MW _{th}	1.4	0.06	[82], [85]
LSHP	M€/MW _{th}	1.7	0.034	[32]
Electric Chiller	M€/MW _c	1.7	0.034	[32]
TES	M€/MWh _{th}	0.008	0	[27]

Technology's capacity market sizes boundaries

To define the feasible design space for the capacity design optimization, physical capacity boundaries must be established for each technology k . These sizing limits are mathematically formulated as the interval $[P_{min,k}, P_{max,k}]$. Within these bounds, the optimization model dynamically determines either the optimal newly installed capacity or, in the case of pre-installed brownfield assets (the BB and CHP), the maximum operational capacity utilization. Among these constraints, the upper bound $P_{max,k}$ is the most critical parameter as it dictates the maximum permissible design capacity size for each candidate technology.

The sizing boundaries are parameterized based on existing municipal infrastructure limits and similar accomplished projects and remain the same for each case study city:

- **CHP and BB:** Since a brownfield modeling framework is also assumed within the scope of this thesis for these technologies, their sizing boundaries are directly constrained by the existing network capacities. Consequently, $P_{max,CHP}$ and $P_{max,BB}$

reflect the maximum capacities of the existing district heating system assets already integrated into the municipal grid.

- **LSHP:** The maximum capacity limit $P_{\max, \text{HP}}$ is benchmarked against Everllence's mega heat pump installations in Denmark, where the largest deployed single-unit configuration has a capacity of 70 MW_{th} [32].
- **TES:** The upper boundary for the storage capacity, $P_{\max, \text{TES}}$, is benchmarked against the seasonal water Pit TES project implemented in Dronninglund, Denmark [27]. This reference facility features an excavation volume of 60,000 m³. Applying the thermodynamic conversion outlined in Equation (4.20) under the district network's operating temperature lift ($\Delta T_{\text{lift}} = 40^\circ\text{C}$), this volumetric boundary yields a maximum thermal energy storage capacity of approximately 3,000 MWh_{th}.

As a result, the technology capacity sizing limits are summarized in Table 4.4.

Table 4.4: Technology capacity sizing limits

Technology	Unit	$P_{\min, k}$	$P_{\max, k}$
BB	MW _{th}	0.05	50
CHP	MW _{th}	0.05	1000
LSHP	MW _{th}	0.05	70
Electric Chiller	MW _c	0.05	70
TES	MWh _{th}	0.05	2779.2

Technology minimum load fractions

To ensure physical and operational realism in modeling the dispatch of each candidate technology, a minimum operational threshold must be enforced whenever an asset is selected to operate by the optimizer at any given time interval. Consequently, during heating mode operation, dispatched technologies are constrained to operate above a baseline minimum idle fraction, as summarized in Table 4.5. These minimum idle load limits are established based on standard single unit utility practices, and the literature reviewed in Chapter 2. In the configuration without the LSHP or the TES enabled in the MES, the minimum load fraction of the CHP is set to 9.5% to prevent model infeasibilities arising from the operational minimum load limits of the BB and CHP.

Conversely, for the cooling mode of the LSHP and the operation of the electric chiller, a minimum part-load fraction of zero is assumed. This modeling assumption is dictated by the nature of the cooling load profiles, which frequently fall below the technical minimums typical of standard heating dispatch. As a result, when selected for cooling operation, these technologies are allowed to modulate down to near-zero production outputs. While this prevents artificial solver infeasibilities during low-demand hours,

Table 4.5: Minimum load fractions by technology during heating operation

Technology	Minimum Load Fraction (δ_k)
Biomass Boiler (BB)	10%
Combined Heat & Power (CHP)	15%
Large-Scale Heat Pump (LSHP)	15%

this unconstrained low-load modulation represents a limitation of the present modeling framework.

4.2 Environmental Perspective: the Assessment of GHG Emissions

To evaluate the ecological footprint of the optimized MES configurations, GHG emissions are selected as the primary metric to quantify their ecological footprint. In order to have a holistic evaluation of their ecological footprint a Life Cycle Analysis (LCA) of all their associated GHG emissions regarding their operation is being examined. As a result, we quantify both the operational and embodied emissions of the candidate technologies across the examined configurations and district networks. To ensure strict methodological coherence and transparency, all GHG emission factors are fetched from the ecoinvent database [86], considering an attributional LCA approach using the system model ‘allocation, cut-off by classification’.

Consequently, this subchapter is organized into three main subsections that diverge from traditional standalone scope boundaries to aggregate total LCA impacts logically. First, the evaluation focuses on the operational emissions stemming from primary input fuels, which consolidates the consumption of electricity, natural gas, and biomass. Crucially, this section couples the direct combustion and electricity consumption impacts (Scopes 1 & 2) with their respective upstream procurement and transportation footprints (Scope 3) into a single operational stream. Within this scope, the biogenic carbon dioxide released by the BB during on-site combustion is treated as net-zero, as the carbon emitted balances the carbon absorbed by the wood fuel during its vegetative growth phase. Second, the framework isolates the embedded emissions of each candidate technology and TES, focusing purely on the Scope 3 lifecycle impacts generated during the raw material extraction, manufacturing, and installation phases of the equipment. Finally, the last subsection presents the unified GHG calculations that aggregate these operational and embedded emission components to establish the total annualized GHG emissions.

4.2.1 Operational GHG Emissions

To parameterize the aggregated operational streams defined in the framework introduction for both examined district networks, LCA emission factors for each primary energy carrier are compiled from search queries in the Ecoinvent database [86] and with the search location being the examined countries. The technology-specific parameterization values for these input energy carriers are defined as follows:

- **BB:** In accordance with the net-zero operational combustion boundaries established for biogenic assets, the emission characterization for the BB isolates the upstream supply chain footprint. Consequently, only the emissions associated with the harvesting, manufacturing, chipping, and distribution of the wood chips are retrieved in $\text{kg CO}_2\text{-eq/kg}_{\text{wood}}$ by inserting the search string “*market for wood chips*”. This mass-based parameter is subsequently converted into an energy-specific intensity factor ($\text{kg CO}_2\text{-eq/kWh}_{\text{biomass}}$) using a reference wet wood chip Lower Heating Value (LHV) of 3.5 kWh/kg sourced from [87].

- **CHP:** The emission factor for the gas-fired CHP unit is derived through a two-step accounting process. First, the upstream emissions corresponding to the processing, manufacturing, and high-pressure distribution of natural gas are retrieved in $\text{kg CO}_2\text{-eq/m}^3$ by inserting the search sting “*market for natural gas high pressure*”. This volumetric parameter is converted into an energy-specific baseline $\text{kg CO}_2\text{-eq/kWh}_{\text{gas}}$ by applying a standard natural gas LHV of 10.55 kWh/m^3 sourced from IEA [88]. Second, because the CHP plant simultaneously generates dual useful streams (electricity and thermal energy), which complicates the direct attribution of emissions per unit of useful localized output, the on-site direct combustion product is imposed onto the supply chain footprint by adding the direct factor of $0.2 \text{ kg CO}_2\text{-eq/kWh}_{\text{gas}}$ [72].
- **LSHP and Chiller:** Because these technologies are in MW-size, they are modeled as industrial loads drawing electricity directly from the high-voltage transmission grid. The corresponding emission intensity factor consolidates both the upstream grid-infrastructure embedded footprint and the emissions associated with the consumption of 1 kWh of electricity by inserting the search sting “*market for electricity high voltage*”. This comprehensive baseline parameter is expressed in $\text{kg CO}_2\text{-eq/kWh}$.

To ensure dimensional compatibility with the annualized energy flows of the core optimization problems, these localized intensity values are transformed into $\text{t CO}_2\text{-eq/MWh}$.

4.2.2 Technology Embedded Emissions

To parameterize the aggregated embedded emissions of the candidate conversion technologies and the TES system outlined in the framework introduction, LCA emission factors are compiled via queries within the Ecoinvent database [86] and with the search location being the examined countries, which are common for both examined case studies. The asset-specific values, and scaling procedures are defined as follows:

- **BB:** The embedded footprint for this generation technology is retrieved utilizing the database search query “*market for furnace, wood chips, with silo, 5000kW*”. The baseline impact factor in $\text{kg CO}_2\text{-eq}$ per $5,000 \text{ kW}$ of reference capacity is linearly scaled to yield a normalized capacity metric expressed in metric tons of carbon dioxide equivalent per megawatt of installed capacity ($\text{t CO}_2\text{-eq/MW}$).
- **CHP:** The embedded emissions for the co-generation infrastructure are extracted via the search string “*heat and power co-generation unit construction, 1 MW electrical, components for heat only*”. The reference value in $\text{kg CO}_2\text{-eq}$ per $1,000 \text{ kW}$ is linearly scaled to establish a standard capacity-specific factor in $\text{t CO}_2\text{-eq/MW}$.
- **LSHP and Chiller:** The manufacturing and installation footprints for these compression units are compiled using the search query “*market for heat pump, 30kW*”. Operating under a linear capacity scaling assumption, the baseline database value in $\text{kg CO}_2\text{-eq}$ per 30 kW is converted into a standardized asset intensity of $\text{t CO}_2\text{-eq/MW}$.

- **Pit TES:** Because the embedded emissions of this storage infrastructure is primarily dominated by civil earthworks and volumetric excavation, its structural footprint is retrieved via the database string “*excavation, hydraulic digger*”. The raw volumetric lifecycle value in $\text{kg CO}_2\text{-eq/m}^3$ is first linearly scaled and subsequently transformed into an energy-specific capacity metric ($\text{t CO}_2\text{-eq/MWh}_{\text{th}}$) by applying Equation (4.20).

4.2.3 Total System Emissions Aggregation

In order to extract the total system LCA emissions across the yearly optimization horizon under the different system configurations and examined district networks, a comprehensive aggregation formulation is applied. This summing mechanism pairs the operational emissions with each technology’s input energy consumption rate, and the embedded LCA emissions with the optimized design capacities.

Let $e_{\text{op},j}^c$ denote the comprehensive LCA operational emission factor for energy carrier $j \in \{\text{NG}, \text{B}, \text{E}\}$ in district network $c \in \{\text{ZH}, \text{AM}\}$ as parameterized in Section 4.2.1. Furthermore, let $e_{\text{em},k}$ and $e_{\text{em},\text{TES}}$ denote the standardized embedded capacity emission factors for each conversion technology $k \in \mathcal{K}$ and the TES system, respectively, as established in Section 4.2.2. Lastly, let n denote the operational and economic lifetime of the MES, established at 20 years as defined in Section 4.1.6.

Accounting for the deterministic first-stage optimization framework, which executes at $\Delta t = 1 \text{ h}$ across the annualized horizon $t \in \mathcal{T}$, the total annualized system greenhouse gas emissions ($\text{GHG}_{\text{total}}^c$) in $\text{t CO}_2\text{-eq}$ are quantified using the following summing formulation:

$$\text{GHG}_{\text{total}}^c = \sum_{t \in \mathcal{T}} \sum_{k \in \mathcal{K}} \sum_{j \in \{\text{NG}, \text{B}, \text{E}\}} \Delta t \cdot (U_{j,k,t} \cdot e_{\text{op},j}^c) + \sum_{k \in \mathcal{K}} \left(\frac{e_{\text{em},k} \cdot P_k}{n} \right) + \frac{e_{\text{em},\text{TES}} \cdot P_{\text{TES}}}{n} \quad (4.21)$$

Accounting for the stochastic second-stage optimization framework, which executes at $\Delta t = 0.25 \text{ h}$ across the annualized horizon $t \in \mathcal{T}_{15}$, the total system expected annualized greenhouse gas emissions ($\mathbb{E}[\text{GHG}_{\text{total}}^c]$) in $\text{t CO}_2\text{-eq}$ are quantified using the following summing formulation:

$$\mathbb{E}[\text{GHG}_{\text{total}}^c] = \sum_{s \in \mathcal{S}} \rho_s \cdot \left[\sum_{t \in \mathcal{T}_{15}} \sum_{k \in \mathcal{K}} \sum_{j \in \{\text{NG}, \text{B}, \text{E}\}} \Delta t \cdot (U_{j,k,t,s} \cdot e_{\text{op},j}^c) \right] + \sum_{k \in \mathcal{K}} \left(\frac{e_{\text{em},k} \cdot P_k}{n} \right) + \frac{e_{\text{em},\text{TES}} \cdot P_{\text{TES}}}{n} \quad (4.22)$$

These formulations for both stage optimizations ensure that all MES configurations can be evaluated on a completely transparent, equivalent LCA basis alongside the economic optimization metrics.

4.3 MES Configurations considered and $LCOE_{DN}$

The performance of the optimized MES is systematically evaluated under the presented two-stage optimization framework under both greenfield and brownfield investment approaches for the district networks of Zurich and Amsterdam.

To maintain strict continuity with the core research objectives, the results are organized into two primary assessment phases based on the stages of the optimization:

1. **Brownfield investment pathway:** The evaluation assumes a pre-existing municipal infrastructure where the conventional units, specifically the CHP plant and the BB, are fully amortized and treated as zero CAPEX assets. Within this phase, the deterministic first-stage optimization evaluates three distinct structural configurations to address the first research questions defined in Chapter 1:
 - *Baseline - fossil dominated (Baseline):* A fossil fuel-based supply strategy where the CHP and BB units are coupled exclusively with a standalone electric chiller to satisfy heating and cooling demands. As a result, the only present technologies in the MES in this configuration are the BB, CHP and the Electric Chiller.
 - *Heat Pump without storage (HP w/o storage):* A decarbonizing strategy featuring the integration of the LSHP in the absolute absence of thermal storage. As a result, the only present technologies in the MES in this configuration are the BB, CHP and the LSHP.
 - *Heat Pump with storage (HP w/ storage):* A fully integrated multi-energy configuration where the LSHP is physically coupled with the TES system. As a result, all depicted technologies of Figure 3.1 are considered in this configuration.

Subsequently, the second-stage stochastic operational optimization isolates the “Heat Pump with storage” configuration to determine to what extent annual expenditures can be further mitigated through active, sub-hourly participation in the BM.

2. **Greenfield Investment Pathway:** The exact multi-configuration, two-stage optimization procedure executed in the brownfield phase is replicated under a greenfield approach. In this paradigm, all candidate technologies are constructed from scratch, thereby incurring full initial capital expenditures.

Finally, to evaluate the unit generation cost of the energy delivered by the district system across these various scenarios, a new economic metric is also introduced for the results analysis; named the Levelized Cost of Energy of the District Network ($LCOE_{DN}$). The $LCOE_{DN}$ (€/MWh) quantifies the system cost incurred per unit of combined thermal and cooling energy delivered to end users across the optimization horizon for each municipal network c . It is formally defined as the ratio of the T(E)AC to the aggregated annual heating and cooling demands:

$$LCOE_{DN}^c = \frac{T(E)AC^c}{D_H^c + D_C^c} \quad (4.23)$$

Chapter 5

Optimization Results & Associated GHG Emissions

This chapter presents the results and quantitative findings derived from solving the two-stage MILP optimization framework, with a MILP gap of less than 1%.

Across both investment pathways and optimization stages, the system-level performance indicators are exported in unified tabular formats. These encompass the minimization metrics of the optimization objective function, specifically the TAC for Stage 1 and the TEAC for Stage 2, alongside the capacity design sizing decision variables (relevant to the first-stage design decisions) and the post-processing ecological footprint of the examined configurations in terms of their LCA GHG emissions footprint.

Furthermore, to evaluate the dynamic operation and physical dispatch logic of the generation fleet, two representative two-week seasonal periods (running chronologically from Monday to Sunday) are extracted from the annual optimization horizon. The first period is selected during January (13.01.2025–26.01.2025) to capture intensive winter thermal peaks, while the second is selected during August (04.08.2025–17.08.2025) to characterize typical summer conditions, which also include cooling demand. These high-resolution profiles illustrate how the aggregate network heat demand is satisfied and which specific technologies are prioritized by the optimizer at each discrete timestep. The localized cooling dispatch dynamics are explicitly scrutinized during the August operational window, matching the major climate-driven space cooling needs of the examined networks. Figures regarding the whole optimization horizon dispatch results are displayed in Appendix 8.5 for the Zurich brownfield approach regarding the “HP w/ storage” configuration as indicative of the whole optimization horizon, although they are not so informative as the ones displayed in this Chapter.

5.1 Brownfield Investment Pathway

Under the brownfield investment pathway, the BB and CHP units are assumed to be pre-existing infrastructure. Within this scope, the optimizer seeks to determine the optimal capacity utilization of these legacy assets, while concurrently evaluating whether newly

integrated candidate technologies should be deployed. This dual assessment is executed across all system configurations to achieve the minimum TAC during the first-stage planning horizon and then quantify the minimum TEAC with BM participation of the designed assets.

5.1.1 First-Stage Optimization Results

The quantitative solutions regarding the asset design capacities, annualized economic objective function, and corresponding life cycle GHG emissions (measured in metric kton) are returned across all configurations and both district networks as defined in Chapters 3 and 4. For the TES, its design variable is reported in m^3 for more tangible representation of results after applying Equation (4.20). Table 5.1 and Table 5.2 summarize these primary design variables, economic outcomes, and environmental metrics for the simulations executed for Zurich and Amsterdam, respectively.

Table 5.1: First stage optimization results for Zurich brownfield approach

Zurich	Baseline	HP w/o storage	HP w/ storage
BB (MW)	16.1	16.1	50.0
CHP (MW)	161.7	108.0	57.8
LSHP or Chiller (MW)	44.5	53.7	44.5
TES (m^3)	-	-	6773.7
TAC (M €)	38.8	37.9	34.3
LCOE _{DN} (€/MWh)	86.30	84.35	76.34
GHG _{total} (kton CO ₂ -eq)	163.8	155.5	120.9

Table 5.2: First stage optimization results for Amsterdam brownfield approach

Amsterdam	Baseline	HP w/o storage	HP w/ storage
BB (MW)	16.1	16.1	50.0
CHP (MW)	169.8	108.0	52.9
LSHP or Chiller (MW)	29.4	61.5	29.4
TES (m^3)	-	-	17440.5
TAC (M €)	35.6	39.4	29.7
LCOE _{DN} (€/MWh)	80.81	89.28	67.3
GHG _{total} (kton CO ₂ -eq)	155.4	135.2	118.2

A comparative analysis of the two urban networks reveals distinct regional economic and environmental trade-offs in integrating low-carbon assets. In the Zurich district network (Table 5.1), investment in a LSHP proves economically and environmentally viable across all configurations. The introduction of the LSHP in “HP w/o storage”

configuration drops the TAC while simultaneously reducing life cycle GHG emissions ($\text{GHG}_{\text{total}}$) compared to the fossil fuel-based supply strategy of “Baseline” configuration.

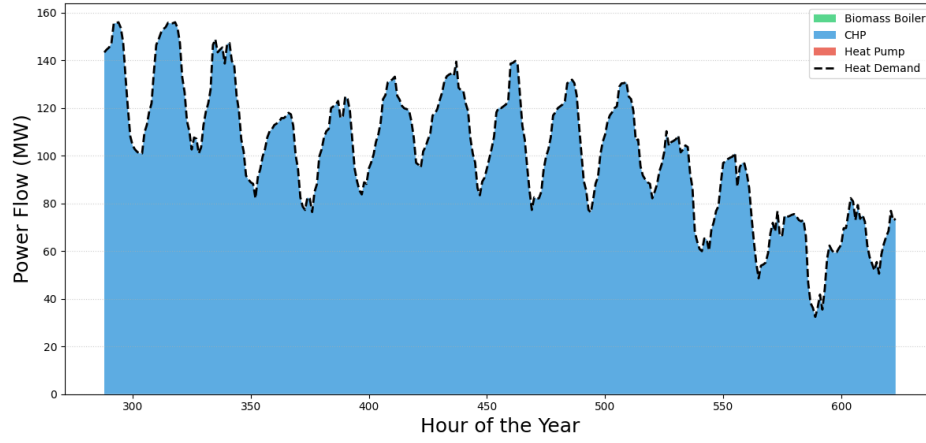
Conversely, the economic viability of the LSHP in the Amsterdam network (Table 5.2) is strictly contingent upon the co-adoption of the TES system. Operating the LSHP without heat storage in Amsterdam (“HP w/o storage”) causes the TAC to increase, rendering a standalone electrification pathway financially disadvantageous. This is justified because of the fact that natural gas prices are competitive compared to DAM electricity prices in the absence of storage where it would have been possible to capitalize on charging the TES during DAM low-price hours and provide heat through TES discharging during DAM high-price hours. However, from an environmental perspective, this transition remains highly effective, successfully lowering $\text{GHG}_{\text{total}}$.

Crucially, the structural integration of the TES in “HP w/ storage” configuration yields an identical sizing optimization pattern in both cities by mitigating the required peak capacity of the LSHP fleet. By providing thermal load-shifting capabilities, the TES allows the optimized LSHP capacity to drop from 53.7 MW to 44.5 MW in Zurich, and more drastically from 61.5 MW to 29.4 MW in Amsterdam.

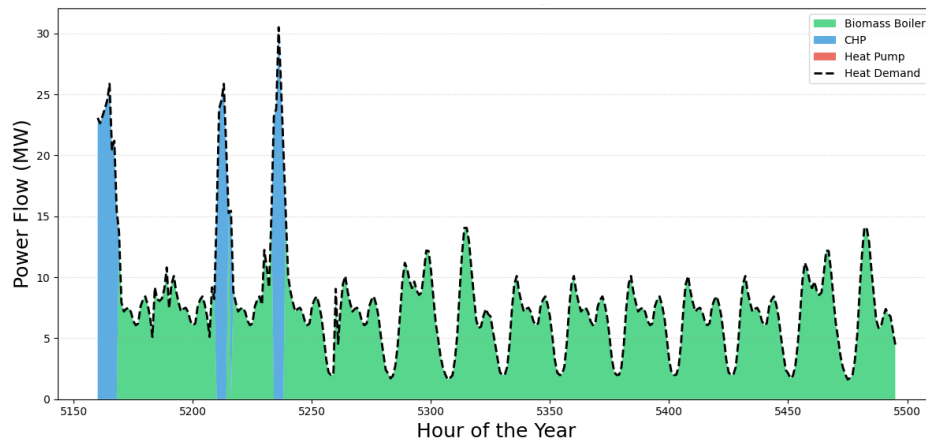
When observing the fully coupled system in “HP w/ storage” configuration, Amsterdam achieves a significantly lower TAC and LCOE_{DN} compared to Zurich. This economic variation is driven by two localized factors. First, Amsterdam exhibits lower space cooling demands, reducing the overall infrastructure needs on the cooling asset fleet. Second, the wider DAM price spreads, characterized by larger intra-day variations between peak and off-peak electricity prices, in the Netherlands serve as the primary economic catalyst for investing in a larger TES capacity. This expanded storage volume allows the system to maximize arbitrage opportunities by charging the thermal reservoir during low-price hours and discharging it during high-price windows, thereby bypassing the requirement to install additional LSHP capacity.

Interestingly, the resulting carbon footprint reductions remain balanced between both municipal regions in “HP w/ storage” configuration. Under this brownfield framework, the district networks have not yet reached the deep threshold of electrification required to expose the vast disparity between the carbon intensities of the national grids. Consequently, the low carbon energy grid mix of Switzerland compared to the more fossil-reliant grid mix of the Netherlands does not manifest as a dominant factor in the aggregate annual lifecycle carbon account.

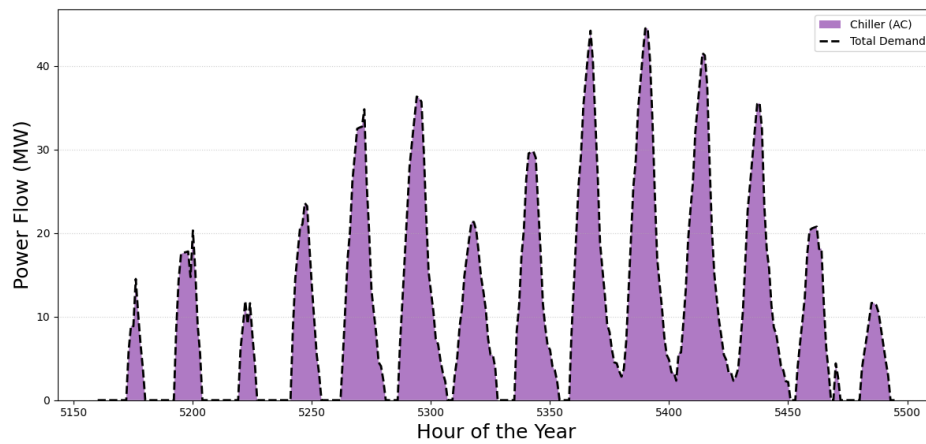
Baseline configuration seasonal heating and cooling dispatch profiles



(a) January heating dispatch profile.

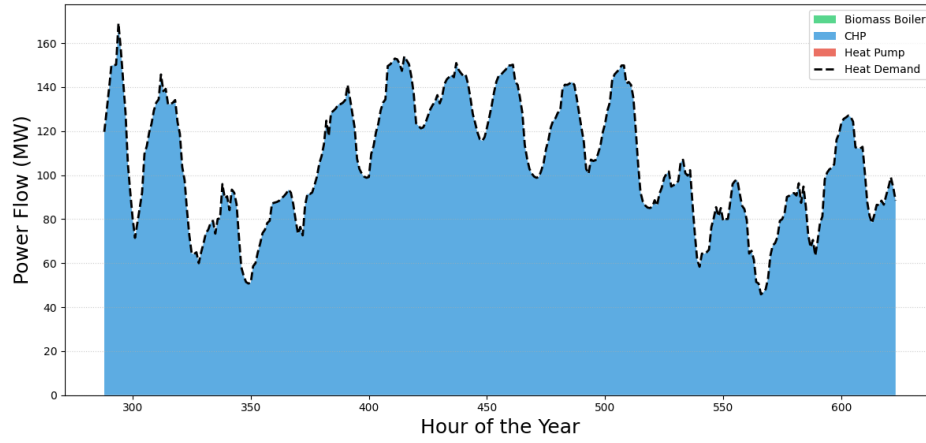


(b) August heating dispatch profile.

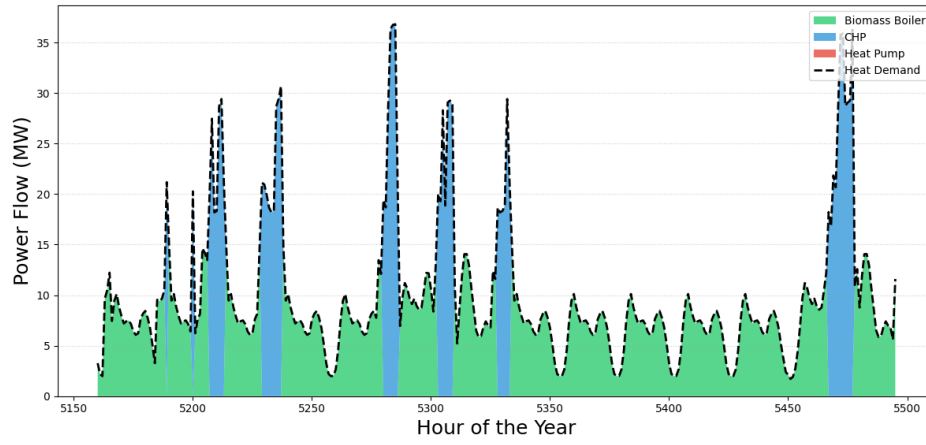


(c) August cooling dispatch profile.

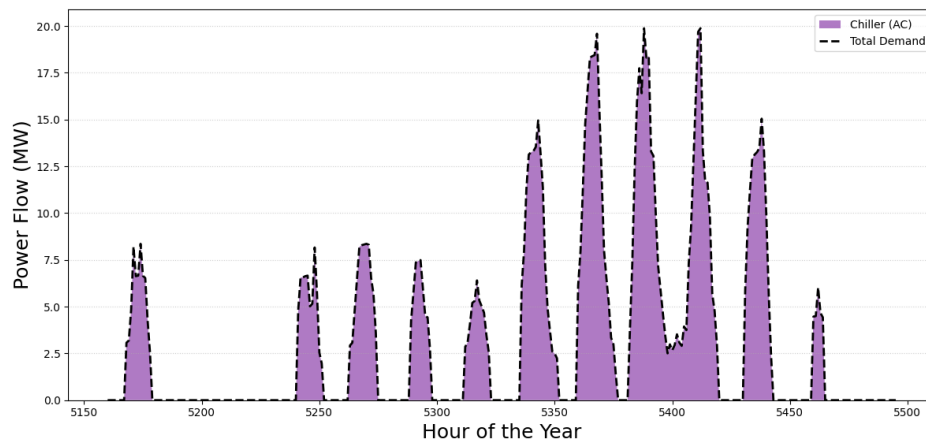
Figure 5.1: Baseline configuration seasonal heating and cooling dispatch operational profiles for the **Zurich** brownfield network approach.



(a) January heating dispatch profile.



(b) August heating dispatch profile.



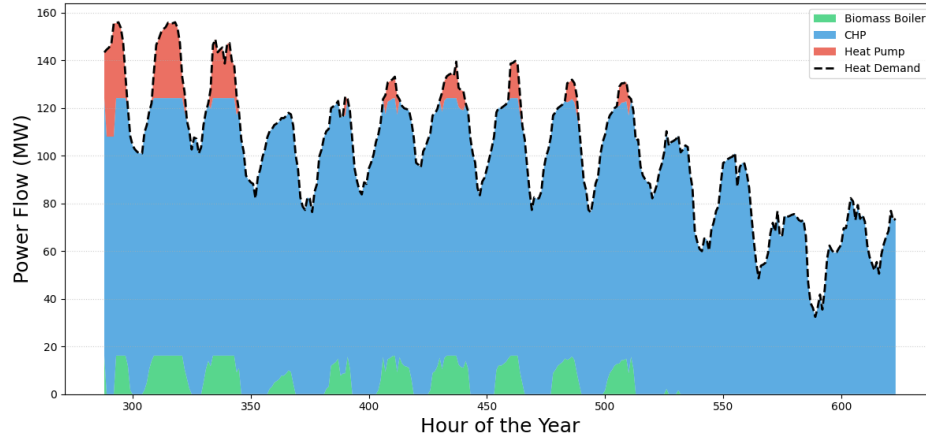
(c) August cooling dispatch profile.

Figure 5.2: Baseline configuration seasonal heating and cooling dispatch operational profiles for the **Amsterdam** brownfield network approach.

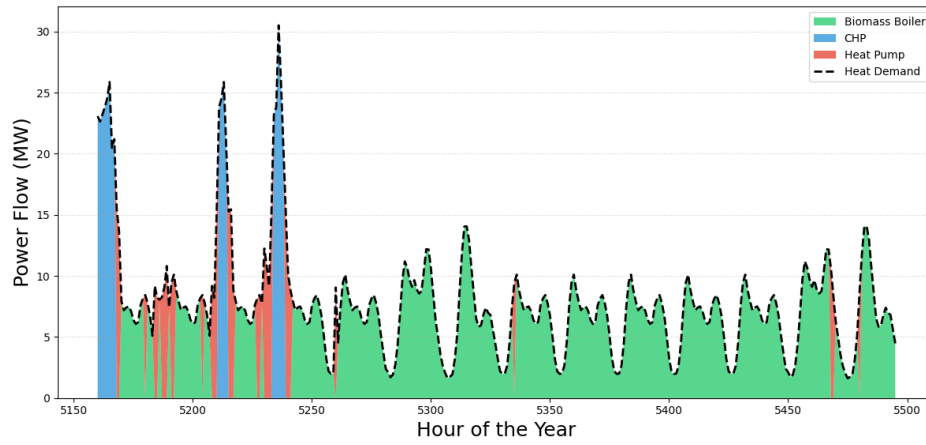
The above Figures illustrate the cost-optimal hourly generation dispatch profiles selected by the optimizer to satisfy the localized heating and cooling demand curves of “Baseline” configuration. The vertical axis (y -axis) quantifies the instantaneous thermal and cooling power flow of the network, while the horizontal axis (x -axis) tracks the sequential hourly timesteps across the selected multi-week periods. Consequently, Figure 5.1 illustrates the seasonal heating and cooling operational dispatch profiles for the Zurich network, while Figure 5.2 displays the corresponding operational results for Amsterdam.

The operational dispatch profiles across both networks reveal highly consistent seasonal behaviors, dominated primarily by CHP utilization in this configuration. During the winter period, the heating demands of both Zurich and Amsterdam are met entirely by the CHP, which operates continuously near its peak output because it yields lower operational costs than operating the BB. In contrast, the summer period, characterized by significantly lower heating peaks, is covered dominantly by the BB across both networks, as the thermal load falls below the CHP’s minimum load fraction (δ_{CHP}).

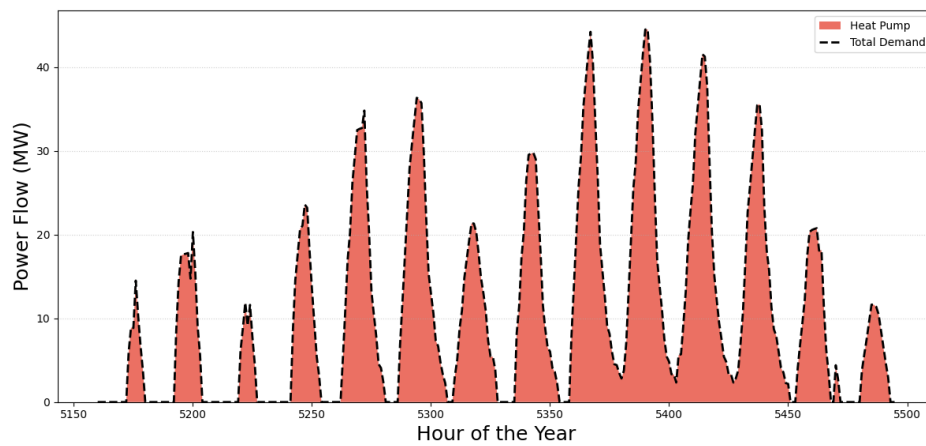
HP w/o storage configuration seasonal heating and cooling dispatch profiles



(a) January heating dispatch profile.

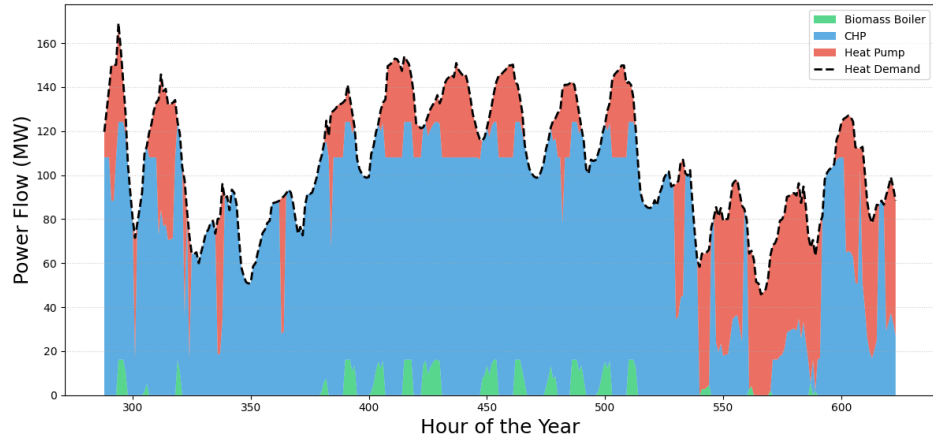


(b) August heating dispatch profile.

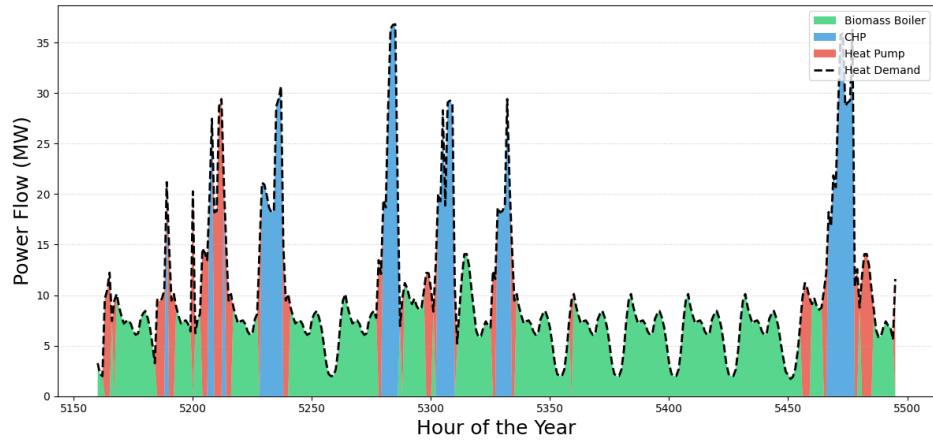


(c) August cooling dispatch profile.

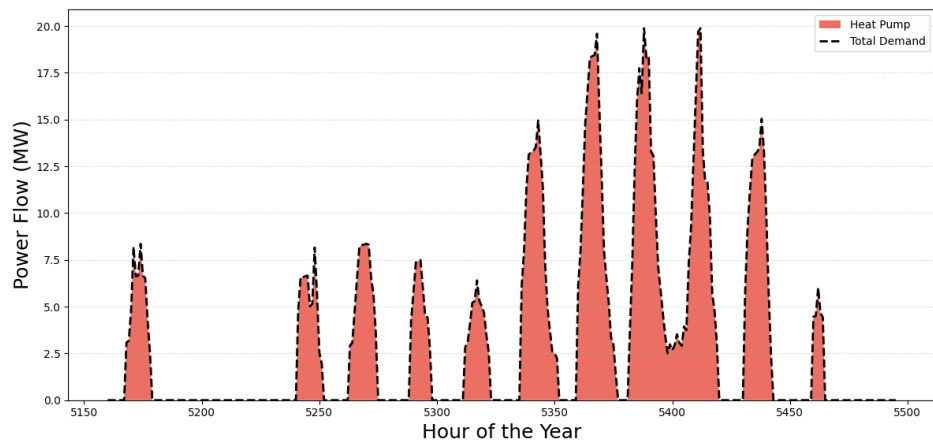
Figure 5.3: HP w/o storage configuration seasonal heating and cooling dispatch operational profiles for the **Zurich** brownfield network approach.



(a) January heating dispatch profile.



(b) August heating dispatch profile.

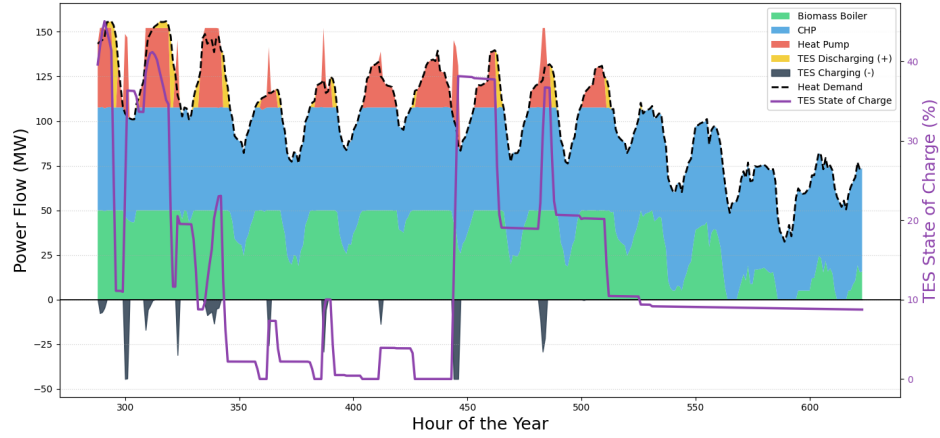


(c) August cooling dispatch profile.

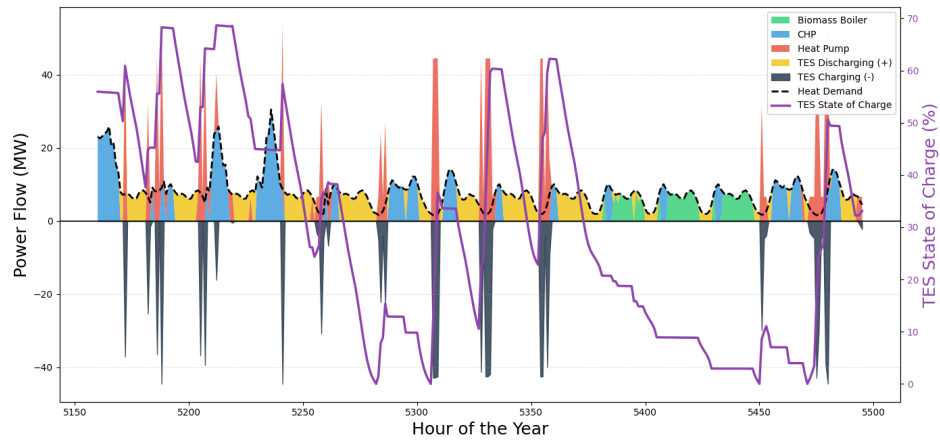
Figure 5.4: HP w/o storage configuration seasonal heating and cooling dispatch operational profiles for the **Amsterdam** brownfield network approach.

The integration of the LSHP in this configuration introduces significant modifications to the operational dispatch dynamics, as detailed in Figure 5.3 for Zurich and Figure 5.4 for Amsterdam. During the winter period, the LSHP actively participates in peak-shaving the thermal demand, noticeably displacing portions of the CHP generation, a trend that is particularly pronounced in the Amsterdam network. In the summer period, the asset's multi-generation capability becomes central; the LSHP completely satisfies the localized cooling demand by operating in chilling mode, while simultaneously providing flexible mid-load heating support to complement the BB.

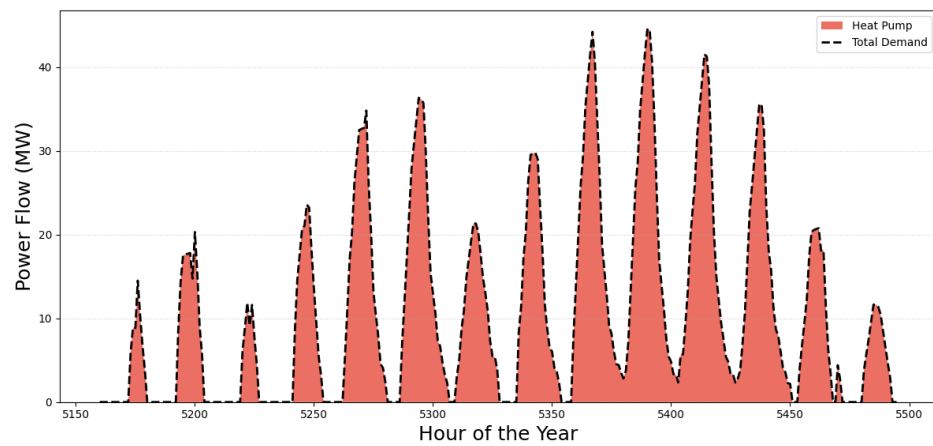
HP w/ storage configuration seasonal heating and cooling dispatch profiles



(a) January heating dispatch profile.

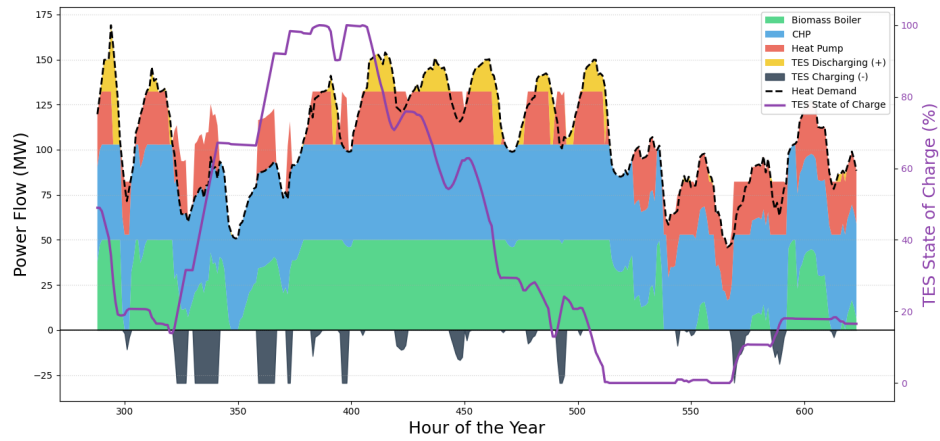


(b) August heating dispatch profile.

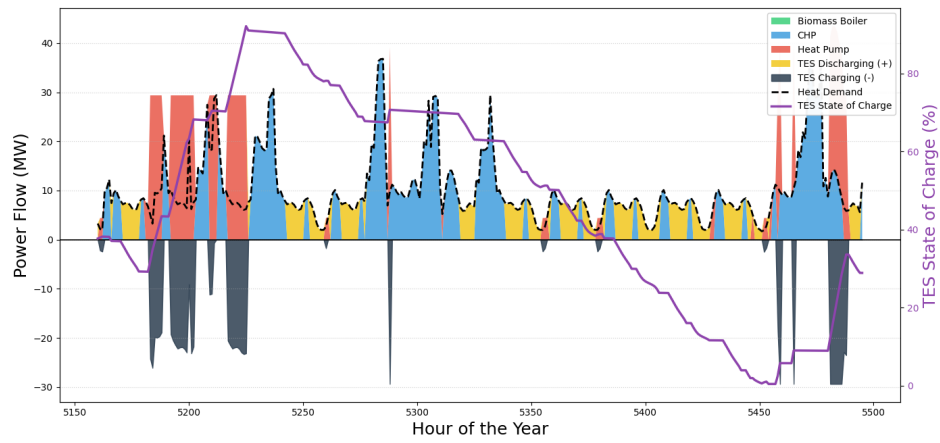


(c) August cooling dispatch profile.

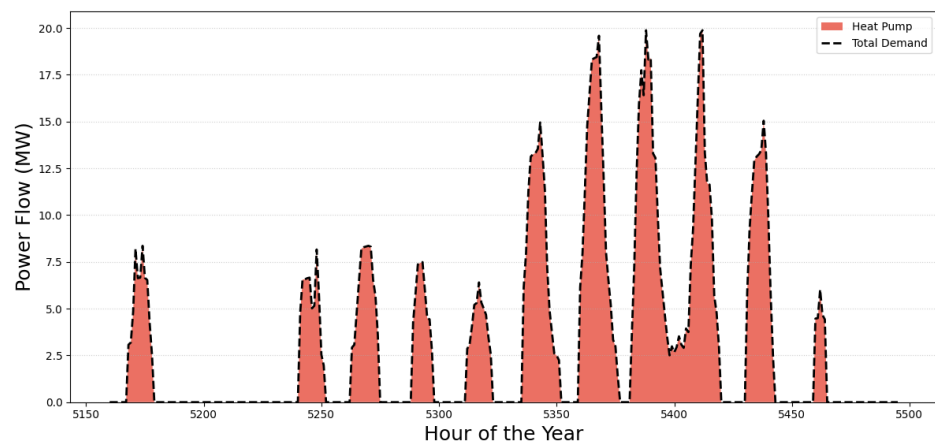
Figure 5.5: HP w/ storage configuration seasonal heating and cooling dispatch operational profiles for the **Zurich** brownfield network approach.



(a) January heating dispatch profile.



(b) August heating dispatch profile.



(c) August cooling dispatch profile.

Figure 5.6: HP w/ storage configuration seasonal heating and cooling dispatch operational profiles for the **Amsterdam** brownfield network approach.

The operational dispatch profiles for “HP w/ storage” configuration are presented in Figure 5.5 for Zurich and Figure 5.6 for Amsterdam. In addition to the thermal power flows measured on the primary left vertical axis (y -axis) and the chronological timesteps on the horizontal axis (x -axis), these combined profiles introduce a secondary right vertical axis. This secondary axis tracks the State of Charge (SoC) of the TES as a percentage (%), while negative values on the primary axis display the active charging rates of the storage unit.

The integration of the TES alongside the LSHP fundamentally transforms the operational flexibility of both energy networks. During the winter period, the system no longer relies on continuous peak-following operation from the CHP. Instead, the storage unit is aggressively charged during off-peak demand intervals or favorable electricity price windows, reflected as negative power flows, and subsequently discharged to satisfy thermal demand during peak hours, along with the LSHP. This peak-shaving capability (more evident in the Amsterdam network) enables the BB to operate almost continuously at its maximum capacity limit of 50.0 MW in both cities. Consequently, winter CHP generation is substantially curtailed compared to prior configurations, shifting its role toward a stable mid-load supply rather than an intermittent peak-activated asset.

In the summer period, the operational dynamics between the LSHP, BB, and Pit TES adapt to seasonal demands. Although the LSHP possesses multi-generation capabilities, the dispatch profiles explicitly indicate that when the asset operates in cooling mode, it functions solely as a standalone chiller without concurrently outputting heat to the thermal network. Crucially, the inclusion of the TES uncouples heat generation from real-time thermal consumption. As it can be seen from Figures 5.5, 5.6 - (b) in Amsterdam, where the optimized TES volume is considerably larger, the storage is charged aggressively during the opening days of the representative August period when favorable DAM electricity prices prevail and cooling loads are absent. This front-loaded charging strategy establishes a high SoC buffer, enabling a sustained, steady TES discharging phase over the majority of the remaining window. This contrasts sharply with the Zurich network, which exhibits much more frequent, tightly balanced TES charging and discharging cycles throughout the same period.

5.1.2 Second-Stage Optimization Results

Building upon the baseline asset sizing established in the first stage optimization for “HP w/ storage” configuration, the second stage stochastic model evaluates the maximum potential reduction in TEAC attainable through BM participation by setting the bid acceptance factors to unity ($\mu = \nu = 1$), as formulated in Chapter 3. The resulting revised economic metrics, and life cycle GHG emissions for Zurich and Amsterdam are systematically summarized in Table 5.3 and Table 5.4, respectively. Additionally, Table 5.5 details the expected volume of upward and downward energy flexibility that the LSHP can provide to the grid across the optimization horizon for both municipal networks, as defined in Section 3.2.5.

Table 5.3: Second stage optimization results for Zurich brownfield approach

Zurich	No BM participation	Fully BM participation
BB (MW)	50.0	50.0
CHP (MW)	57.8	57.8
LSHP (MW)	44.5	44.5
TES (m ³)	6773.7	6773.7
TEAC (M €)	34.3	29.9
LCOE _{DN} (€/MWh)	76.34	66.41
EGHG _{total} (K ton CO ₂ -eq)	120.9	89.3

Table 5.4: Second stage optimization results for Amsterdam brownfield approach

Amsterdam	No BM participation	Fully BM participation
BB (MW)	50.0	50.0
CHP (MW)	52.9	52.9
LSHP (MW)	29.4	29.4
TES (m ³)	17440.5	17440.5
TEAC (M €)	29.7	26.1
LCOE _{DN} (€/MWh)	67.30	59.15
EGHG _{total} (K ton CO ₂ -eq)	118.2	108.7

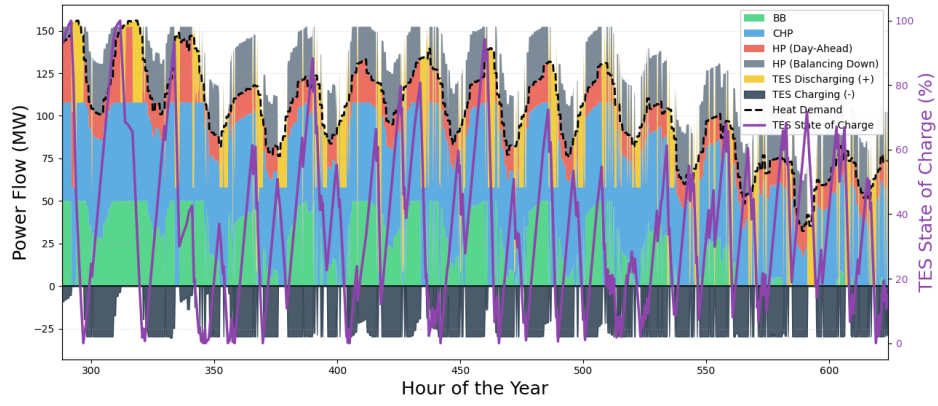
Table 5.5: Expected operational flexibility provided by the LSHP across the optimization horizon under a brownfield approach.

Flexibility Metric	Zurich (GWh)	Amsterdam (GWh)
Expected upward flexibility	7.4	4.2
Expected downward flexibility	37.9	29.2
Total expected flexibility	45.3	33.4

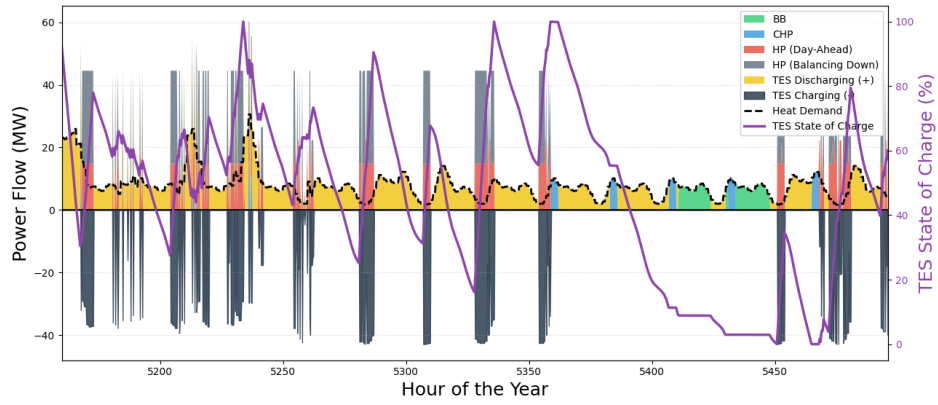
Comparing the second stage optimization outcomes between Table 5.3 and Table 5.4 demonstrates, as anticipated, the tangible economic gains unlocked through BM participation in both cities. In Zurich, participation in the BM yields a TEAC reduction of 4.4 M€, whereas in Amsterdam, the corresponding cost decrease is limited to 3.6 M€. Notably, Zurich achieves higher absolute monetary savings despite deploying a significantly smaller TES volume than Amsterdam, a trend further evidenced by the higher total balancing volumes delivered in Zurich according to Table 5.5. This divergence is primarily attributed to the structurally higher and more remunerative BM activation prices prevailing across the Swiss bidding zone compared to the Dutch electricity market.

Furthermore, downward balancing flexibility strongly prevails over upward flexibility in both cities due to significantly more favorable market remuneration rates. This dominance is driven by frequent occurrences of negative downward BM prices across both bidding zones, where the MES operator is effectively compensated for drawing power from the grid to charge the TES. Consequently, this negative pricing regime provides a powerful economic incentive to prioritize downward over upward balancing operations.

Regarding life cycle GHG impacts ($\text{EGHG}_{\text{total}}$), transitioning to full BM participation significantly reduces total emissions across both municipal networks. This reduction is driven by prevailing downward balancing activations, which increase the electricity consumption and thermal generation of the LSHP. The resulting surplus heat charges the TES more frequently, thereby allowing TES discharging to cover a larger fraction of the municipal heat demand. Consequently, this operation directly displaces part of the original thermal dispatch of the BB and CHP units of the “HP w/ storage” configuration, which exhibit higher emission intensities, resulting in a net decrease in overall emissions. This emission reduction is significant higher in Zurich because of a lower carbon electricity grid intensity compared to Amsterdam.

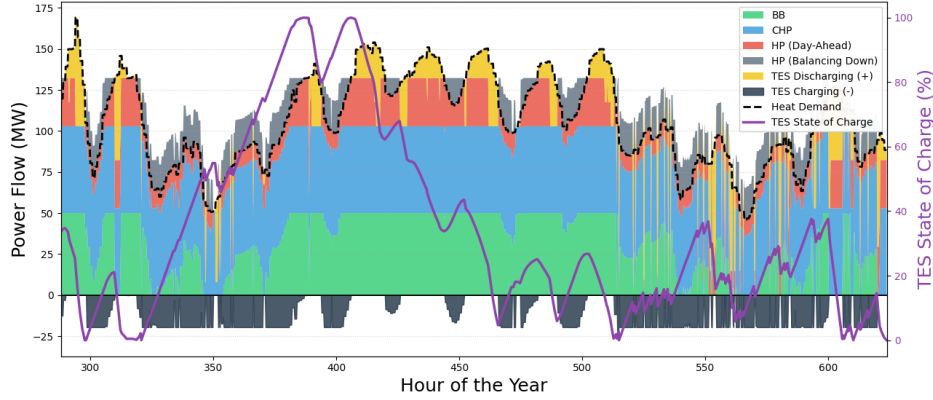


(a) January heat dispatch profile for Sc. 1.

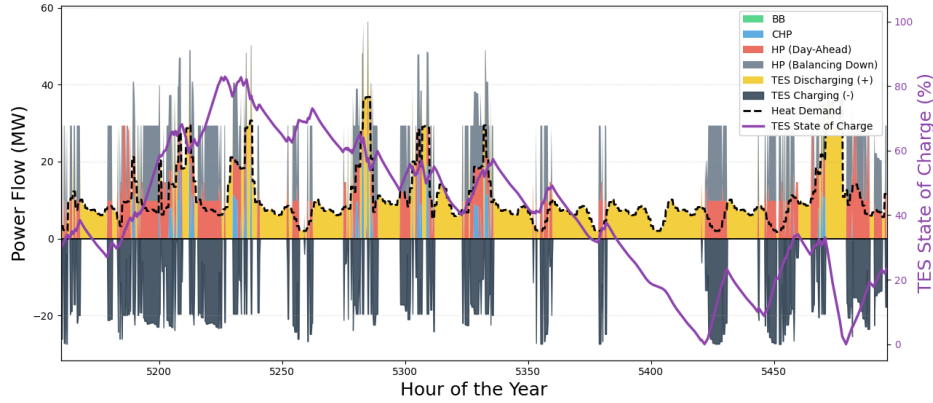


(b) August heat dispatch profile for Sc. 1.

Figure 5.7: Seasonal heat dispatch profiles of Scenario 1 for **Zurich** with full participation in the BM during a brownfield network approach.



(a) January heat dispatch profile for Sc. 1.



(b) August heat dispatch profile for Sc. 1.

Figure 5.8: Seasonal heat dispatch profiles of Scenario 1 for **Amsterdam** with full participation in the BM during a brownfield network approach.

Figures 5.7 and 5.8 illustrate the operational heat dispatch profiles under full BM participation for the optimal asset capacities determined in “HP w/ storage” configuration. Because the second-stage optimization framework is stochastic, these operational timeseries correspond specifically to Scenario 1 (Sc. 1), which represents the historical 2024 balancing market price cluster for both municipal networks. Following the conventions established in Figures 5.5 and 5.6, the primary y -axis denotes thermal power flow [MW], the secondary y -axis tracks the SoC of the TES [%], and the x -axis indicates the hour of the year.

The colored stacked bars in the dispatch plots represent the actual operational heat output from all considered MES technologies, in accordance with the dispatch plots of “HP w/ storage” configuration, with the specific exception of the LSHP. For the LSHP, the red bars (HP (Day-Ahead)) indicates the anticipated heat production based on scenario-specific DA electricity commitments, which can be curtailed in real time to

provide upward flexibility, while the gray bar (HP (Balancing Down)) represents real-time heat generation resulting from downward balancing participation.

A visual examination of these profiles validates the physical and operational mechanics encoded in the optimization model. Any thermal output exceeding instantaneous heat demand is attributable either to heat produced from DA electricity procurement (HP (Day-Ahead)) or, far more commonly, to downward balancing flexibility provision (HP (Balancing Down)) that generates surplus thermal energy. In strict alignment with model assumptions, this surplus heat is absorbed exclusively by thermal storage, as evidenced by the exact mirroring between excess heat generation and the charging profile (TES Charging (-)).

Conversely, minor visual discrepancies occur in intervals where storage discharge (TES Discharging (+)) appears to exceed end-user heat demand. This occurs because the scenario-specific baseline DA LSHP output (HP (Day-Ahead)) reflects the initial scheduled commitment prior to real-time adjustments. Consequently, any apparent excess in storage discharge directly corresponds to periods where the MES provides upward balancing flexibility, substituting for the heat pump capacity curtailed during real-time execution.

Finally, a comparative assessment between these profiles and those of “HP w/ storage” configuration reveals no substantial shifts in the dispatch merit order. Instead, the primary influence of BM participation, as anticipated, lies in how an operator strategically leverages the available TES capacity. This is visually evidenced by the distinct SoC trajectories observed across identical time intervals between the first and second stage optimization dispatch graphs. Furthermore, the strong economic incentive of downward balancing prices drives the optimizer to maximize downward flexibility provision. As illustrated in Figures 5.7 and 5.8 corresponding to 2024 BM prices (Sc. 1), more favorable downward flexibility remuneration in Zurich prompts a noticeably more aggressive downward flexibility bidding strategy compared to Amsterdam in the winter period. Cooling dispatch profiles are omitted from this secondary assessment, as BM participation exerts no influence on cooling operations relative to “HP w/ storage” configuration.

5.2 Greenfield Investment Pathway

Under the greenfield investment pathway, all district network technologies are assumed to be constructed entirely from scratch, representing the development of a brand-new DHCN without pre-existing assets. Within this scope, the optimization framework determines the optimal technology selection and asset capacities while accounting for each single ($CAPEX_k$) alongside operational costs across all candidate technologies. Mirroring the methodology of the brownfield pathway, the deterministic first stage optimization minimizes the TAC to establish baseline sizing and operational dispatch. Subsequently, the second stage stochastic optimization minimizes the TEAC to evaluate system operation under BM participation.

5.2.1 First Stage Optimization Results

Analogous to the brownfield assessment, Table 5.6 and Table 5.7 systematically summarize the key design variables, economic performance metrics, and environmental impacts for the greenfield optimizations executed for Zurich and Amsterdam, respectively.

Table 5.6: First stage optimization results for Zurich greenfield approach.

Zurich	Baseline	HP w/o storage	HP w/ storage
BB (MW)	16.1	16.0	50.0
CHP (MW)	161.7	108.0	24.6
LSHP or Chiller (MW)	44.5	53.9	44.5
TES (m ³)	-	-	25488.7
TAC (M €)	69.9	58.9	42.9
LCOE _{DN} (€/MWh)	155.39	131.11	95.46
GHG _{total} (K ton CO ₂ -eq)	163.8	155.4	69.5

Table 5.7: First stage optimization results for Amsterdam greenfield approach.

Amsterdam	Baseline	HP w/o storage	HP w/ storage
BB (MW)	16.1	16.1	50.0
CHP (MW)	169.8	108.0	37.1
LSHP or Chiller (MW)	29.4	61.5	34.9
TES (m ³)	-	-	38142.2
TAC (M €)	68.2	60.4	41.1
LCOE _{DN} (€/MWh)	154.70	136.93	93.20
GHG _{total} (K ton CO ₂ -eq)	155.4	135.2	103.4

A comparative analysis between the greenfield optimization outcomes (Table 5.6 and Table 5.7) and the brownfield approach reveals consistent systemic trends regarding optimal asset sizing, cost reduction, and decarbonization trajectories.

First, the capacity allocations across “Baseline” configuration remain identical and nearly identical for the “HP w/o storage” configuration to those obtained under the brownfield framework. This demonstrates that the optimal sizing of conversion technologies in the absence of TES is structurally independent of the initial CAPEX_k associated with the conventional units (CHP and BB). However, a distinct shift emerges in “HP w/o storage” configuration: in a brand-new network development, integrating a LSHP without TES becomes economically viable across all examined regions. Unlike the brownfield pathway, this configuration yields a net reduction in TAC in Amsterdam, confirming that LSHP deployment is universally beneficial under greenfield investment parameters.

Second, the inclusion of the TES in “HP w/ storage” configuration creates a substantial compounding benefit. The synergy between the LSHP and TES accelerates the reduction of TAC at nearly twice the rate observed during the transition from “Baseline” configuration to “HP w/o storage” configuration. To maximize this economic advantage, the optimizer allocates significantly larger TES storage volumes in both cities compared to the brownfield pathway. This expanded installed TES capacity reduces the need to size the CHP to a capacity similar to the one sized in the brownfield approach.

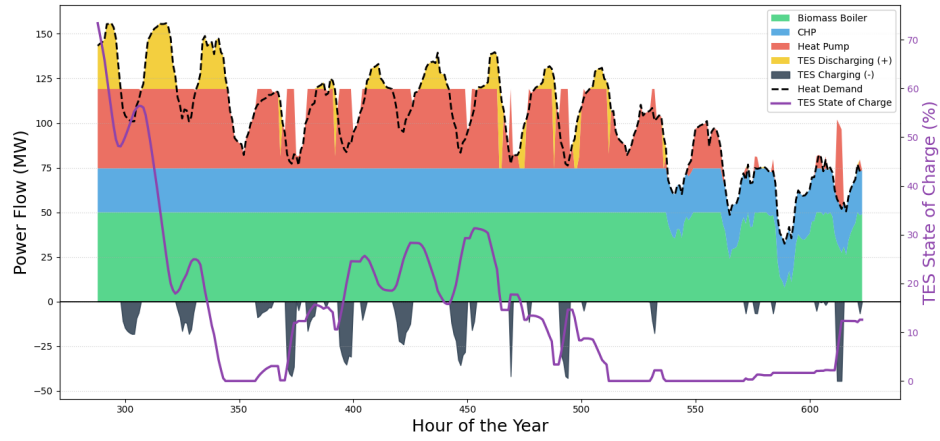
Finally, the progressive electrification of the DHCN delivers concurrent environmental gains alongside costs reductions. LCA GHG emissions ($EGHG_{total}$) drop systematically when transitioning from the fossil fuel-based supply strategy of “Baseline” configuration through to the fully integrated “HP w/ storage” configuration, reinforcing the alignment between economic optimization and decarbonization goals in modern district networks.

Baseline and HP w/o storage configurations seasonal heating and cooling dispatch profiles

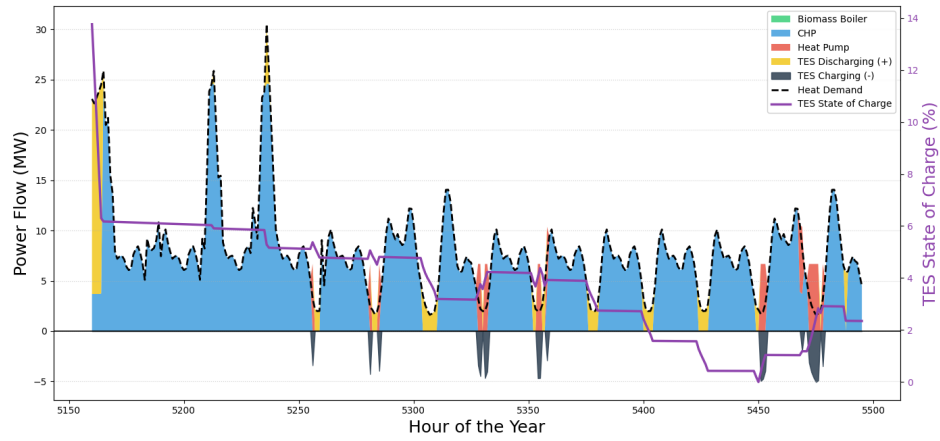
The operational dispatch figures for “Baseline” and “HP w/o storage” configurations are omitted here, as their seasonal dispatch profiles are identical to those observed under the brownfield investment pathway. This consistency highlights that operational technology allocation is primarily driven by service delivery requirements and physical operational constraints rather than initial capital expenditures. As anticipated, the CHP unit remains the dominant heat generation source despite its high investment costs. This outcome is governed by its dual-generation capability, which enables the asset to capture stable revenue streams by feeding also power into the electrical grid.

HP w/ storage configuration seasonal heating and cooling dispatch profiles

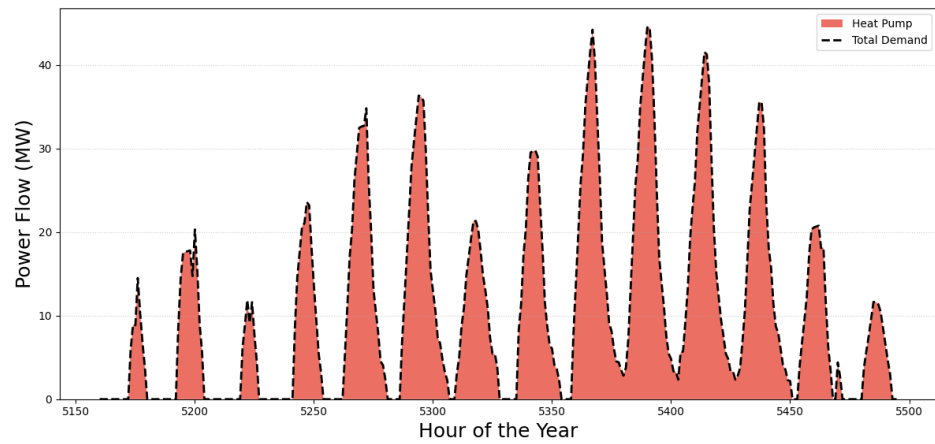
Following the structure established in the brownfield assessment, Figure 5.9 and Figure 5.10 present the seasonal heating and cooling operational dispatch profiles for “HP w/ storage” configuration.



(a) January heating dispatch profile.

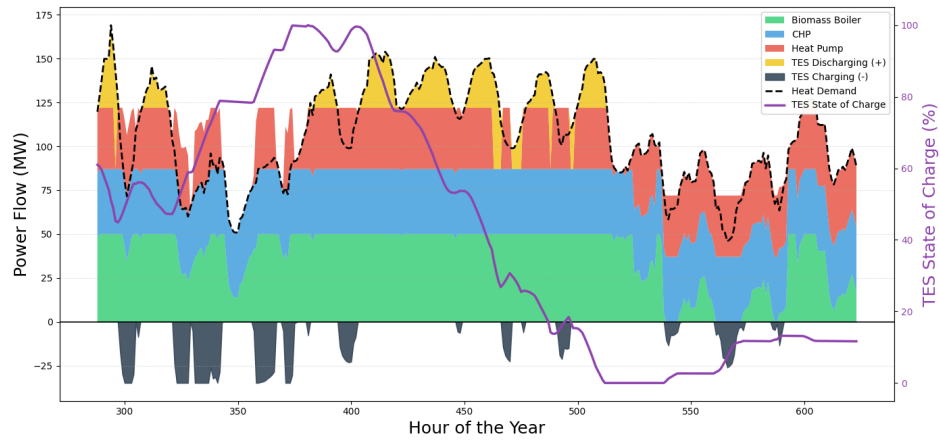


(b) August heating dispatch profile.

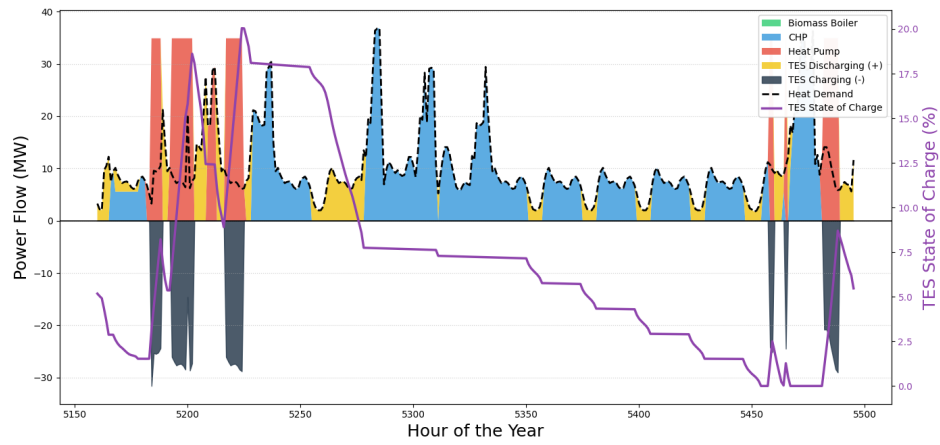


(c) August cooling dispatch profile.

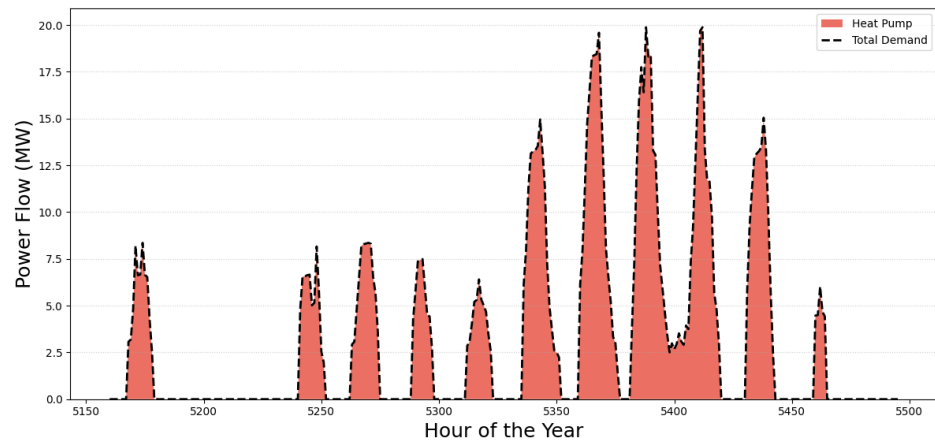
Figure 5.9: “HP w/ storage” configuration seasonal heating and cooling dispatch operational profiles for the **Zürich** greenfield network approach.



(a) January heating dispatch profile.



(b) August heating dispatch profile.



(c) August cooling dispatch profile.

Figure 5.10: “HP w/ storage” configuration seasonal heating and cooling dispatch operational profiles for the **Amsterdam** greenfield network approach.

The operational dispatch strategy under the greenfield investment pathway follows a similar logic to that of the brownfield approach, wherein the LSHP and the TES discharging collaboratively cover peak demand periods in the winter, while the CHP and BB units primarily provide baseload thermal supply for the same period.

However, two distinct operational shifts emerge when comparing these results to the brownfield pathway:

1. **Reduced CHP generation share:** The thermal output share of the CHP unit within the overall energy mix is significantly diminished. This displacement is directly enabled by the expanded TES capacity, which allows the system to exploit low-cost thermal energy more effectively and reduce reliance on thermal generation from the CHP.
2. **Transition toward seasonal thermal storage:** During August, the TES operates across lower SoC levels, exhibiting a operational shift toward seasonal thermal energy management. Owing to its larger physical volume, the TES effectively stores surplus heat to cover summer domestic hot water demand without requiring high-frequency full charge-discharge cycles. This operational pattern is particularly pronounced in the August dispatch profile for Amsterdam.

5.2.2 Second Stage Optimization Results

Following again the methodology established in the brownfield assessment, Tables 5.8, 5.9, and 5.10 summarize the second stage optimization results. These simulations incorporate the optimal asset capacities designed in “HP w/ storage” configuration under the identical full BM participation assumption of $\mu = \nu = 1$.

Table 5.8: Second stage optimization results for Zurich greenfield approach

Zurich	No BM participation	Fully BM participation
BB (MW)	50.0	50.0
CHP (MW)	24.6	24.6
LSHP (MW)	44.5	44.5
TES (m ³)	25488.7	25488.7
TEAC (M €)	42.9	38.2
LCOE _{DN} (€/MWh)	95.46	84.85
EGHG _{total} (K ton CO ₂ -eq)	69.5	51.5

Table 5.9: Second stage optimization results for Amsterdam greenfield approach

Amsterdam	No BM participation	Fully BM participation
BB (MW)	50.0	50.0
CHP (MW)	37.1	37.1
LSHP (MW)	34.9	34.9
TES (m ³)	38142.2	38142.2
TEAC (M €)	41.1	37.1
LCOE _{DN} (€/MWh)	93.20	84.08
EGHG _{total} (K ton CO ₂ -eq)	103.4	93.2

Table 5.10: Expected operational flexibility provided by the LSHP across the optimization horizon under a greenfield approach.

Flexibility Metric	Zurich (GWh)	Amsterdam (GWh)
Expected upward flexibility	7.5	4.8
Expected downward flexibility	36.9	31.5
Total expected flexibility	44.4	36.3

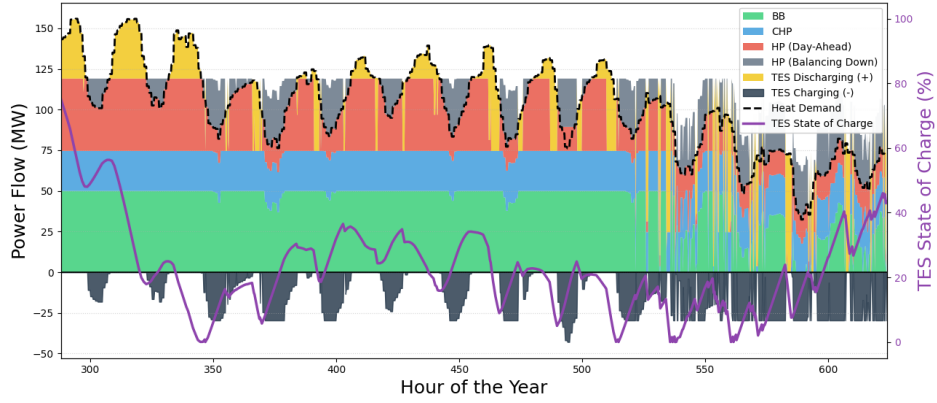
Comparing the second stage optimization outcomes across Table 5.8 and Table 5.9 again validates the strong potential of a district network system to diminish its TEAC through fully participation in BM. These findings align with the trends established in the brownfield assessment, with Zurich achieving more economic cost reductions than Amsterdam, although by a relatively close margin.

Interestingly, both cities capture slightly higher annual cost savings under the greenfield investment pathway, with Amsterdam having a slight further increase over Zurich in this approach with (0.4 M€ for Amsterdam) compared to (0.3 M€ for Zurich) of further costs saving over the brownfield approach. This behavior is attributed to the dual role of TES: although a larger TES is sized under the greenfield framework in “HP w/ storage” configuration, its capacity expansion is tailored primarily toward optimizing baseline thermal dispatch rather than unlocking incremental balancing flexibility. This interpretation is directly supported by the flexibility offer volumes presented in Table 5.5 and Table 5.10, which exhibit only marginal volume decrease for Zurich and increase for Amsterdam under the greenfield pathway. Consequently, these offered flexibility volumes approach the upper bound attainable by any system configuration optimized primarily to meet deterministic heating and cooling loads.

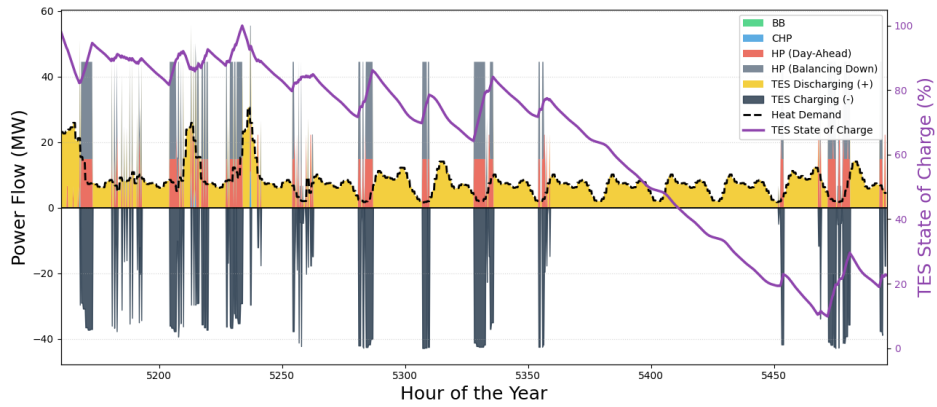
Regarding the life cycle GHG emissions, the environmental divergence relative to “HP w/ storage” configuration stems almost entirely from shifts in the LSHP’s net sub-hourly electricity intake as it adjusts operations to accommodate BM signals, similar to the brownfield approach.

Figures 5.11 and 5.12 illustrate the real-time operational heat dispatch profiles under full BM participation for the optimal asset capacities determined in “HP w/ storage”

configuration. Consistent with the brownfield analysis, these dispatch timeseries correspond to Sc. 1, representing the historical 2024 BM price cluster. The operational outcomes converge closely with the brownfield results, demonstrating that the primary systemic impact of market participation lies in the strategic operational dispatch of the TES, leveraging its thermal storage capacity to offer power grid flexibility alongside end-user thermal supply. Cooling dispatch profiles are omitted again from this assessment, as balancing market participation exerts no operational influence on cooling network operations.

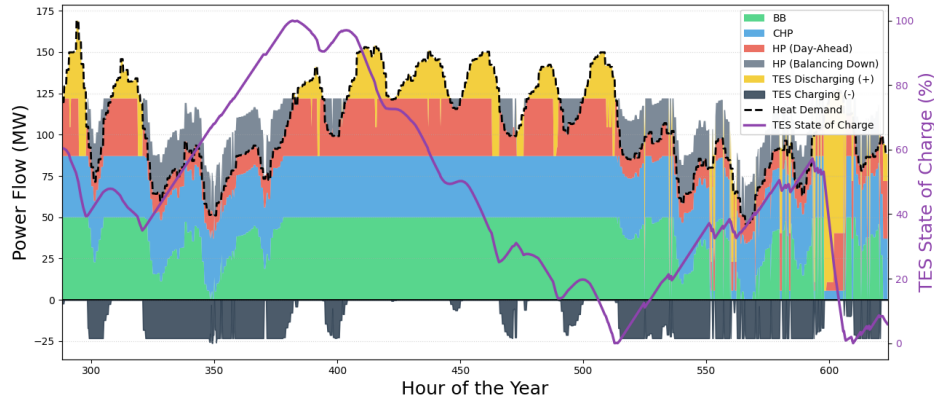


(a) January heat dispatch profile for Sc. 1.

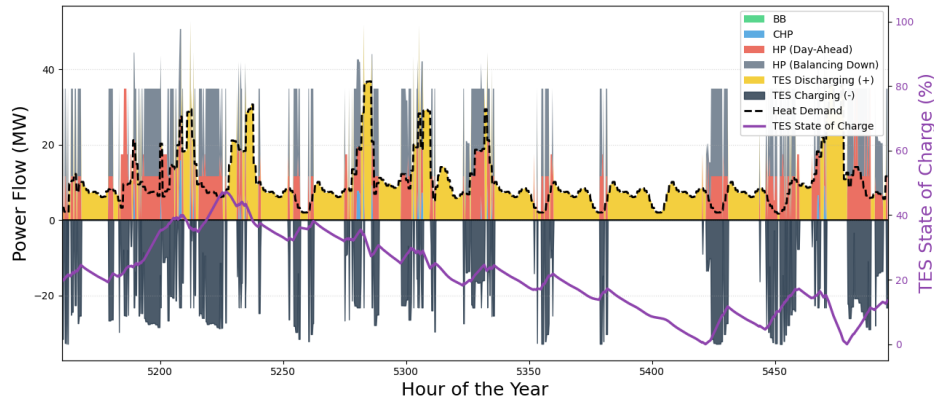


(b) August heat dispatch profile for Sc. 1.

Figure 5.11: Seasonal heat dispatch profiles of Scenario 1 for **Zurich** with full participation in the BM during a greenfield network approach.



(a) January heat dispatch profile for Sc. 1.



(b) August heat dispatch profile for Sc. 1.

Figure 5.12: Seasonal heat dispatch profiles of Scenario 1 for **Amsterdam** with full participation in the BM during a greenfield network approach.

Notably, from the above figures under the greenfield approach with 2024 BM prices, summer heat demand in both the Zurich and Amsterdam networks is satisfied exclusively by the LSHP and the TES. This operational behavior is enabled by the larger TES capacity relative to the brownfield approach, allowing the optimizer to fully capitalize on lucrative downward flexibility prices and DAM prices for this summer period.

Chapter 6

Discussion

Based on the results presented in Chapter 5, this chapter synthesizes the primary insights within the defined optimization framework. It begins by reflecting on the core techno-economic and environmental takeaways, comparing the cross-regional dynamics of the Zurich and Amsterdam case studies, and benchmarking the findings against existing literature. Subsequently, the limitations of this modeling framework are addressed in a critical way. Finally, concrete recommendations for future research are outlined to advance the scope and insights of this study.

6.1 Results Reflection

The results from the first-stage deterministic optimization demonstrate that integrating LSHP and TES can reduce TAC across the examined configurations for both Zurich and Amsterdam networks. Moving from the fossil fuel-based baseline configuration to a LSHP-integrated MES without storage, and ultimately to a fully coupled configuration with all candidate technologies available (including TES), yields marked economic improvements across both district networks under greenfield and brownfield approaches.

The only exception occurs for the Amsterdam brownfield configuration, where integrating a LSHP without TES fails to lower the TAC relative to the fossil fuel-based supply. This outcome is primarily driven by the relative prices of natural gas and electricity: in Amsterdam, natural gas prices remain competitive against DA wholesale electricity prices. Without TES, the LSHP has limited ability to decouple electricity consumption from the timing of heat demand and therefore cannot exploit low-price electricity hours as effectively.

Regarding GHG emissions, the progressive electrification of heating and cooling is consistently associated with lower life-cycle CO₂ emissions, underscoring the benefits of deploying LSHP-TES configurations in mitigating climate change impacts. This decarbonization effect is noticeably more pronounced in Zurich due to the substantially lower carbon intensity of the Swiss electricity generation mix relative to the Dutch one.

These first-stage findings align closely with trends reported in the literature for comparable urban energy systems. For instance, Cheng et al. [26] in an effort to propose

a DHN configuration that will meet China’s carbon neutral target demonstrated that retrofitting a DHN with HPs yielded an economic benefit of 362 million Chinese Yuan alongside an annual abatement of 219,000 tonnes of CO₂. Similarly, Bach et al. [10] verified that coupling HPs and TES with a DHN provides a cost-effective alternative to conventional heat generation by shaving winter peak heat demand, a dispatch dynamic directly validated also by the “HP w/ storage” configuration results and figures across both approaches and particularly into effect for the greenfield approach and the Amsterdam network. Furthermore, Kontu et al. [28] showed that optimizing HP capacity within DHNs reduces average operational expenditures by up to 25%, with the optimal configuration heavily dictated by TES availability and electricity and natural gas prices. This dependency is reflected in the cross-network comparison observed in this work, where the optimizer dimensions a larger TES and favors higher LSHP–TES utilization in Amsterdam compared to Zurich. Finally, García et al. [89] investigated a brownfield integration of a HP into a pre-existing Combined Cycle Gas Turbine (CCGT)–DHN infrastructure, proving that hybrid operation remains profitable across diverse market environments, which substantiates the economic viability demonstrated in the Zurich “HP w/o storage” configuration.

Moving to the second-stage stochastic optimization, the results reveal that full participation in the aFRR energy market achieves an additional 9–13% reduction in TAC across all examined cases. The Zurich brownfield configuration achieves the largest relative economic gains, driven by more lucrative BM prices in the Swiss bidding zone compared to the Netherlands.

These results align with literature on electricity market-integrated heating assets. Pesola et al. [30] observed a 24% reduction in operational expenditure when providing ancillary services through HPs; the higher savings relative to this thesis results stem from participation in more remunerative FCR BM segment for both capacity and energy provisions. Similarly, Tan et al. [22] demonstrated that strategic participation in reserve markets significantly reduces total investment and operational costs, prioritizing investments in TES to unlock flexibility, a trend validated also by the second stage optimization findings in this thesis. Finally, Javanshir et al. [29] in a study regarding power-to-heat sector coupling with P2H technologies (like HPs) and thermal storage established that co-optimizing these P2H assets across DAM, intraday market, and BM provides lucrative revenue streams that enhance investment bankability. This conclusion is reinforced by our results, where BM-integrated configurations consistently achieve the lowest TAC across all examined configurations.

6.2 Limitations of the Study

Notwithstanding the insights provided by the proposed two-stage optimization framework, several methodological and modeling limitations should be acknowledged. First, this study does not conduct explicit sensitivity analysis on key exogenous parameters, such as CAPEX_k of each technology, natural gas prices, or DA electricity prices. Instead, the analysis relies on a cross-regional comparison between Zurich and Amsterdam, eval-

uating how differences in thermal and cooling demand profiles, and prevailing DA and BM prices as well as GHG emissions differences shape system performance across both greenfield and brownfield approaches. This thesis focuses on evaluating different MES configurations that are linked to DHCNs to examine the influence of the presence of different technologies and their BM participation potential instead of performing different sensitivity analysis to exogenous parameters.

Another limitation is related to the scope of the examined electricity markets interactions. This operational co-optimization framework exclusively models the interaction between the DAM and aFRR energy activation. More complex multi-market trading strategies, such as participation in continuous intraday markets and other BM segments such as tertiary reserves and capacity availability markets are not incorporated. Evaluating the simultaneous allocation of flexible capacity and energy across these additional markets could potentially unlock further TAC reductions and operational synergies. Crucially, multi-market synergies across DAM, intraday market, and various BM segments which include also capacity bidding enable DHCN operators to continuously optimize electricity procurement and stack further ancillary service revenues. As demonstrated by [30], such coordinated market participation unlocks cost reductions beyond those realized in this work, where bidding is restricted to the interaction of DAM and energy aFRR activation optimization.

Furthermore, balancing energy dispatch is subject to structural modeling restrictions regarding direct heat delivery. In the second stage stochastic formulation, downward balancing electricity consumed by the LSHP is strictly directed toward charging the TES rather than serving instantaneous network heat demand directly. Although this assumption aligns with this model requirement that baseline heating schedule must be secured through prior scenario-specific DA commitments, it excludes the possibility of downward balancing activations directly contributing to the heat demand.

An additional pivotal limitation is that the BM formulation assumes full bid acceptance ratios for energy activation ($\mu = 1$, $\nu = 1$) whenever LSHP submits an energy activation bid to the TSO. In reality, TSO activation depends on dynamic grid imbalance states and merit-order positioning throughout the year. Nonetheless, the findings of this thesis provide an optimistic assumption about actual activation and they are intended to provide indicative bounds on TAC reductions within the assumptions of the model: the Stage 1 deterministic optimization represents the lower bound (no-BM participation), while the Stage 2 stochastic optimization provides an optimistic upper benchmark for TAC reduction under full bid acceptance. Under a more realistic acceptance bid ratios it is estimated that the reduction in TAC would end up in the middle of this thesis indicative cost reduction boundaries. In practical BM operations by demand-response assets (such as the LSHP-TES coupling), downward flexibility often exhibits acceptance ratios approaching unity. Increasing consumption incurs minimal opportunity cost, as surplus thermal energy is readily absorbed by charging further the TES, allowing these bids to be submitted at highly competitive clearing prices. Conversely, upward flexibility requires curtailing electricity consumption, which risks end-consumer demand not to be served and introduces significant opportunity costs. Consequently, upward bids are

priced high on the merit order with tangible risks of not being included in it, resulting in substantially lower real-world activation ratios.

Finally, several technical and methodological limitations should be acknowledged. The first-stage design optimization is executed deterministically over a single representative year (2025 data), neglecting long-term price uncertainties over the 20-year asset lifetime and as a result proposing investment decisions based on the latest available market conditions. Thermodynamically, the LSHP's COP is modeled as capacity-independent to avoid computationally complex MINLP formulations, varying solely with hourly source water temperatures (see Appendix 8.2), while remaining technologies assume static conversion efficiencies. Furthermore, dynamic operating limits such as ramp rates and minimum up/down times are omitted; while this poses minor discrepancies for fast-responding MW-size LSHP participating in aFRR [29], it represents an optimistic assumption for the higher thermal inertia of the CHP and BB. In addition, asset degradation over the multi-decade lifespan is not captured, potentially understating long-term energy conversion performances. Lastly, minimum load fraction thresholds are relaxed, as strict operational minimums induced solver infeasibilities during low-demand summer periods.

6.3 Recommendations for Further Research

This work can be extended in several directions to build on the current findings. First, the optimization framework itself can be expanded in scope and technical depth. Price uncertainty for natural gas, biomass, and electricity may be incorporated directly into the first-stage design formulation but is likely to result in additional model complexity. The physical representation of assets could also be enriched by adding ramp-up/down constraints and capacity-dependent conversion efficiencies, transitioning the formulation toward an MINLP model. For the second-stage operational model, integrating continuous ID trading alongside DAM and aFRR markets would allow operators to refine electricity positions closer to real time and potentially unlock additional economic value. Finally, the environmental evaluation can be broadened beyond carbon metrics. Future studies could incorporate land-use footprint and water consumption into a generic LCA of the environmental burden of each technology configuration. In addition, multi-objective optimization could be implemented to minimize both TAC and system emissions simultaneously, generating Pareto fronts to explicitly map the techno-economic trade-offs for urban energy planners.

Chapter 7

Conclusions and Outlook

This thesis develops an optimization framework to assess the economic and environmental benefits of integrating LSHP into a DHCN through a MES model. The model is applied to two real-world case studies: Zurich and Amsterdam. The findings can provide practical insights for municipalities with active heat transition plans, DHCN operators, and urban energy investors planning future technology deployments.

Addressing the primary research question, to what extent LSHP-coupled district networks can simultaneously achieve cost-effectiveness and decarbonization, the first-stage optimization results demonstrate that both objectives can be achieved across the examined networks when LSHP is appropriately integrated with TES. In the brownfield approach, the cost-optimal configuration achieves a 16% reduction in TAC for Amsterdam and an 11% reduction for Zurich compared to the fossil fuel-based supply strategy. Both cities achieve a comparable decrease in GHG emissions of approximately 25%, with Zurich seeing slightly higher gains. In the greenfield approach, where generation assets are designed from scratch, TAC decreases by roughly 38% in both cities, with Amsterdam achieving slightly higher cost savings. In terms of emissions, the greenfield configuration lead to a 33% reduction in Amsterdam and a 57% reduction in Zurich. These differences reflect the local energy-system conditions: Amsterdam benefits from lower DAM electricity prices and wider electricity price spreads, creating stronger cost reduction potential, while Zurich’s lower electricity carbon intensity, enables greater emissions reductions.

Addressing the sub-research question regarding the extent to which thermal storage influences the integration of LSHP within DHCNs, the optimization results reveal that TES plays a catalytic role within the examined MES. Across all evaluated configurations, coupling the LSHP with TES systematically reduces TAC and carbon emissions compared to standalone deployments. Notably, for the Amsterdam brownfield network, integrating a LSHP without TES actually increases the TAC, demonstrating that heat storage flexibility is required to make the electrification of this network economically viable with pre-existing CHP and BB assets in the system. Furthermore, TES is the technology that enables the system to reduce its TAC even further through BM participation, where its role is pivotal in storing excess heat in order for the MES to capitalize on the lucrative remuneration by providing both upward and downward flexibility.

Regarding the sub-research question on how location-specific conditions influence the integration and economic benefits of LSHP, pronounced regional divergences emerge between the two municipal networks. In Amsterdam, lower annual cooling demand coupled with more favorable DAM prices and spreads drive lower overall TAC and LCOE_{DN} ; this dynamic incentivizes the optimizer to dimension a larger TES capacity to actively exploit this intraday spreads arbitrage. Conversely, Zurich benefits from more lucrative BM clearing prices and a significantly lower carbon intensity in the Swiss power mix. Consequently, the Zurich network achieves higher system costs reductions through BM participation and greater decarbonization potential when deploying the LSHP–TES configuration within the district network.

Lastly, addressing the sub-research question regarding the potential of the MES model to provide grid flexibility, the second-stage optimization results show that participating in the aFRR energy activation may further reduce the TEAC by 9–13% across both brownfield and greenfield cases. As mentioned before, Zurich achieves a slightly higher cost reduction than Amsterdam due to more favorable BM energy clearance prices.

Overall, coupling LSHP and TES in a MES model within a DHCN provides substantial economic and environmental value, which can be further enhanced by monetizing demand-side flexibility in power balancing markets.

Chapter 8

Appendix

8.1 Literature review Tables

The Tables 8.1, 8.2, 8.3, 8.4 and 8.5 summarize the literature review conducted on the optimization techniques reviewed for the examined thesis topic. The columns of the Tables include, from left to right: the paper title, the main author, the geographical scope of the study, the publication year, the objective function optimized, the type of optimization applied, the technologies considered, whether the study focuses on design, operation, or both, and finally, the time resolution.

Table 8.1: Literature review summary (Papers 1-11)

Paper name	Main Author	Geographical scope	Year	Objective function	Type of optimization	Technologies considered	Feature	Time steps
Early cross-sector decarbonisation for Europe's hard-to-abate sectors: Insights from Denmark's 2030 target	F. Fristed	Denmark & Europe	2025	Min system cost (CAPEX + OPEX)	LP	HP, Wind/Solar, Electrolyzers, CCS	Design & Operational	8760 (Hourly) [90]
Integration of pit thermal energy storages into district heating networks - a techno-economic case study	I. Sifinaos	Denmark (Copenhagen)	2025	Min Levelized Cost of Heat (LCOH)	MILP	Industrial HP, TES, Solar Thermal, CHP	Design & Operational	8760 (Hourly) [91]
Validation and optimization of a solar heating plant with a large-scale heat pump	C. Zhan	Denmark (Høje Taastrup)	2025	Max Solar Fraction & Min Electricity usage	Rule-based optimization (RBO)	LSHP, Solar Thermal, Storage	Operational	8760 (Hourly) [92]
Cost-optimal dimensioning of hybrid heat pump systems utilizing waste heat from hydrogen production for a kindergarten in cold climate	Y. Ju	Finland (Espoo)	2025	Min Life Cycle Cost (LCC)	Genetic Algorithm (GA)	HP, Electrolyzers (Waste Heat), TES	Design	8760 (Hourly) [93]
Techno-economic feasibility of producers in urban district heating systems utilising low-temperature industrial excess heat	U. Traubert	Germany (Kassel)	2025	Max Present Value (NPV)	Merit Order optimization	LSHP, TES	Operational	Variable [94]
Techno-economic analysis of flexible sector coupling between electrical and thermal sectors	R. Calvo Garcia	Sweden (Stockholm)	2025	Max operational profit & grid flexibility	MILP	LSHP, TES	Operational	8760 (Hourly) [95]
Integrating renewable energy and thermal storage in district heating networks: A design optimization approach	A. Ceruti	Germany (Bavaria)	2025	Min TAC & CO2 emissions	Multi-objective MILP	HP, Geothermal, TES, Biomass	Design	8760 (Hourly) [96]
Evaluating the impact of thermal energy storage on grid flexibility in district thermal source networks	A. Etemad	Ireland (Dublin)	2025	Max Grid Flexibility & Min Peak Electricity Demand	Multi-period Simulation	TES, HP	Operational	8760 (Hourly) [97]
Cost-optimization modeling of local level cross-vector flexibility in coupled district heating and electricity networks: A case study of Kungsbacka Sweden	O. Ly-senko	Sweden (Kungsbacka)	2025	Min system operational cost	MILP	TES, HP, CHP, Electric boilers	Operational	8760 (Hourly) [98]
An economic assessment of integrating a pit thermal energy storage in a district heating system	I. Sifinaos	Denmark (Copenhagen)	2024	Min TAC	MILP	Industrial HP, Waste Heat, CHP	Operational	8760 (Hourly) [99]
Economic viability and scalability of a novel domestic hot water substation for 4th generation district heating: A case study of temperature optimization in the Viborg district heating network	M. Tunzi	Denmark (Viborg District Heating)	2024	Minimization of Total Cost of Ownership (TCO)	Multi-parameter Temperature Optimization	DHW Substations, Heat Exchangers	Design & Operational	8760 (Hourly) [100]

Table 8.2: Literature review summary (Papers 12-20)

Paper name	Main Author	Geographical scope	Year	Objective function	Type of optimization	Technologies considered	Feature	Time steps
Dispatch strategies for large-scale heat pump based district heating under high renewable share and risk-aversion: A multistage stochastic optimization approach	M.B. Siddique	Denmark (Western Denmark)	2024	Min of Expected Cost	Multistage Stochastic Optimization	LSHPs, TES, Wind/Solar, Electric Boilers	Operational	8760 (Hourly) [101]
Thermal design and optimization of high-temperature heat pump integrated with district heating benchmarked in Denmark for process heat supply	M.Sadeghi	Denmark	2024	Min TAC & Max COP	Multi-variable Thermodynamic Optimization	High-Temperature HP (HTHP)	Design & Operational	52,560 (10-min) [41]
Dual-objective optimization of large-scale solar heating systems integrated with water-to-water heat pumps for improved technological performance	R. Zhang	China (Western region)	2023	Min of LCC & Max of Solar Fraction	Multi-objective MILP Optimization	Water-to-water HP, Solar Thermal, Water Storage Tanks	Design	8760 (Hourly) [102]
Cost-optimization model to design and operate hybrid heating systems - Case study of district heating system with decentralized heat pumps in Finland	A. Pesola	Finland (Espoo/Helsinki)	2023	Min LCC	MILP	Decentralized HPs, Heat Exchangers, Storage	Design & Operational	8760 (Hourly) [30]
MILP based approach for the preliminary investigation of thermal networks in urban areas	F. Belfiore	Switzerland	2022	Min of TAC & CO2 Emissions	MILP	HPs	Design	8760 (Hourly) [103]
Energy Planning of Beijing Towards Low-carbon, Clean and Efficient Development in 2035	C. Li	China (Beijing)	2022	Min of Total System Cost & Carbon Emissions	Multi-period LP	Air-sourced LSHPs, Geothermal HPs, Gas CHP, Solar/Wind, Evs	Design	8760 (Hourly) [104]
Strategic investment for district heating systems participating in energy and reserve markets using heat flexibility	J. Tan	Denmark	2022	Max of Total Profit (Energy Revenues) + Reserve Revenues	Stochastic MILP	LSHPs, Electric Boilers, TES, CHP	Design & Operational	35,040 (15-min) [22]
Evaluation and Optimization of heat Pump Combined District Heating System: A Case Study of China	L. Cheng	China (Northern China / Tianjin)	2022	Min of Primary Energy Consumption & Total Cost	Multi-objective Pareto Optimization (NSGA-II)	Absorption Heat Pumps (AHP), Electric HPs, Gas Boilers, TES	Design & Operational	8760 (Hourly) [26]
Is Electrified Low-Carbon District Heating Able to Manage Electricity Price Shocks?	N. Javanshir	Finland (Helsinki)	2022	Min of total system cost	LP	LSHPs, Electric Boilers, TES, CHP	Operational	8760 (Hourly) [105]

Table 8.3: Literature review summary (Papers 21-29)

Paper name	Main Author	Geographical scope	Year	Objective function	Type of optimization	Technologies considered	Feature	Time steps
Integration of flexibility potentials of district heating systems into electricity markets: A review	H. Golmohammadi	Europe	2022	Review paper	Review paper	LSHPs, Electric Boilers, TES, CHP, P2H	N/A	N/A (Review) [21]
Operational optimization of a 4th generation DH network with MIQCP	D. Herding	Germany	2022	Min of total operational costs (Electricity + Gas)	Mixed-Integer Quadratically Constrained Programming (MIQCP)	HPs, Electric Boilers, CHP, TES	Operational	72 (3 hourly sample days) [106]
Optimal Design and Dispatch of Electrically Driven Heat Pumps and Chillers for a New Development Area	H. Piep	Denmark (Greater Copenhagen Area)	2021	Min of TAC	MILP	HPs, Chillers, Thermal Storage (Heat & Cold), Seawater heat source	Design & Operational	8760 (Hourly) [107]
Experimental investigations and multi-objective optimization of an air-source absorption heat pump for residential district heating	Z. Wu	China (Tianjin)	2021	Min TAC & Max COP	Multi-objective Pareto Optimization (NSGA-II)	Air-Source Absorption Heat Pumps (ASAHF), Ammonia-Water cycle	Design & Operational	288 (12 hourly typical days) [108]
Thermodynamic performance analysis and modified thermoeconomic cost optimization of a hybrid district heating system considering energy levels	Y. Chen	China	2021	Min of Modified Thermoeconomic Cost (MTEC)	Exergy Analysis optimization	LSHPs, Solar Thermal, Gas CHP, TES	Design & Operational	8760 (Hourly) [109]
District heating in 100% renewable energy systems: Combining industrial excess heat and heat pumps	M. Yuan	Denmark	2021	Min of TAC	Scenario-based Modeling (EnergyPLAN)	LSHPs, Industrial Excess Heat (IEH), Wind/Solar, TES	Design	8760 (Hourly) [110]
Techno-Economic Optimization of a Combined Cycle Combined Heat and Power Plant With Integrated Heat Pump and Low-Temperature Thermal Energy Storage	J. Garcia	Sweden	2020	Max NPV	MILP	CCGT, Industrial HP, Low-Temp TES	Design & Operational	8760 (Hourly) [89]
Technical, economic and environmental optimization of district heating expansion in an urban agglomeration	D. F. Dominkovic	Croatia (Zagreb Urban Area)	2020	Min of Total System Cost & CO2 Emissions	MILP	LSHPs, CHP, TES, Gas Peak Boilers	Design (Expansion Planning)	8760 (Hourly) [111]
Thermo-economic analysis of composite district heating substation with absorption heat pump	Z. Wu	China	2020	Min of TAC	Multi-objective Pareto Optimization (NSGA-II)	Absorption Heat Pump (AHP), Plate Heat Exchanger	Design & Operational	8760 (Hourly) [112]

Table 8.4: Literature review summary (Papers 30-40)

Paper name	Main Author	Geographical scope	Year	Objective function	Type of optimization	Technologies considered	Feature	Time steps
Reducing electricity demand and enhancing heat supply flexibility through energy efficiency and district heating	H. Kauko	Europe	2020	Min System Cost & Min Energy Demand	LP	LSHPs, TES	Design & Operational	8760 (Hourly) [113]
Minimizing Wind Power Curtailment and Carbon Emissions by Power to Heat Sector Coupling-A Stackelberg Game Approach	A. A. Bashir	Finland	2020	Max Profit for Power Grid (Leader) & Min Cost for Program-Heat Grid (Follower)	Stackelberg Game / Bi-level Programming	LSHPs, Electric Boilers, Wind Power, TES	Operational	8760 (Hourly) [114]
Optimal placement and sizing of heat pumps and heat only boilers in a coupled electricity and heating networks	Getnet T. Ayele	France	2019	Min of Total System Cost (Investment + Operational)	MILP	ASHP, Heat Only Boilers (HOB)	Design	8760 (Hourly) [115]
Exergy analysis and thermoeconomic optimization of a district heating network with solar- photovoltaic and heat pumps	Getnet T. Ayele	France	2019	Min of Total Revenue Requirement (TRR)	MILP	Solar-PV, ASHP, TES	Design & Operational	8760 (Hourly) [116]
Modelling framework for integration of large-scale heat pumps in district heating using low-temperature heat sources: A case study of Tallinn, Estonia	H. Pieper	Estonia (Tallinn)	2019	Min Marginal Heat Production Cost	MILP	LSHPs, CHP	Operational	8760 (Hourly) [117]
Introducing modern heat pumps to existing district heating systems - Global lessons from viable decarbonizing of district heating in Finland	K. Kontu	Finland	2019	Min of Socio-economic and Private-economic costs	Scenario-based Energy System Analysis	LSHPs, Excess Heat, Biomass, CHP	Design	8760 (Hourly) [28]
Pseudo-dynamic simulation on a district energy system made of coupling technologies	Getnet T. Ayele	France	2018	Min of daily operational costs & CO2 emissions	Multi-period LP	HPs, HOB, CHP, TES	Operational	8760 (Hourly) [118]
Monte carlo simulation of district heating system short-term operation	I. Dimoulkas	Sweden (Stockholm)	2017	Min of expected operational costs	Stochastic Optimization (Monte Carlo)	CHP, LSHPs, HOB, TES	Operational	24 (Daily) [119]
Barriers for district heating as a source of flexibility for the electricity system	K. Skytte	Scandinavia (Denmark, Norway, Sweden)	2017	N/A	Overview paper	LSHPs, Electric Boilers, CHP, TES	N/A	N/A [120]
Optimal Placement of a Heat Pump in an Integrated Power and Heat Energy System	S. Klyapovskiy	Denmark (Bornholm Island)	2017	Min of Total System Operation and Investment Cost	MILP	LS-ASHP	Design	8760 (Hourly) [121]
A MINLP model of optimal scheduling for a district heating and cooling system: A case study of an energy station in Tianjin	N. Deng	China (Tianjin)	2017	Min of daily operational cost & CO2 emissions	MINLP	HP, Ice Storage, Water Gas Boilers	Operational	24 (Daily) [122]

Table 8.5: Literature review summary (Papers 41-43)

Paper name	Main Author	Geographical scope	Year	Objective function	Type of optimization	Technologies considered	Feature	Time steps
Integration of large-scale heat pumps in the DH systems of Greater Copenhagen	B. Bach	Denmark (Copenhagen)	2016	Min of economic and Private-economic costs	Scenario-based Energy System Modeling (Energy-PLAN)	LSHPs, CHP, Biomass, Wind	Design	8760 (Hourly) [10]
Multi-Objectives, Multi-Period Optimization of district heating networks Using Evolutionary Algorithms and Mixed Integer Linear Programming (MILP)	S. Fazlollahi	France	2012	Min of Total System Cost & CO2 emissions	MILP	HPs, CHP, Biomass, Boilers, TES	Design & Operational	288 hourly typical days) [123]
Environomic multi-objective optimisation of a district heating network considering centralized and decentralized heat pumps	A. Molyneux	Switzerland	2010	Min of LCC & Environmental Impact	Multi-Objective Genetic Algorithm	HPs, Gas Boilers, CHP	Design	288 hourly typical days) [124]

8.2 Lake/Seawater monthly temperature fluctuations for the calculation of COP

The following Table 8.6 represents the monthly temperature values used as the $T_{s,t}$ vector for the modeling of the performance of the COP. These values steam from the the report of the Danish Energy Agency on the “Generation of Electricity and District heating” [59] and refer to the seawater monthly fluctuations of the Copenhagen greater area.

Table 8.6: Seawater temperatures in Denmark

Month	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Average T (°C) (Bay and shore)	4	2	5	7	12	14	17	18	15	11	7	4

8.3 TTF natural gas prices

The following Table 8.7 represents the monthly TTF natural gas wholesale prices of 2025 used as a cost component ($\Lambda_{\text{gas},t}$) of the CHP’s input fuel price and are retrieved from [69].

Table 8.7: TTF monthly natural gas wholesale prices for 2025

Month	Natural Gas TTF Price [€/MWh]
January	45.058
February	48.140
March	47.140
April	41.960
May	35.622
June	35.340
July	36.697
August	33.847
September	32.869
October	32.343
November	31.946
December	30.884

8.4 K-Medoids scenario clustering formulation

To capture the stochastic price uncertainty of electricity markets, characterized by high volatility, extreme price spikes, and non-normal distributions, a K -medoids clustering

algorithm is employed to reduce 10 years of historical 15-minute price time series (2016–2025, each containing 35,040 intervals) into $K = 5$ representative operational scenarios ($K \leq 10$). Unlike K -means, which computes synthetic cluster centroids, K -medoids selects actual historical years as cluster centers (medoids), thereby preserving the physical autocorrelation and peak dynamics of real market data while exhibiting superior robustness against extreme outliers [79, 80].

The clustering problem is formulated as follows according to [79, 80]:

$$\min_{\{m_j\}, \{z_{ij}\}} \sum_{i=1}^N \sum_{j=1}^K z_{ij} d(x_i, m_j) \quad (8.1)$$

$$\text{subject to } \sum_{j=1}^K z_{ij} = 1, \quad \forall i \in \{1, \dots, N\} \quad (8.2)$$

$$z_{ij} \in \{0, 1\}, \quad \forall i \in \{1, \dots, N\}, j \in \{1, \dots, K\} \quad (8.3)$$

where:

- N is the total number of historical yearly price profiles ($N = 10$).
- K is the target number of representative clusters/scenarios ($K = 5$).
- $x_i \in \mathbb{R}^{35040}$ represents the full 15-minute price trajectory of historical year i .
- $m_j \in \{x_1, \dots, x_N\}$ is the designated medoid (representative year) for cluster j .
- $d(x_i, m_j) = \|x_i - m_j\|$ denotes the standard Euclidean distance metric.
- z_{ij} is a binary assignment variable equal to 1 if historical year i is assigned to cluster j , and 0 otherwise.

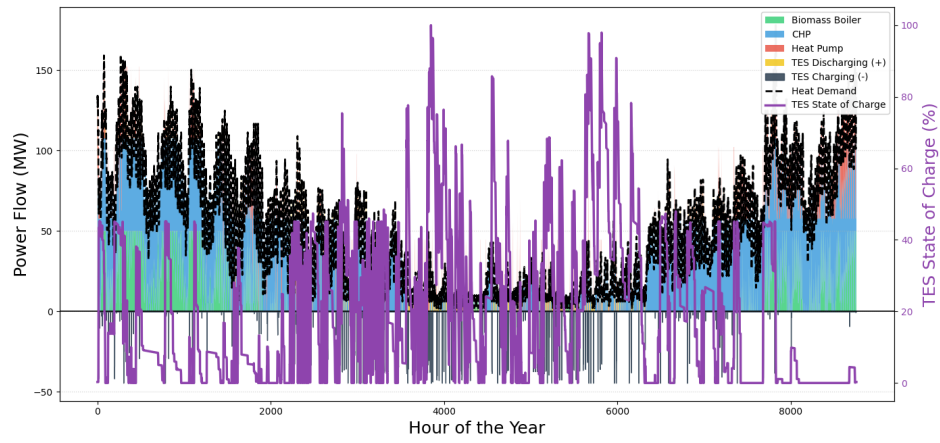
Constraint (8.2) ensures that each historical year is strictly assigned to exactly one cluster medoid. The occurrence probability ρ_s of each representative stochastic scenario $s \in \{1, \dots, K\}$ used in the Stage 2 optimization is calculated directly from its empirical cluster frequency:

$$\rho_s = \frac{N_s}{N} \quad (8.4)$$

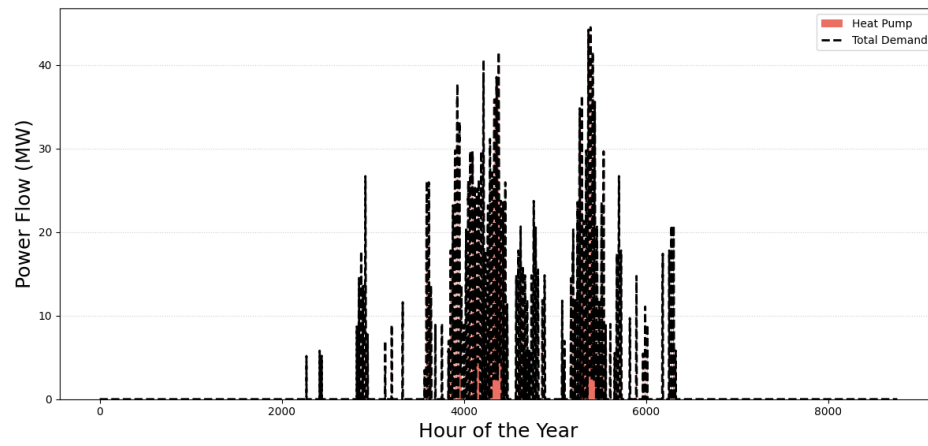
where $N_s = \sum_{i=1}^N z_{is}$ is the number of historical years assigned to medoid s .

8.5 Full year optimization results plots

The following Figure 8.1 represents the full year optimization results for the Zurich brownfield “HP w/ storage” configuration as indicative plots regarding the heating and cooling dispatch of the whole optimization horizon examined.



(a) Full year heating dispatch profile.



(b) Full year cooling dispatch profile.

Figure 8.1: HP w/ storage configuration full year heating and cooling dispatch operational profiles for the **Zurich** brownfield network approach.

Bibliography

- [1] IPCC, “Climate change: a threat to human wellbeing and health of the planet. taking action now can secure our future -press release.”
- [2] N. Adamo, N. Al-Ansari, and V. Sissakian, “Review of climate change impacts on human environment: Past, present and future projections,” vol. 13, no. 11, pp. 605–630.
- [3] Climate Watch, “Greenhouse gas (GHG) emissions by sector.”
- [4] European Council, “Fit for 55.”
- [5] Eurostat, “Energy consumption in households.”
- [6] European Commission, “Heating and cooling.”
- [7] D. Wang, X. Li, J. Marquant, J. Carmeliet, and K. Orehounig, “Advancing the thermal network representation for the optimal design of distributed multi-energy systems,” vol. 9, p. 668124.
- [8] R. Lund, D. D. Ilic, and L. Trygg, “Socioeconomic potential for introducing large-scale heat pumps in district heating in denmark,” vol. 139, pp. 219–229.
- [9] H. Averfalk and S. Werner, “Economic benefits of fourth generation district heating,” vol. 193, p. 116727.
- [10] B. Bach, J. Werling, T. Ommen, M. Münster, J. M. Morales, and B. Elmegaard, “Integration of large-scale heat pumps in the district heating systems of greater copenhagen,” vol. 107, pp. 321–334.
- [11] T. Terlouw, T. Alskaif, C. Bauer, and W. van Sark, “Optimal energy management in all-electric residential energy systems with heat and electricity storage,” vol. 254, p. 113580.
- [12] H. Lund, S. Werner, R. Wiltshire, S. Svendsen, J. E. Thorsen, F. Hvelplund, and B. V. Mathiesen, “4th generation district heating (4gdh): Integrating smart thermal grids into future sustainable energy systems,” vol. 68, pp. 1–11.

- [13] A. R. Mazhar, S. Liu, and A. Shukla, "A state of art review on the district heating systems," vol. 96, pp. 420–439.
- [14] International Energy Agency, "Global energy and CO2 status report 2018,"
- [15] W. Gang, S. Wang, F. Xiao, and D.-c. Gao, "District cooling systems: Technology integration, system optimization, challenges and opportunities for applications," vol. 53, pp. 253–264.
- [16] A. Inayat and M. Raza, "District cooling system via renewable energy sources: A review," vol. 107, pp. 360–373.
- [17] F. Schlosser, M. Jesper, J. Vogelsang, T. G. Walmsley, C. Arpagaus, and J. Hesselbach, "Large-scale heat pumps: Applications, performance, economic feasibility and industrial integration," vol. 133, p. 110219.
- [18] A. Tosatto, A. Dahash, and F. Ochs, "Simulation-based performance evaluation of large-scale thermal energy storage coupled with heat pump in district heating systems," vol. 61, p. 106721.
- [19] H. Lund, P. A. Ostergaard, D. Connolly, and B. V. Mathiesen, "Smart energy and smart energy systems," vol. 137, pp. 556–565.
- [20] H. Pieper, I. Krupenski, W. Brix Markussen, T. Ommen, A. Siirde, and A. Volkova, "Method of linear approximation of COP for heat pumps and chillers based on thermodynamic modelling and off-design operation," vol. 230, p. 120743.
- [21] H. Golmohamadi, K. G. Larsen, P. G. Jensen, and I. R. Hasrat, "Integration of flexibility potentials of district heating systems into electricity markets: A review," vol. 159, p. 112200.
- [22] J. Tan, Q. Wu, and M. Zhang, "Strategic investment for district heating systems participating in energy and reserve markets using heat flexibility," vol. 137, p. 107819.
- [23] J. Janssen, A. Zocca, B. Zwart, and J. Kazempour, "Dynamic dimensioning of frequency containment reserves: The case of the nordic grid," vol. 41, no. 3, pp. 1744–1755.
- [24] G. A. Lunde, E. V. Damm, P. A. V. Gade, and J. Kazempour, "Aggregator of electric vehicles bidding in nordic FCR-d markets: A chance-constrained program,"
- [25] European Parliament and Council of the European Union, "Regulation (EU) 2019/943 on the internal market for electricity." *Usr_lan*: EN.
- [26] L. Cheng, Z. Yu, S. Xia, S. Li, Y. Li, H. Zhang, B. Li, S. Zhang, Z. Liu, and W. Zheng, "Evaluation and optimization of heat pump combined district heating system: A case study of china," vol. 15, no. 20, p. 7622.

- [27] I. Sifnaios, G. Gauthier, D. Trier, J. Fan, and A. R. Jensen, “Dronninglund water pit thermal energy storage dataset,” vol. 251, pp. 68–76.
- [28] K. Kontu, S. Rinne, and S. Junnila, “Introducing modern heat pumps to existing district heating systems – global lessons from viable decarbonizing of district heating in finland,” vol. 166, pp. 862–870.
- [29] N. Javanshir, S. Syri, S. Tervo, and A. Rosin, “Operation of district heat network in electricity and balancing markets with the power-to-heat sector coupling,” vol. 266, p. 126423.
- [30] A. Pesola, “Cost-optimization model to design and operate hybrid heating systems – case study of district heating system with decentralized heat pumps in finland,” vol. 281, p. 128241.
- [31] Gurobi, “Gurobi optimization.”
- [32] MAN Energy Solutions, “Mega heat pump delivers first heat in esbjerg.”
- [33] I. Sifnaios and A. R. Jensen, “Performance analysis of the høje taastrup water pit thermal energy storage,” vol. 131, p. 117504.
- [34] I. Sifnaios, S. Furbo, and A. R. Jensen, “Monitoring data of the høje taastrup water pit thermal energy storage,” vol. 58, p. 111305.
- [35] Helen, “Eiranranta heat pump plant.”
- [36] Smart City Sweden, “Stockholm innovates district heating with new solutions and renewable sources | news.”
- [37] K. Hognabba, F. Pettersson, and R. Zevenhoven, “Optimizing the integration of power-to-heat technologies in district energy infrastructures in turku, finland, using an MILP formulation,” vol. 14, pp. 3641–3656.
- [38] T. Terlouw, P. Gabrielli, T. AlSkaif, C. Bauer, R. McKenna, and M. Mazzotti, “Optimal economic and environmental design of multi-energy systems,” vol. 347, p. 121374.
- [39] A. Delangle, R. S. Lambert, N. Shah, S. Acha, and C. N. Markides, “Modelling and optimising the marginal expansion of an existing district heating network,” vol. 140, pp. 209–223.
- [40] T. Mertz, S. Serra, A. Henon, and J.-M. Reneaume, “A MINLP optimization of the configuration and the design of a district heating network: Academic study cases,” vol. 117, pp. 450–464.
- [41] M. Sadeghi, T. Petersen, Z. Yang, B. Zühlsdorf, and K. S. Madsen, “Thermal design and optimization of high-temperature heat pump integrated with district heating benchmarked in denmark for process heat supply,” vol. 159, pp. 356–370.

- [42] X. Zhang, G. Strbac, F. Teng, and P. Djapic, “Economic assessment of alternative heat decarbonisation strategies through coordinated operation with electricity system – UK case study,” vol. 222, pp. 79–91.
- [43] “Demand-side frequency response based on district heating system integrated with heat pump.”
- [44] M. Zhang, Q. Wu, J. Wen, B. Zhou, Q. Guan, J. Tan, Z. Lin, and F. Fang, “Day-ahead stochastic scheduling of integrated electricity and heat system considering reserve provision by large-scale heat pumps,” vol. 307, p. 118143.
- [45] Axpo, “10 answers about axpo’s energy trading.”
- [46] S. W. Werner, “Annual heat losses in district heating networks,” vol. 11:6.
- [47] S. Werner, “District heating and cooling,” in *Reference Module in Earth Systems and Environmental Sciences*, p. B9780124095489010940, Elsevier.
- [48] Swiss Energy Research Center, “Ehub modeling and optimization of ERZ district heating network in zurich.”
- [49] City of Amsterdam, “New amsterdam climate - roadmap amsterdam climate neutral 2050 - march 2020,”
- [50] P. Lukasova, A. Kubin, and R. Prochazka, “Dispatch optimization and sizing of molten-salt power-to-heat systems in balancing electricity markets for industrial heat electrification,” vol. 364, p. 121706.
- [51] F. Glover, “Improved linear integer programming formulations of nonlinear integer problems,” vol. 22, no. 4, pp. 455–460.
- [52] SWM Infrastruktur, “Lastprofil warmepumpe.”
- [53] BDEW, “Standardlastprofile strom.”
- [54] European Union, “Hotmaps open data repositories.”
- [55] M. N. Mistry, “Historical global gridded degree-days: A high-spatial resolution database of CDD and HDD,” vol. 6, no. 2, pp. 214–221.
- [56] REPSOL, “Ideal air conditioning temperature at home in winter and summer.”
- [57] L. M. Hofmann, A. Clerjon, F. Perdu, R. Baviere, and P. Azais, “How much district heating in a future sector-coupled energy system? a comprehensive methodology for mapping and clustering the infrastructure costs on an entire territory.”
- [58] International Energy Agency, “Energy efficiency 2025,”
- [59] Danish Energy Agency, “Technology data for generation of electricity and district heating | energistyrelsen.”

- [60] M. Rama and K. Sipila, “Transition to low temperature distribution in existing systems,” vol. 116, pp. 58–68.
- [61] Eurostat, “Reporting instructions for completing the district heating and district cooling template for data reporting under article 24(6) of directive 2012/27/EU,”
- [62] EWZ, “Seewasserverbund CoolCity ewz.”
- [63] M. Jesper, F. Schlosser, F. Pag, T. G. Walmsley, B. Schmitt, and K. Vajen, “Large-scale heat pumps: Uptake and performance modelling of market-available devices,” vol. 137, p. 110646.
- [64] ASHRAE, “HVAC equipment minimum efficiency requirements - HVAC systems encyclopedia.”
- [65] M. Doninelli, G. Di Marcoberardino, C. Invernizzi, and P. Iora, “An efficient small-scale biomass cogeneration plant with silicon tetrachloride as working fluid: Techno-economic analysis,” vol. 261, p. 125272.
- [66] Future Market Insights, “Combined heat and power (CHP) systems market : Global industry analysis and opportunity assessment, 2036.”
- [67] A. Dahash, F. Ochs, and A. Tosatto, “Techno-economic and exergy analysis of tank and pit thermal energy storage for renewables district heating systems,” vol. 180, pp. 1358–1379.
- [68] E. Guelpa and V. Verda, “Thermal energy storage in district heating and cooling systems: A review,” vol. 252, p. 113474.
- [69] Protergia, “TTF prices per month | protergia.”
- [70] ACER, “Network tariffs | ACER.”
- [71] Trading Economics, “EU carbon permits - price - chart - historical data - news.”
- [72] V. Quaschnig, “Specific carbon dioxide emissions of various fuels.”
- [73] ENTSO-E, “Transparency platform.”
- [74] EKZ, “Aktuelle tarife fur geschäftskunden ab 100 MWh.”
- [75] TenneT, “Electricity grid tariffs in the netherlands.”
- [76] M. Cermak, J. Malatakova, J. Malatak, M. Aniszewska, and A. Gendek, “Regional wood chip quality parameters decomposition and price linkage with impact on polish energy sustainability: Time frequency analysis between 2013 and 2019,” vol. 10, no. 13, p. e33322.
- [77] SwissGrid, “Swiss electricity balancing market.”

- [78] TenneT, “Download page transparency.”
- [79] A. Jhones Passos Nascimento, M. Berrehil El Kattel, J. Antonio Fernandes De Macedo, and F. L. M. Antunes, “Integrated time series analysis, clustering, and forecasting for energy efficiency optimization and tariff management,” vol. 13, pp. 59309–59325.
- [80] Y. Sun, M. Li, I. Valiyeva, and Z. Xakimov, “Uncovering residential electricity consumption patterns using an integrated time-series clustering approach,” vol. 65, p. 102214.
- [81] JASM - Data platform, “Energy conversion technologies in STEM.”
- [82] Organisation for Economic Co-operation and Development, *Projected Costs of Generating Electricity*. OECD Publishing.
- [83] W. Salameh, J. Faraj, E. Harika, R. Murr, and M. Khaled, “On the optimization of electrical water heaters: Modelling simulations and experimentation,” vol. 14, no. 13, p. 3912.
- [84] “Biomass boiler costs UK 2025: Complete price guide & payback times | trecO.”
- [85] T. Terlouw, N. Savvakis, C. Bauer, R. McKenna, and G. Arampatzis, “Designing multi-energy systems in mediterranean regions towards energy autonomy,” vol. 377, p. 124458.
- [86] G. Wernet, C. Bauer, B. Steubing, J. Reinhard, E. Moreno-Ruiz, and B. Weidema, “The ecoinvent database version 3 (part i): overview and methodology,” vol. 21, no. 9, pp. 1218–1230.
- [87] Forest Research, “Typical calorific values of fuels.”
- [88] International Energy Agency, *Energy Statistics Manual*. OECD.
- [89] J. Garcia, V. Smet, R. Guedez, and A. Sorce, “Techno-economic optimization of a combined cycle combined heat and power plant with integrated heat pump and low-temperature thermal energy storage,” American Society of Mechanical Engineers Digital Collection.
- [90] F. Fristed, S. Tonning, Z. Xie, L. Langer, and G. Bruun Andresen, “Early cross-sector decarbonisation for europe’s hard-to-abate sectors: Insights from denmark’s 2030 target,” vol. 401.
- [91] I. Sifnaios, J. Fan, and A. R. Jensen, “Integration of pit thermal energy storages into district heating networks – a techno-economic case study,” vol. 279, p. 127770.
- [92] C. Zhan, Y. Xu, J. Fan, M. Gao, W. Kong, J. Wu, D. Wang, and Z. Tian, “Validation and optimization of a solar heating plant with a large-scale heat pump,” vol. 319, p. 134898.

- [93] Y. Ju, X. Hu, J. Jokisalo, R. Kosonen, T. Xue, A. Meriläinen, and A. Kosonen, “Cost-optimal dimensioning of hybrid heat pump systems utilizing waste heat from hydrogen production for a kindergarten in cold climate,” vol. 332, p. 115430.
- [94] U. Trabert, F. Pag, J. Orozalieva, U. Jordan, and K. Vajen, “Techno-economic feasibility of prosumers in urban district heating systems utilising low-temperature industrial excess heat,” vol. 330, p. 136583.
- [95] R. Calvo García, J. M. Marín Arcos, S. Kumar, S. N. Gunasekara, and J. Thakur, “Techno-economic analysis of flexible sector coupling between electrical and thermal sectors,” vol. 27, p. 101145.
- [96] A. Ceruti, J. Lambert, and H. Spliethoff, “Integrating renewable energy and thermal storage in district heating networks: A design optimization approach,” vol. 345, p. 120323.
- [97] A. Etemad, A. Trousseau, D. Finn, J. O’Shea, A. Maccarini, A. Afshari, and J. O’Donnell, “Evaluating the impact of thermal energy storage on grid flexibility in district thermal source networks,” vol. 3140, no. 6, p. 062003.
- [98] O. Lysenko, M. Mata, and P. Sobha, “Cost-optimization modeling of local level cross-vector flexibility in coupled district heating and electricity networks: A case study of kungsbacka sweden,” no. 4, pp. 8–17.
- [99] I. Sifnaios, J. Fan, and A. R. Jensen, “An economic assessment of integrating a pit thermal energy storage in a district heating system,” vol. 103, p. 114266.
- [100] M. Tunzi, Q. Yang, J. B. Olesen, T. Diget, and L. C. Fournel, “Economic viability and scalability of a novel domestic hot water substation for 4th generation district heating: A case study of temperature optimization in the viborg district heating network,” vol. 313, p. 134010.
- [101] M. B. Siddique, D. Keles, F. Scheller, and P. S. Nielsen, “Dispatch strategies for large-scale heat pump based district heating under high renewable share and risk-aversion: A multistage stochastic optimization approach,” vol. 136, p. 107764.
- [102] R. Zhang, D. Wang, Z. Yu, Y. Sun, H. Wan, Y. Liu, Q. Jiao, M. Gao, J. Fan, and B. Lan, “Dual-objective optimization of large-scale solar heating systems integrated with water-to-water heat pumps for improved techno-economic performance,” vol. 296, p. 113281.
- [103] F. Belfiore, T. Rey, J. Page, and F. Maréchal, “MILP based approach for the preliminary investigation of thermal networks in urban areas,” vol. 17 of *Building Simulation*, pp. 581–588, IBPSA.
- [104] C. Li, N. Wang, X. Shen, Y. Zhang, Z. Yang, X. Tong, F. Maréchal, L. Wang, and Y. Yang, “Energy planning of beijing towards low-carbon, clean and efficient development in 2035,” vol. 10, no. 3, pp. 913–928.

- [105] N. Javanshir, P. Hiltunen, and S. Syri, “Is electrified low-carbon district heating able to manage electricity price shocks?,” in *2022 18th International Conference on the European Energy Market (EEM)*, pp. 1–5, IEEE.
- [106] D. Hering, M. R. Faller, A. Xhonneux, and D. Müller, “Operational optimization of a 4th generation district heating network with mixed integer quadratically constrained programming,” vol. 250, p. 123766.
- [107] H. Piep, “Optimal design and dispatch of electrically driven heat pumps and chillers for a new development area,”
- [108] Z. Wu, S. You, H. Zhang, Y. Wang, Y. Jiang, Z. Liu, L. Sha, and S. Wei, “Experimental investigations and multi-objective optimization of an air-source absorption heat pump for residential district heating,” vol. 240, p. 114267.
- [109] Y. Chen, H. Hua, J. Wang, and P. D. Lund, “Thermodynamic performance analysis and modified thermo-ecological cost optimization of a hybrid district heating system considering energy levels,” vol. 224, p. 120067.
- [110] M. Yuan, J. Z. Thellufsen, P. Sorknæs, H. Lund, and Y. Liang, “District heating in 100% renewable energy systems: Combining industrial excess heat and heat pumps,” vol. 244, p. 114527.
- [111] D. F. Dominković, G. Stunjek, I. Blanco, H. Madsen, and G. Krajačić, “Technical, economic and environmental optimization of district heating expansion in an urban agglomeration,” vol. 197, p. 117243.
- [112] Z. Wu, Y. Wang, S. You, H. Zhang, X. Zheng, J. Guo, and S. Wei, “Thermo-economic analysis of composite district heating substation with absorption heat pump,” vol. 166, p. 114659.
- [113] H. Kauko, B. M. Delgado, S. Backe, and I. Sartori, “Reducing electricity demand and enhancing heat supply flexibility through energy efficiency and district heating,” vol. 322, p. 135310.
- [114] A. A. Bashir, A. Lund, M. Pourakbari-Kasmaei, and M. Lehtonen, “Minimizing wind power curtailment and carbon emissions by power to heat sector coupling—a stackelberg game approach,” vol. 8, pp. 211892–211911.
- [115] G. T. Ayele, M. T. Mabrouk, P. Haurant, B. Laumert, and B. Lacarrière, “Optimal placement and sizing of heat pumps and heat only boilers in a coupled electricity and heating networks,” vol. 182, pp. 122–134.
- [116] G. T. Ayele, M. T. Mabrouk, P. Haurant, B. Laumert, B. Lacarrière, and M. Santarelli, “Exergy analysis and thermo-economic optimization of a district heating network with solar- photovoltaic and heat pumps,” in *ECOS 2019 proceedings*, ECOS 2019 proceedings.

- [117] H. Pieper, V. Masatin, A. Volkova, T. Ommen, B. Elmegaard, and W. B. Markussen, “Modelling framework for integration of large-scale heat pumps in district heating using low-temperature heat sources: A case study of tallinn, estonia,” vol. 20.
- [118] G. T. Ayele, M. T. Mabrouk, P. Haurant, B. Laumert, and B. Lacarrière, “Pseudodynamic simulation on a district energy system made of coupling technologies,”
- [119] I. Dimoukas and M. Amelin, “Monte carlo simulation of district heating system short-term operation in electricity markets,” vol. 63, no. 3.
- [120] K. Skytte, O. J. Olsen, and E. R. Soysal, “Barriers for district heating as a source of flexibility for the electricity system,”
- [121] S. Klyapovskiy, S. You, H. W. Bindner, and H. Cai, “Optimal placement of a heat pump in an integrated power and heat energy system,” in *2017 Ninth Annual IEEE Green Technologies Conference (GreenTech)*, pp. 246–253. ISSN: 2166-5478.
- [122] N. Deng, R. Cai, Y. Gao, Z. Zhou, G. He, D. Liu, and A. Zhang, “A MINLP model of optimal scheduling for a district heating and cooling system: A case study of an energy station in tianjin,” vol. 141, pp. 1750–1763.
- [123] S. Fazlollahi, S. L. Bungener, G. Becker, and F. Maréchal, “Multi-objectives, multi-period optimization of district heating networks using evolutionary algorithms and mixed integer linear programming (MILP),” in *22 European Symposium on Computer Aided Process Engineering* (I. D. L. Bogle and M. Fairweather, eds.), vol. 30 of *Computer Aided Chemical Engineering*, pp. 262–266, Elsevier.
- [124] A. Molyneaux, G. Leyland, and D. Favrat, “Environomic multi-objective optimisation of a district heating network considering centralized and decentralized heat pumps,” vol. 35, no. 2, pp. 751–758.



Eidgenössische Technische Hochschule Zürich
Swiss Federal Institute of Technology Zurich

Declaration of originality

The signed declaration of originality is a component of every written paper or thesis authored during the course of studies. **In consultation with the supervisor**, one of the following two options must be selected:

- ☐ I hereby declare that I authored the work in question independently, i.e. that no one helped me to author it. Suggestions from the supervisor regarding language and content are excepted. I used no generative artificial intelligence technologies¹.
- ☒ I hereby declare that I authored the work in question independently. In doing so I only used the authorised aids, which included suggestions from the supervisor regarding language and content and generative artificial intelligence technologies. The use of the latter and the respective source declarations proceeded in consultation with the supervisor.

Title of paper or thesis:

System-level optimization for the integration of heat pumps in district heating and cooling

Authored by:

If the work was compiled in a group, the names of all authors are required.

Last name(s):

Fourlaris

First name(s):

Konstantinos

With my signature I confirm the following:

- I have adhered to the rules set out in the [Citation Guidelines](#).
- I have documented all methods, data and processes truthfully and fully.
- I have mentioned all persons who were significant facilitators of the work.

I am aware that the work may be screened electronically for originality.

Place, date

Zurich, 02.09.2026

Signature(s)

If the work was compiled in a group, the names of all authors are required. Through their signatures they vouch jointly for the entire content of the written work.

¹ For further information please consult the ETH Zurich websites, e.g. <https://ethz.ch/en/the-eth-zurich/education/ai-in-education.html> and <https://library.ethz.ch/en/researching-and-publishing/scientific-writing-at-eth-zurich.html> (subject to change).